31792 -- Advanced Optimization and Game Theory for Energy Systems

Lecture 6: Robust approaches for market clearing

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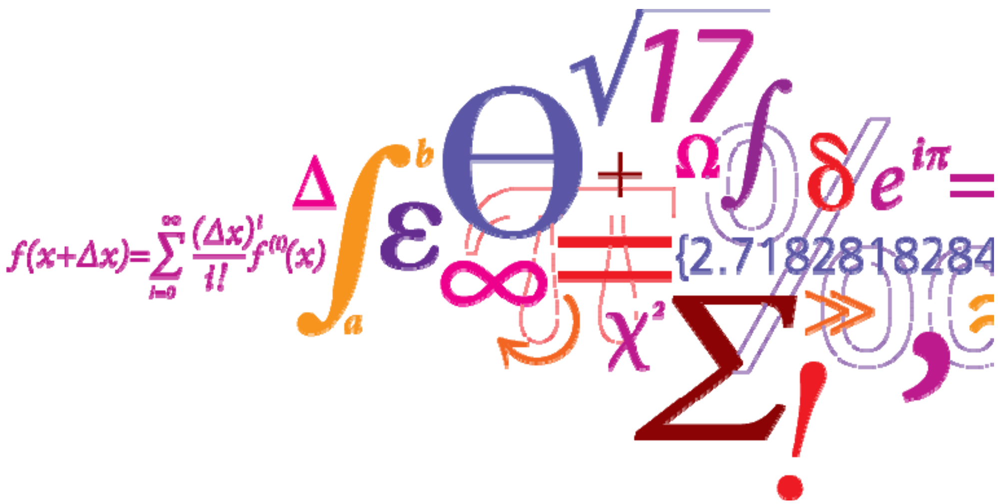
(with an acknowledgment to Dr. Christos Ordoudis for his contribution to prepare this lecture)



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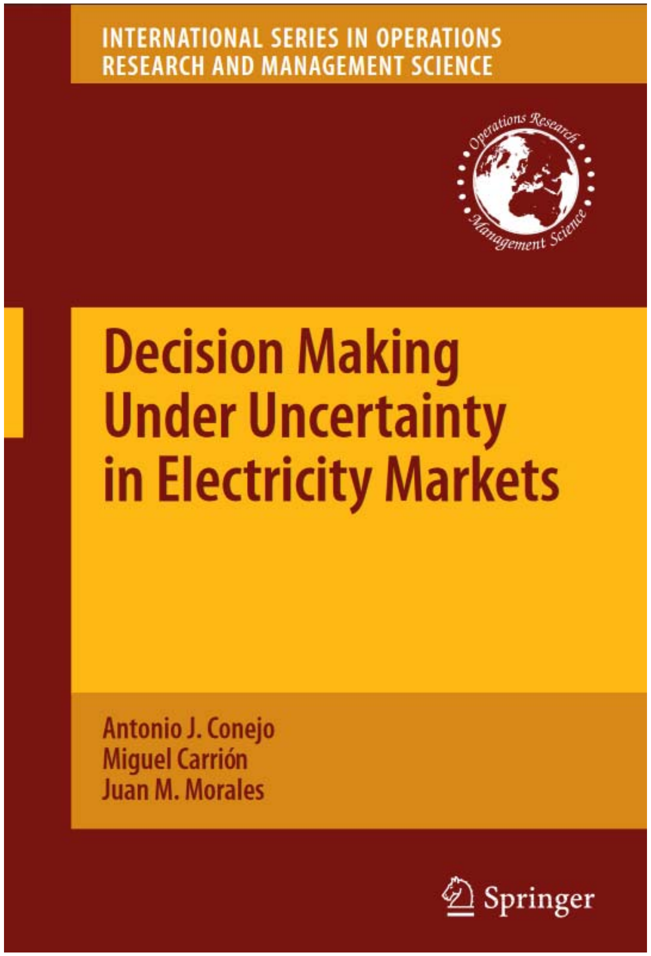
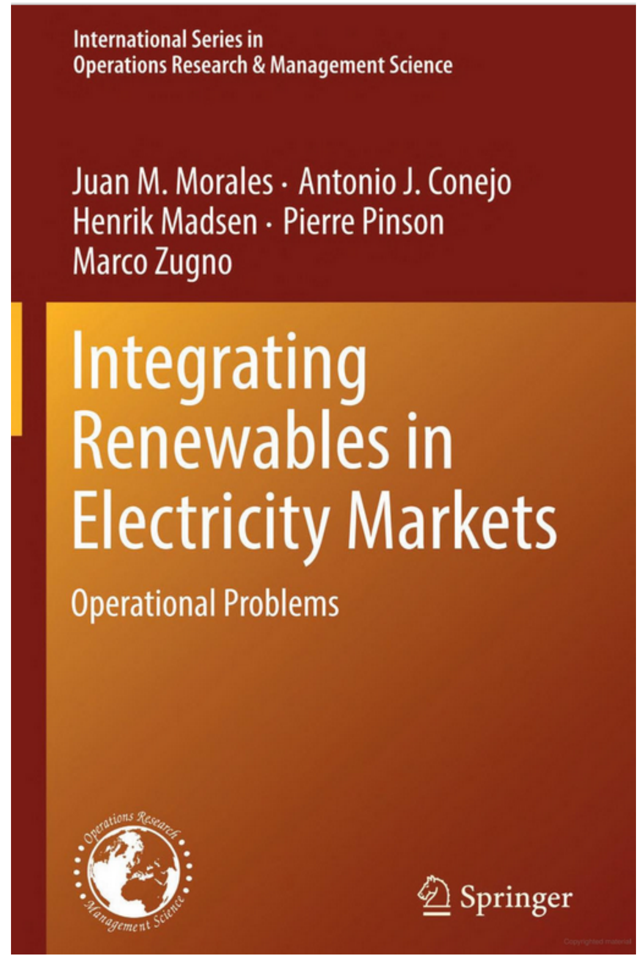
DTU Electrical Engineering
Department of Electrical Engineering

Learning objectives

- Define robust optimization problems
- Explain three different approaches to tackle robust optimization problems
- Solve robust optimization problems
- Define chance-constrained programs and describe the differences with robust optimization and stochastic programming
- Identify distributionally robust optimization problems and describe two different types of ambiguity sets

References


INTERNATIONAL SERIES IN OPERATIONS RESEARCH AND MANAGEMENT SCIENCE
Operations Research Management Science
Decision Making Under Uncertainty in Electricity Markets
Antonio J. Conejo
Miguel Carrión
Juan M. Morales
Springer


International Series in Operations Research & Management Science
Juan M. Morales · Antonio J. Conejo
Henrik Madsen · Pierre Pinson
Marco Zugno
Integrating Renewables in Electricity Markets
Operational Problems
Operations Research Management Science
Springer

More references

Lectures on Robust Convex Optimization

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H. Milton Stewart School of Industrial and Systems Engineering
Georgia Institute of Technology, Atlanta Georgia 30332-0205 USA

https://www2.isye.gatech.edu/~nemirovs/RO_LN.pdf

LECTURES ON STOCHASTIC PROGRAMMING

MODELING AND THEORY

Alexander Shapiro
Georgia Institute of Technology
Atlanta, Georgia

Darinka Dentcheva
Stevens Institute of Technology
Hoboken, New Jersey

Andrzej Ruszczyński
Rutgers University
New Brunswick, New Jersey

https://www2.isye.gatech.edu/people/faculty/Alex_Shapiro/SPbook.pdf

Recap from the previous lecture

Stochastic market clearing: optimization

Minimize (total operational cost[1] of the system in DA) +

(total expected operational cost of the system in RT)

subject to:

- Production limits (in DA and RT)
- Transmission network limits (in DA and RT)
- Load shedding limits (in RT)
- Nodal power balances (in DA and RT)

[1] We assume loads are inelastic to price, otherwise the social welfare should be maximized.

DA: day-ahead

RT: real-time

Recall this example from the last lecture

G1

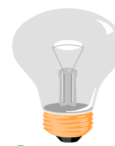
- Capacity: 100 MW
- Production cost: $10/MWh
- **<u>Inflexible</u>** in real time (RT)

G2

- Capacity: 30 MW
- Production cost: $20/MWh
- **<u>Fully flexible</u>** in RT

WP

- Capacity: 70 MW
- Production cost: $0/MWh
- Uncertain production in DA



Load

- 120 MW (inelastic and deterministic)
- Curtailment cost: $80/MWh

Recall this example from the last lecture


G1

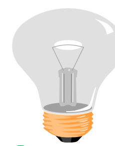
- Capacity: 100 MW
- Production cost: $10/MWh
- **<u>Inflexible</u>** in real time (RT)


G2

- Capacity: 30 MW
- Production cost: $20/MWh
- **<u>Fully flexible</u>** in RT


WP

- Capacity: 70 MW
- Production cost: $0/MWh
- Uncertain production in DA


Load

- 120 MW (inelastic and deterministic)
- Curtailment cost: $80/MWh

A slight change with respect to the example in the previous lecture:

Let's assume the wind power production under 4 equiprobable scenarios are 30 MW, 60 MW, 70 MW and 40 MW, respectively.

Stochastic market clearing outcomes

	DA	RT (scenario 1) Wind power realization: 30 MW	RT (scenario 2) Wind power realization: 60 MW	RT (scenario 3) Wind power realization: 70 MW	RT (scenario 4) Wind power realization: 40 MW
G1 [MW]	60	Unchanged	Unchanged	Unchanged	Unchanged
G2 [MW]	30	Unchanged	-30	-30	-10
WP [MW]	30	Unchanged	+30	+30 (10 spilled)	+10
Load [MW]	120	Not curtailed	Not curtailed	Not curtailed	Not curtailed
Corresponding operational cost [$]	1200	0	-600	-600	-200

Stochastic market clearing outcomes

	DA	RT (scenario 1) Wind power realization: 30 MW	RT (scenario 2) Wind power realization: 60 MW	RT (scenario 3) Wind power realization: 70 MW	RT (scenario 4) Wind power realization: 40 MW
G1 [MW]	60	Unchanged	Unchanged	Unchanged	Unchanged
G2 [MW]	30	Unchanged	-30	-30	-10
WP [MW]	30	Unchanged	+30	+30 (10 spilled)	+10
Load [MW]	120	Not curtailed	Not curtailed	Not curtailed	Not curtailed
Corresponding operational cost [$]	1200	0	-600	-600	-200

- Total expected operational cost of the system = 1200 – 0.25*[0+600+600+200] = $850

Stochastic market clearing outcomes

	DA	RT (scenario 1) Wind power realization: 30 MW	RT (scenario 2) Wind power realization: 60 MW	RT (scenario 3) Wind power realization: 70 MW	RT (scenario 4) Wind power realization: 40 MW
G1 [MW]	60	Unchanged	Unchanged	Unchanged	Unchanged
G2 [MW]	30	Unchanged	-30	-30	-10
WP [MW]	30	Unchanged	+30	+30 (10 spilled)	+10
Load [MW]	120	Not curtailed	Not curtailed	Not curtailed	Not curtailed
Corresponding operational cost [$]	1200	0	-600	-600	-200

- Total <u>expected</u> operational cost of the system = 1200 – 0.25*[0+600+600+200] = $850
- If scenario 1 realizes, the total operational cost of the system will be $1200.
- If either scenario 2 or 3 realizes, the total operational cost of the system will be $600.
- If scenario 4 realizes, the total operational cost of the system will be $1000.

Robust (not stochastic!) market clearing

Robust (scenario-based) market clearing

Let's assume that uncertainty is represented by the same set of scenarios, namely, w1, w2, w3 and w4!

Robust (scenario-based) market clearing

Let's assume that uncertainty is represented by the same set of scenarios, namely, w1, w2, w3 and w4!

Minimize (total operational cost[1] of the system in DA) +

(total operational cost of the system in RT under the worst-case scenario)

subject to:

- Production limits (in DA and RT)
- Transmission network limits (in DA and RT)
- Load shedding limits (in RT)
- Nodal power balances (in DA and RT)

[1] We assume loads are inelastic to price, otherwise the social welfare should be maximized.

Robust (scenario-based) market clearing

$$\min_{\substack{p^{\mathrm{G1,DA}},p^{\mathrm{G2,DA}},p^{\mathrm{WP,DA}},p^{\mathrm{G2,RT}}_{\omega_1},p^{\mathrm{G2,RT}}_{\omega_2},p^{\mathrm{G2,RT}}_{\omega_3},p^{\mathrm{G2,RT}}_{\omega_4},\\ p^{\mathrm{shed}}_{\omega_1},p^{\mathrm{shed}}_{\omega_2},p^{\mathrm{shed}}_{\omega_3},p^{\mathrm{shed}}_{\omega_4},p^{\mathrm{spill}}_{\omega_1},p^{\mathrm{spill}}_{\omega_2},p^{\mathrm{spill}}_{\omega_3},p^{\mathrm{spill}}_{\omega_4},\beta}} [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \beta$$

s.t.

$$0 \leq p^{\mathrm{G1,DA}} \leq 100$$
$$0 \leq p^{\mathrm{G2,DA}} \leq 30$$
$$0 \leq p^{\mathrm{WP,DA}} \leq 70$$
$$p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120$$
$$0 \leq (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}_{\omega_1}) \leq 30$$
$$-30 \leq p^{\mathrm{G2,RT}}_{\omega_1} \leq 30$$
$$0 \leq p^{\mathrm{spill}}_{\omega_1} \leq 30$$
$$0 \leq p^{\mathrm{shed}}_{\omega_1} \leq 120$$
$$p^{\mathrm{G2,RT}}_{\omega_1} + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}_{\omega_1}) + p^{\mathrm{shed}}_{\omega_1} = 0$$
$$20p^{\mathrm{G2,RT}}_{\omega_1} + 80p^{\mathrm{shed}}_{\omega_1} \leq \beta$$
$$20p^{\mathrm{G2,RT}}_{\omega_2} + 80p^{\mathrm{shed}}_{\omega_2} \leq \beta$$
$$20p^{\mathrm{G2,RT}}_{\omega_3} + 80p^{\mathrm{shed}}_{\omega_3} \leq \beta$$
$$20p^{\mathrm{G2,RT}}_{\omega_4} + 80p^{\mathrm{shed}}_{\omega_4} \leq \beta$$

Robust (scenario-based) market clearing

$$\min_{\boxed{p^{\mathrm{G1,DA}}, p^{\mathrm{G2,DA}}, p^{\mathrm{WP,DA}}}, p^{\mathrm{G2,RT}}_{\omega_1}, p^{\mathrm{G2,RT}}_{\omega_2}, p^{\mathrm{G2,RT}}_{\omega_3}, p^{\mathrm{G2,RT}}_{\omega_4}, p^{\mathrm{shed}}_{\omega_1}, p^{\mathrm{shed}}_{\omega_2}, p^{\mathrm{shed}}_{\omega_3}, p^{\mathrm{shed}}_{\omega_4}, p^{\mathrm{spill}}_{\omega_1}, p^{\mathrm{spill}}_{\omega_2}, p^{\mathrm{spill}}_{\omega_3}, p^{\mathrm{spill}}_{\omega_4}, \beta} [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \beta$$

s. t.

$$0 \leq p^{\mathrm{G1,DA}} \leq 100$$
$$0 \leq p^{\mathrm{G2,DA}} \leq 30$$
$$0 \leq p^{\mathrm{WP,DA}} \leq 70$$
$$p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120$$
$$0 \leq (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}_{\omega_1}) \leq 30$$
$$-30 \leq p^{\mathrm{G2,RT}}_{\omega_1} \leq 30$$
$$0 \leq p^{\mathrm{spill}}_{\omega_1} \leq 30$$
$$0 \leq p^{\mathrm{shed}}_{\omega_1} \leq 120$$
$$p^{\mathrm{G2,RT}}_{\omega_1} + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}_{\omega_1}) + p^{\mathrm{shed}}_{\omega_1} = 0$$
$$20p^{\mathrm{G2,RT}}_{\omega_1} + 80p^{\mathrm{shed}}_{\omega_1} \leq \beta$$
$$20p^{\mathrm{G2,RT}}_{\omega_2} + 80p^{\mathrm{shed}}_{\omega_2} \leq \beta$$
$$20p^{\mathrm{G2,RT}}_{\omega_3} + 80p^{\mathrm{shed}}_{\omega_3} \leq \beta$$
$$20p^{\mathrm{G2,RT}}_{\omega_4} + 80p^{\mathrm{shed}}_{\omega_4} \leq \beta$$

DA variables: schedules of G1, G2 and WP

Robust (scenario-based) market clearing

$$\min_{\substack{p^{\mathrm{G1,DA}},p^{\mathrm{G2,DA}},p^{\mathrm{WP,DA}},\boxed{p_{\omega_1}^{\mathrm{G2,RT}},p_{\omega_2}^{\mathrm{G2,RT}},p_{\omega_3}^{\mathrm{G2,RT}},p_{\omega_4}^{\mathrm{G2,RT}}}\\ p_{\omega_1}^{\mathrm{shed}},p_{\omega_2}^{\mathrm{shed}},p_{\omega_3}^{\mathrm{shed}},p_{\omega_4}^{\mathrm{shed}},p_{\omega_1}^{\mathrm{spill}},p_{\omega_2}^{\mathrm{spill}},p_{\omega_3}^{\mathrm{spill}},p_{\omega_4}^{\mathrm{spill}},\beta}} [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \beta$$

s. t.

$$0 \le p^{\mathrm{G1,DA}} \le 100$$
$$0 \le p^{\mathrm{G2,DA}} \le 30$$
$$0 \le p^{\mathrm{WP,DA}} \le 70$$
$$p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120$$
$$0 \le (p^{\mathrm{G2,DA}} + p_{\omega_1}^{\mathrm{G2,RT}}) \le 30$$
$$-30 \le p_{\omega_1}^{\mathrm{G2,RT}} \le 30$$
$$0 \le p_{\omega_1}^{\mathrm{spill}} \le 30$$
$$0 \le p_{\omega_1}^{\mathrm{shed}} \le 120$$
$$p_{\omega_1}^{\mathrm{G2,RT}} + (30 - p^{\mathrm{WP,DA}} - p_{\omega_1}^{\mathrm{spill}}) + p_{\omega_1}^{\mathrm{shed}} = 0$$
$$20p_{\omega_1}^{\mathrm{G2,RT}} + 80p_{\omega_1}^{\mathrm{shed}} \le \beta$$
$$20p_{\omega_2}^{\mathrm{G2,RT}} + 80p_{\omega_2}^{\mathrm{shed}} \le \beta$$
$$20p_{\omega_3}^{\mathrm{G2,RT}} + 80p_{\omega_3}^{\mathrm{shed}} \le \beta$$
$$20p_{\omega_4}^{\mathrm{G2,RT}} + 80p_{\omega_4}^{\mathrm{shed}} \le \beta$$

RT variables: flexible unit G2's adjustment under scenarios w1 to w4

Robust (scenario-based) market clearing

$$\min_{\substack{p^{\mathrm{G1,DA}},p^{\mathrm{G2,DA}},p^{\mathrm{WP,DA}},p^{\mathrm{G2,RT}}_{\omega_1},p^{\mathrm{G2,RT}}_{\omega_2},p^{\mathrm{G2,RT}}_{\omega_3},p^{\mathrm{G2,RT}}_{\omega_4},\\ \boxed{p^{\mathrm{shed}}_{\omega_1},p^{\mathrm{shed}}_{\omega_2},p^{\mathrm{shed}}_{\omega_3},p^{\mathrm{shed}}_{\omega_4}},p^{\mathrm{spill}}_{\omega_1},p^{\mathrm{spill}}_{\omega_2},p^{\mathrm{spill}}_{\omega_3},p^{\mathrm{spill}}_{\omega_4},\beta}} [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \beta$$

s.t.

$$0 \le p^{\mathrm{G1,DA}} \le 100$$
$$0 \le p^{\mathrm{G2,DA}} \le 30$$
$$0 \le p^{\mathrm{WP,DA}} \le 70$$
$$p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120$$
$$0 \le (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}_{\omega_1}) \le 30$$
$$-30 \le p^{\mathrm{G2,RT}}_{\omega_1} \le 30$$
$$0 \le p^{\mathrm{spill}}_{\omega_1} \le 30$$
$$0 \le p^{\mathrm{shed}}_{\omega_1} \le 120$$
$$p^{\mathrm{G2,RT}}_{\omega_1} + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}_{\omega_1}) + p^{\mathrm{shed}}_{\omega_1} = 0$$
$$20p^{\mathrm{G2,RT}}_{\omega_1} + 80p^{\mathrm{shed}}_{\omega_1} \le \beta$$
$$20p^{\mathrm{G2,RT}}_{\omega_2} + 80p^{\mathrm{shed}}_{\omega_2} \le \beta$$
$$20p^{\mathrm{G2,RT}}_{\omega_3} + 80p^{\mathrm{shed}}_{\omega_3} \le \beta$$
$$20p^{\mathrm{G2,RT}}_{\omega_4} + 80p^{\mathrm{shed}}_{\omega_4} \le \beta$$

RT variables: load curtailment
under scenarios w1 to w4

Robust (scenario-based) market clearing

$$\min_{\substack{p^{\mathrm{G1,DA}}, p^{\mathrm{G2,DA}}, p^{\mathrm{WP,DA}}, p^{\mathrm{G2,RT}}_{\omega_1}, p^{\mathrm{G2,RT}}_{\omega_2}, p^{\mathrm{G2,RT}}_{\omega_3}, p^{\mathrm{G2,RT}}_{\omega_4}, \\ p^{\mathrm{shed}}_{\omega_1}, p^{\mathrm{shed}}_{\omega_2}, p^{\mathrm{shed}}_{\omega_3}, p^{\mathrm{shed}}_{\omega_4}, \boxed{p^{\mathrm{spill}}_{\omega_1}, p^{\mathrm{spill}}_{\omega_2}, p^{\mathrm{spill}}_{\omega_3}, p^{\mathrm{spill}}_{\omega_4}}, \beta}} [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \beta$$

s. t.

$$0 \leq p^{\mathrm{G1,DA}} \leq 100$$
$$0 \leq p^{\mathrm{G2,DA}} \leq 30$$
$$0 \leq p^{\mathrm{WP,DA}} \leq 70$$
$$p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120$$
$$0 \leq (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}_{\omega_1}) \leq 30$$
$$-30 \leq p^{\mathrm{G2,RT}}_{\omega_1} \leq 30$$
$$0 \leq p^{\mathrm{spill}}_{\omega_1} \leq 30$$
$$0 \leq p^{\mathrm{shed}}_{\omega_1} \leq 120$$
$$p^{\mathrm{G2,RT}}_{\omega_1} + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}_{\omega_1}) + p^{\mathrm{shed}}_{\omega_1} = 0$$
$$20p^{\mathrm{G2,RT}}_{\omega_1} + 80p^{\mathrm{shed}}_{\omega_1} \leq \beta$$
$$20p^{\mathrm{G2,RT}}_{\omega_2} + 80p^{\mathrm{shed}}_{\omega_2} \leq \beta$$
$$20p^{\mathrm{G2,RT}}_{\omega_3} + 80p^{\mathrm{shed}}_{\omega_3} \leq \beta$$
$$20p^{\mathrm{G2,RT}}_{\omega_4} + 80p^{\mathrm{shed}}_{\omega_4} \leq \beta$$

RT variables: wind power spillage
under scenarios w1 to w4

Robust (scenario-based) market clearing

$$\min_{\substack{p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p^{G2,RT}_{\omega_1}, p^{G2,RT}_{\omega_2}, p^{G2,RT}_{\omega_3}, p^{G2,RT}_{\omega_4}, \\ p^{shed}_{\omega_1}, p^{shed}_{\omega_2}, p^{shed}_{\omega_3}, p^{shed}_{\omega_4}, p^{spill}_{\omega_1}, p^{spill}_{\omega_2}, p^{spill}_{\omega_3}, p^{spill}_{\omega_4}, \beta}} \boxed{[10p^{G1,DA} + 20p^{G2,DA}]} + \beta$$

Total operational cost of the system in DA

s. t.

DA constraints

$$0 \leq p^{G1,DA} \leq 100$$
$$0 \leq p^{G2,DA} \leq 30$$
$$0 \leq p^{WP,DA} \leq 70$$
$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120$$

$$0 \leq (p^{G2,DA} + p^{G2,RT}_{\omega_1}) \leq 30$$
$$-30 \leq p^{G2,RT}_{\omega_1} \leq 30$$
$$0 \leq p^{spill}_{\omega_1} \leq 30$$
$$0 \leq p^{shed}_{\omega_1} \leq 120$$
$$p^{G2,RT}_{\omega_1} + (30 - p^{WP,DA} - p^{spill}_{\omega_1}) + p^{shed}_{\omega_1} = 0$$
$$20p^{G2,RT}_{\omega_1} + 80p^{shed}_{\omega_1} \leq \beta$$
$$20p^{G2,RT}_{\omega_2} + 80p^{shed}_{\omega_2} \leq \beta$$
$$20p^{G2,RT}_{\omega_3} + 80p^{shed}_{\omega_3} \leq \beta$$
$$20p^{G2,RT}_{\omega_4} + 80p^{shed}_{\omega_4} \leq \beta$$

Robust (scenario-based) market clearing

$$\min_{\substack{p^{\text{G1,DA}}, p^{\text{G2,DA}}, p^{\text{WP,DA}}, p^{\text{G2,RT}}_{\omega_1}, p^{\text{G2,RT}}_{\omega_2}, p^{\text{G2,RT}}_{\omega_3}, p^{\text{G2,RT}}_{\omega_4}, \\ p^{\text{shed}}_{\omega_1}, p^{\text{shed}}_{\omega_2}, p^{\text{shed}}_{\omega_3}, p^{\text{shed}}_{\omega_4}, p^{\text{spill}}_{\omega_1}, p^{\text{spill}}_{\omega_2}, p^{\text{spill}}_{\omega_3}, p^{\text{spill}}_{\omega_4}, \beta}} [10p^{\text{G1,DA}} + 20p^{\text{G2,DA}}] + \beta$$

s. t.

$$0 \leq p^{\text{G1,DA}} \leq 100$$
$$0 \leq p^{\text{G2,DA}} \leq 30$$
$$0 \leq p^{\text{WP,DA}} \leq 70$$
$$p^{\text{G1,DA}} + p^{\text{G2,DA}} + p^{\text{WP,DA}} = 120$$
$$0 \leq (p^{\text{G2,DA}} + p^{\text{G2,RT}}_{\omega_1}) \leq 30$$
$$-30 \leq p^{\text{G2,RT}}_{\omega_1} \leq 30$$
$$0 \leq p^{\text{spill}}_{\omega_1} \leq 30$$
$$0 \leq p^{\text{shed}}_{\omega_1} \leq 120$$
$$p^{\text{G2,RT}}_{\omega_1} + (30 - p^{\text{WP,DA}} - p^{\text{spill}}_{\omega_1}) + p^{\text{shed}}_{\omega_1} = 0$$
$$20p^{\text{G2,RT}}_{\omega_1} + 80p^{\text{shed}}_{\omega_1} \leq \beta$$
$$20p^{\text{G2,RT}}_{\omega_2} + 80p^{\text{shed}}_{\omega_2} \leq \beta$$
$$20p^{\text{G2,RT}}_{\omega_3} + 80p^{\text{shed}}_{\omega_3} \leq \beta$$
$$20p^{\text{G2,RT}}_{\omega_4} + 80p^{\text{shed}}_{\omega_4} \leq \beta$$

RT constraints under scenario w1 (similar constraints are needed for scenarios w2 to w4)

Robust (scenario-based) market clearing

$$\min_{\substack{p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p^{G2,RT}_{\omega_1}, p^{G2,RT}_{\omega_2}, p^{G2,RT}_{\omega_3}, p^{G2,RT}_{\omega_4},\\ p^{shed}_{\omega_1}, p^{shed}_{\omega_2}, p^{shed}_{\omega_3}, p^{shed}_{\omega_4}, p^{spill}_{\omega_1}, p^{spill}_{\omega_2}, p^{spill}_{\omega_3}, p^{spill}_{\omega_4}, \boxed{\beta}}} [10p^{G1,DA} + 20p^{G2,DA}] + \boxed{\beta}$$

s. t.

$$0 \leq p^{G1,DA} \leq 100$$
$$0 \leq p^{G2,DA} \leq 30$$
$$0 \leq p^{WP,DA} \leq 70$$
$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120$$
$$0 \leq (p^{G2,DA} + p^{G2,RT}_{\omega_1}) \leq 30$$
$$-30 \leq p^{G2,RT}_{\omega_1} \leq 30$$
$$0 \leq p^{spill}_{\omega_1} \leq 30$$
$$0 \leq p^{shed}_{\omega_1} \leq 120$$
$$p^{G2,RT}_{\omega_1} + (30 - p^{WP,DA} - p^{spill}_{\omega_1}) + p^{shed}_{\omega_1} = 0$$
$$\boxed{\begin{aligned} 20p^{G2,RT}_{\omega_1} + 80p^{shed}_{\omega_1} &\leq \beta \\ 20p^{G2,RT}_{\omega_2} + 80p^{shed}_{\omega_2} &\leq \beta \\ 20p^{G2,RT}_{\omega_3} + 80p^{shed}_{\omega_3} &\leq \beta \\ 20p^{G2,RT}_{\omega_4} + 80p^{shed}_{\omega_4} &\leq \beta \end{aligned}}$$

Total operational cost
of the system under the
worst-case scenario

Robust (scenario-based) market clearing

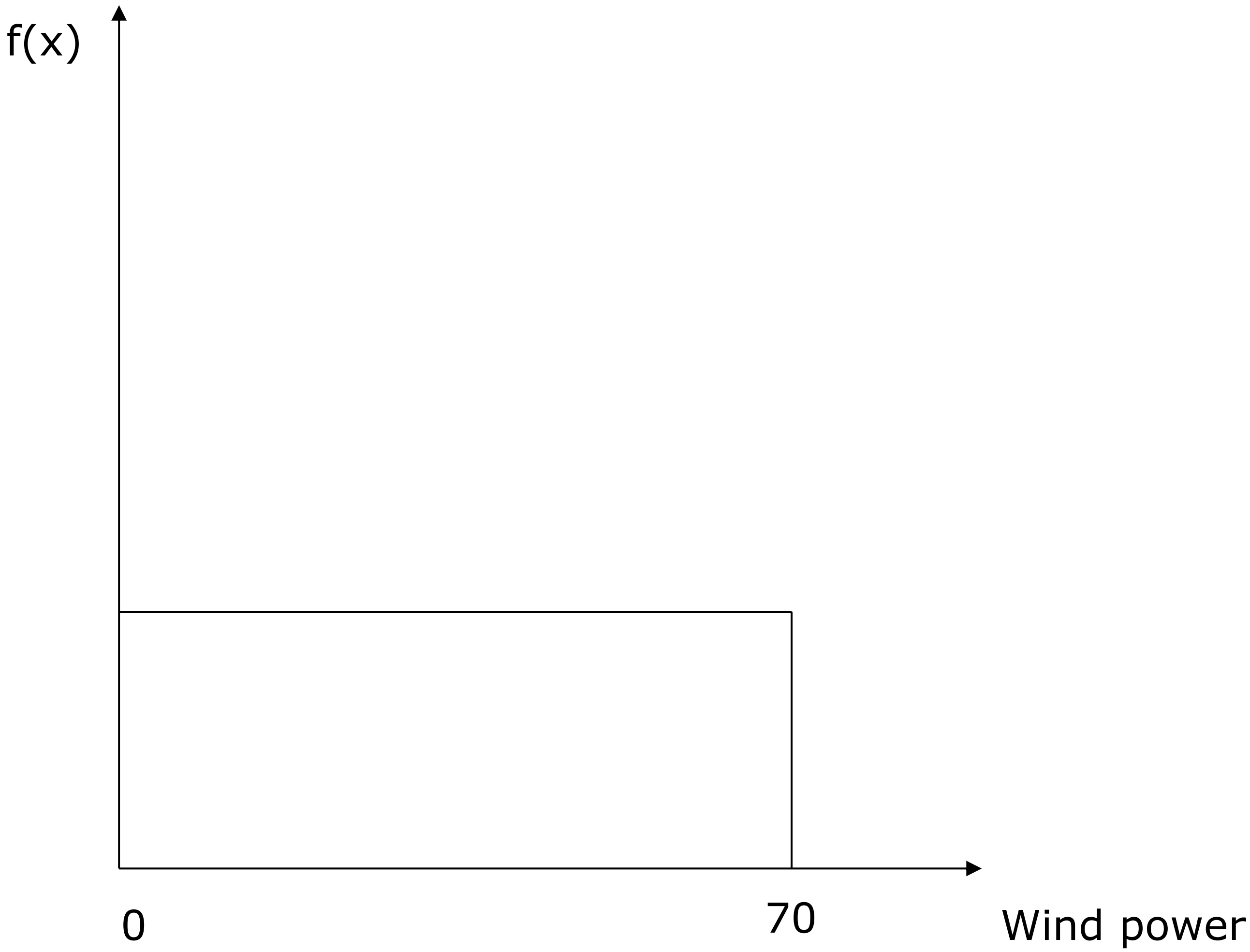
	DA	RT (scenario 1) Wind power realization: 30 MW	RT (scenario 2) Wind power realization: 60 MW	RT (scenario 3) Wind power realization: 70 MW	RT (scenario 4) Wind power realization: 40 MW
G1 [MW]	90	Unchanged	Unchanged	Unchanged	Unchanged
G2 [MW]	0	Unchanged	Unchanged	Unchanged	Unchanged
WP [MW]	30	Unchanged	Unchanged (30 spilled)	Unchanged (40 spilled)	Unchanged (10 spilled)
Load [MW]	120	Not curtailed	Not curtailed	Not curtailed	Not curtailed
Corresponding operational cost [$]	900	0	0	0	0

Robust (scenario-based) market clearing

	DA	RT (scenario 1) Wind power realization: 30 MW	RT (scenario 2) Wind power realization: 60 MW	RT (scenario 3) Wind power realization: 70 MW	RT (scenario 4) Wind power realization: 40 MW
G1 [MW]	90	Unchanged	Unchanged	Unchanged	Unchanged
G2 [MW]	0	Unchanged	Unchanged	Unchanged	Unchanged
WP [MW]	30	Unchanged	Unchanged (30 spilled)	Unchanged (40 spilled)	Unchanged (10 spilled)
Load [MW]	120	Not curtailed	Not curtailed	Not curtailed	Not curtailed
Corresponding operational cost [$]	900	0	0	0	0

- Total expected operational cost of the system = 900 – 0.25*[0+0+0+0] = $900
- If either w1, w2, w3 or w4 realizes, the total operational cost of the system will be $900.

Uncertainty characterization

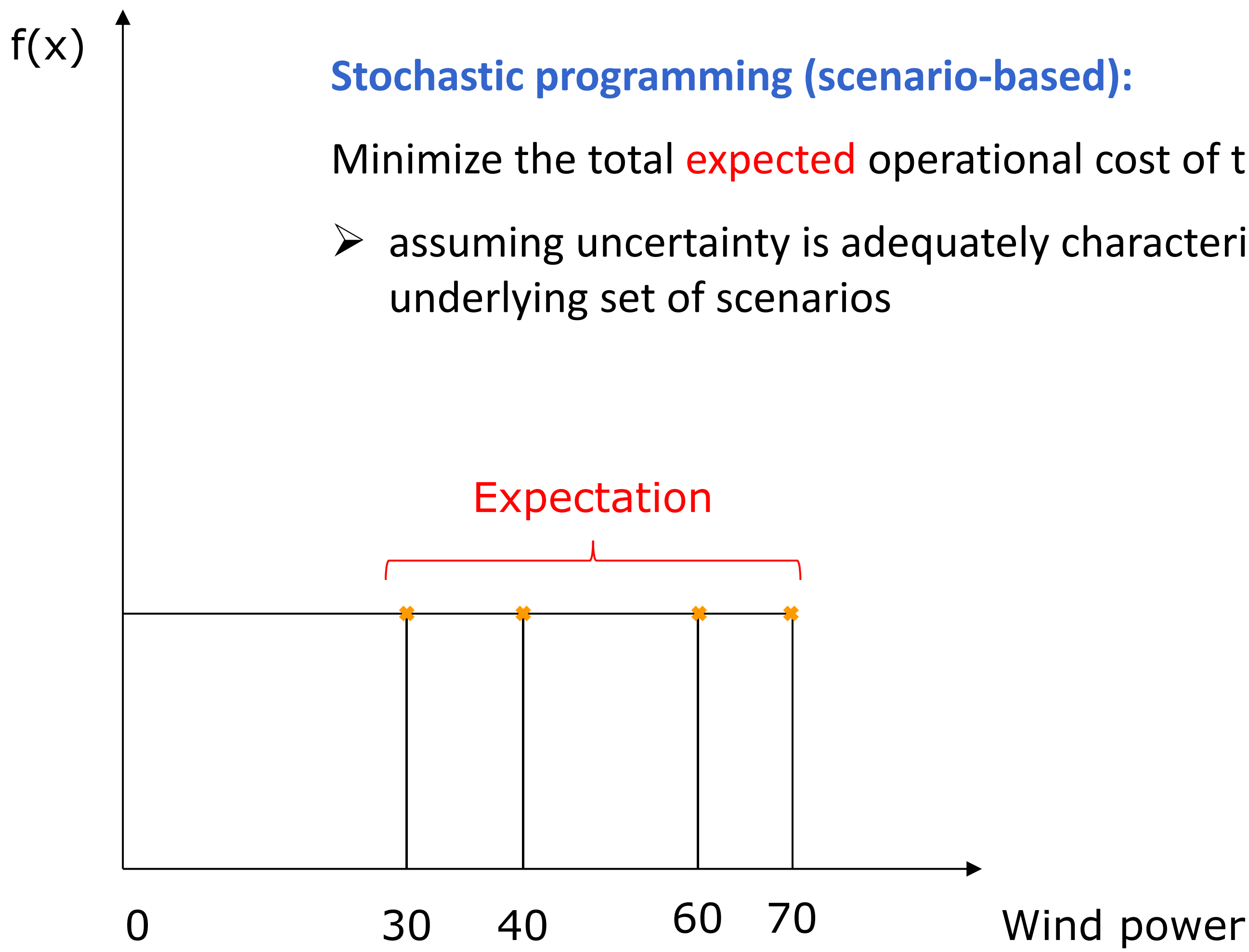


Uncertainty characterization

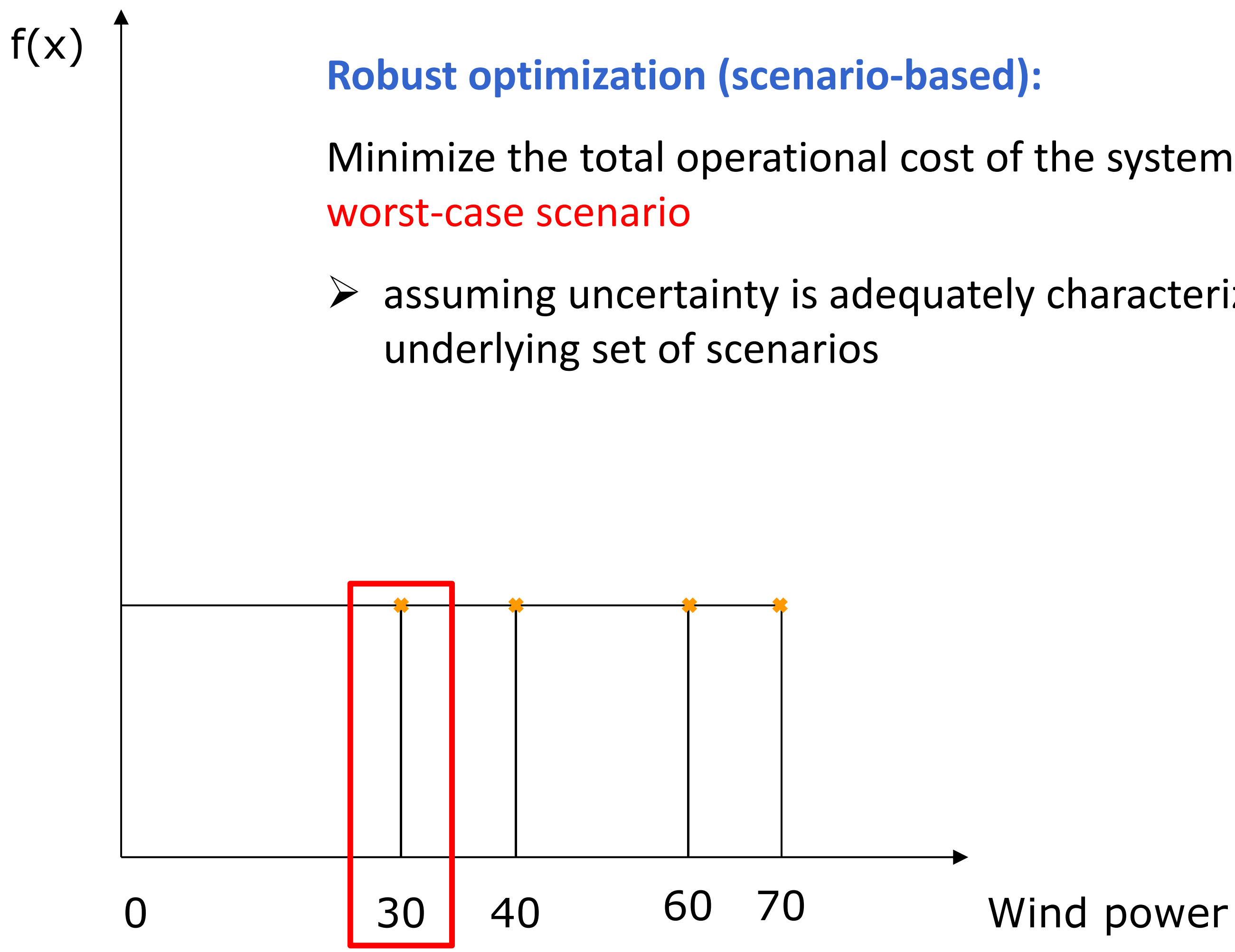
Stochastic programming (scenario-based):

Minimize the total expected operational cost of the system

- assuming uncertainty is adequately characterized by the underlying set of scenarios



Uncertainty characterization



Robust optimization (scenario-based):

Minimize the total operational cost of the system under the worst-case scenario

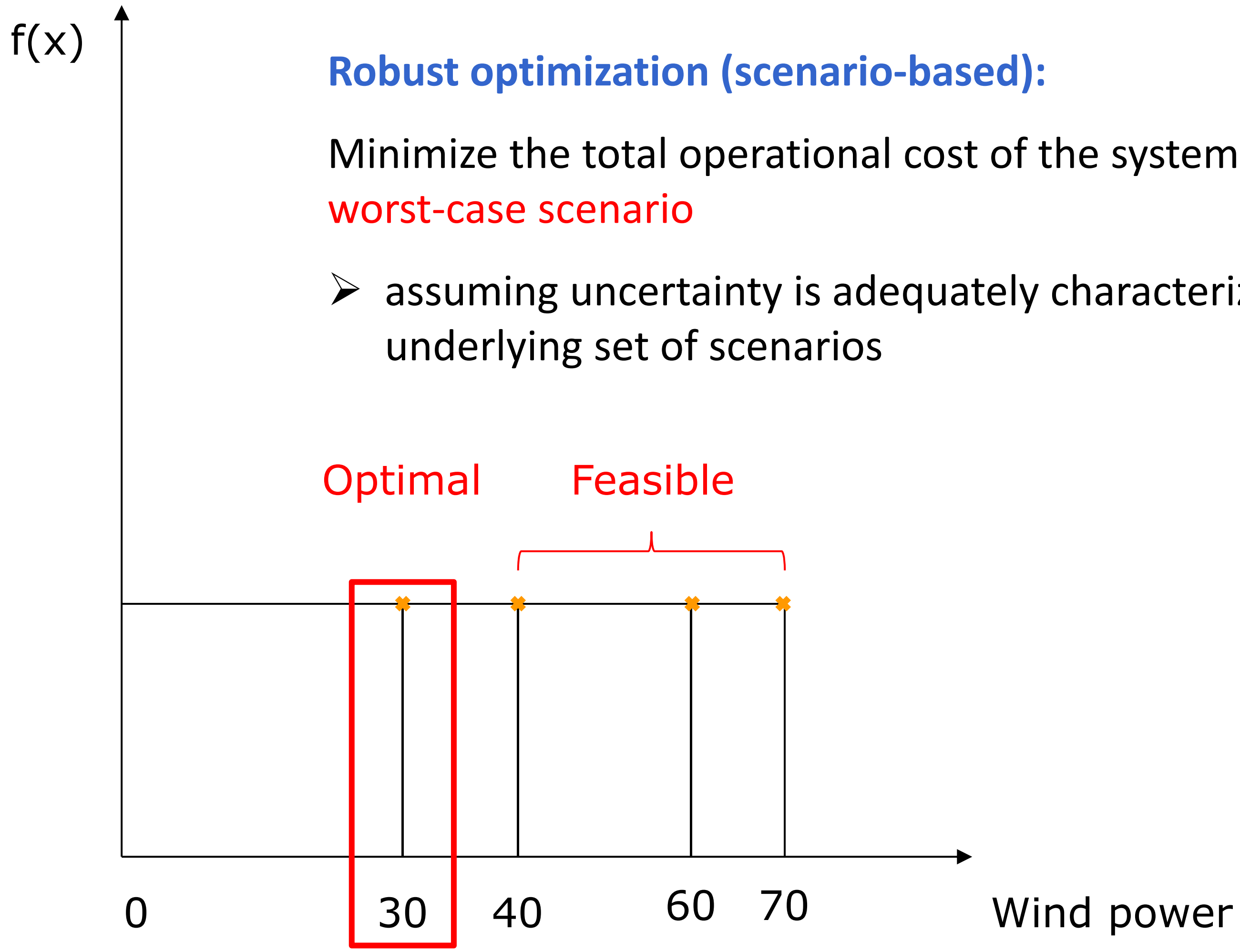
- assuming uncertainty is adequately characterized by the underlying set of scenarios

Uncertainty characterization

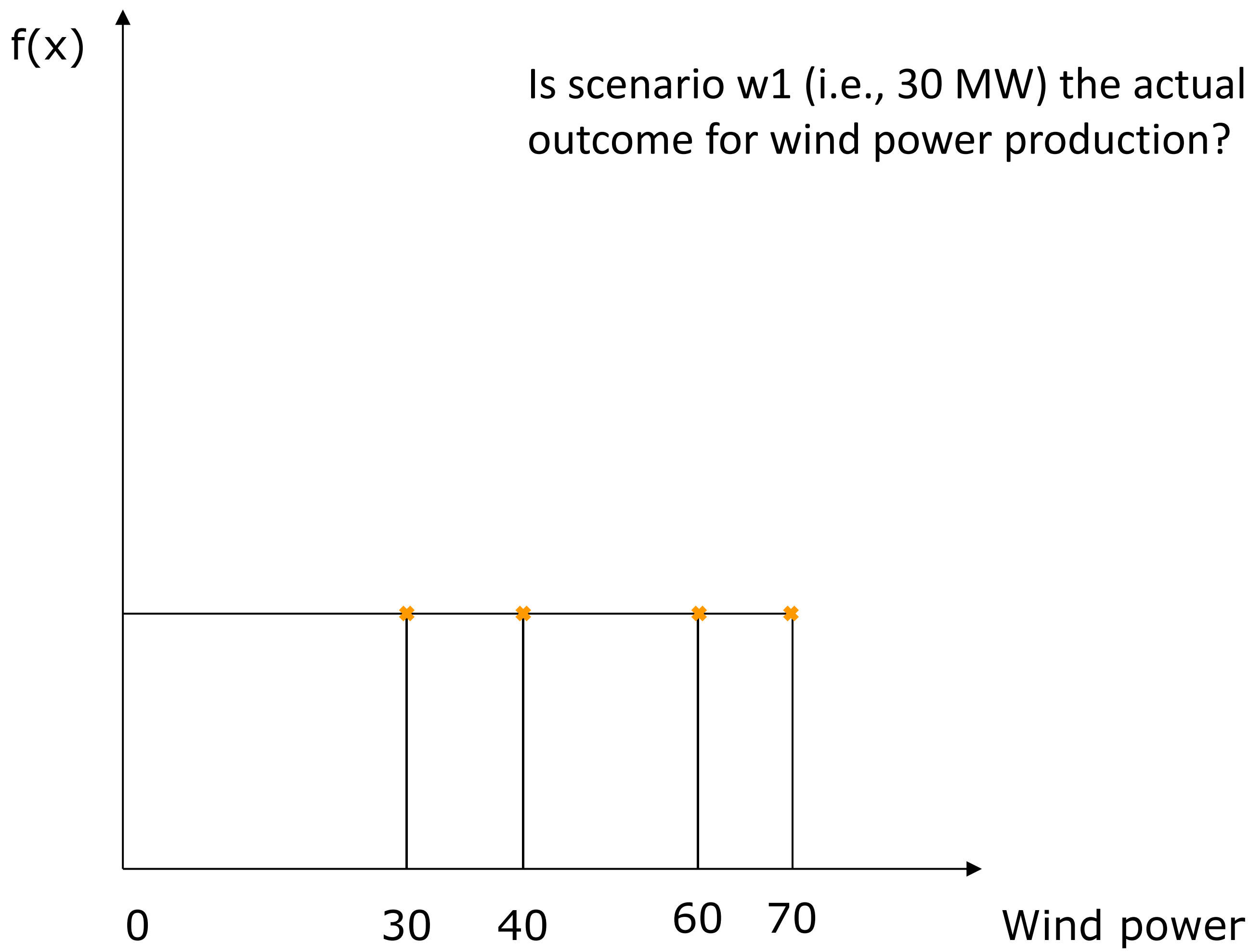
Robust optimization (scenario-based):

Minimize the total operational cost of the system under the worst-case scenario

- ➢ assuming uncertainty is adequately characterized by the underlying set of scenarios

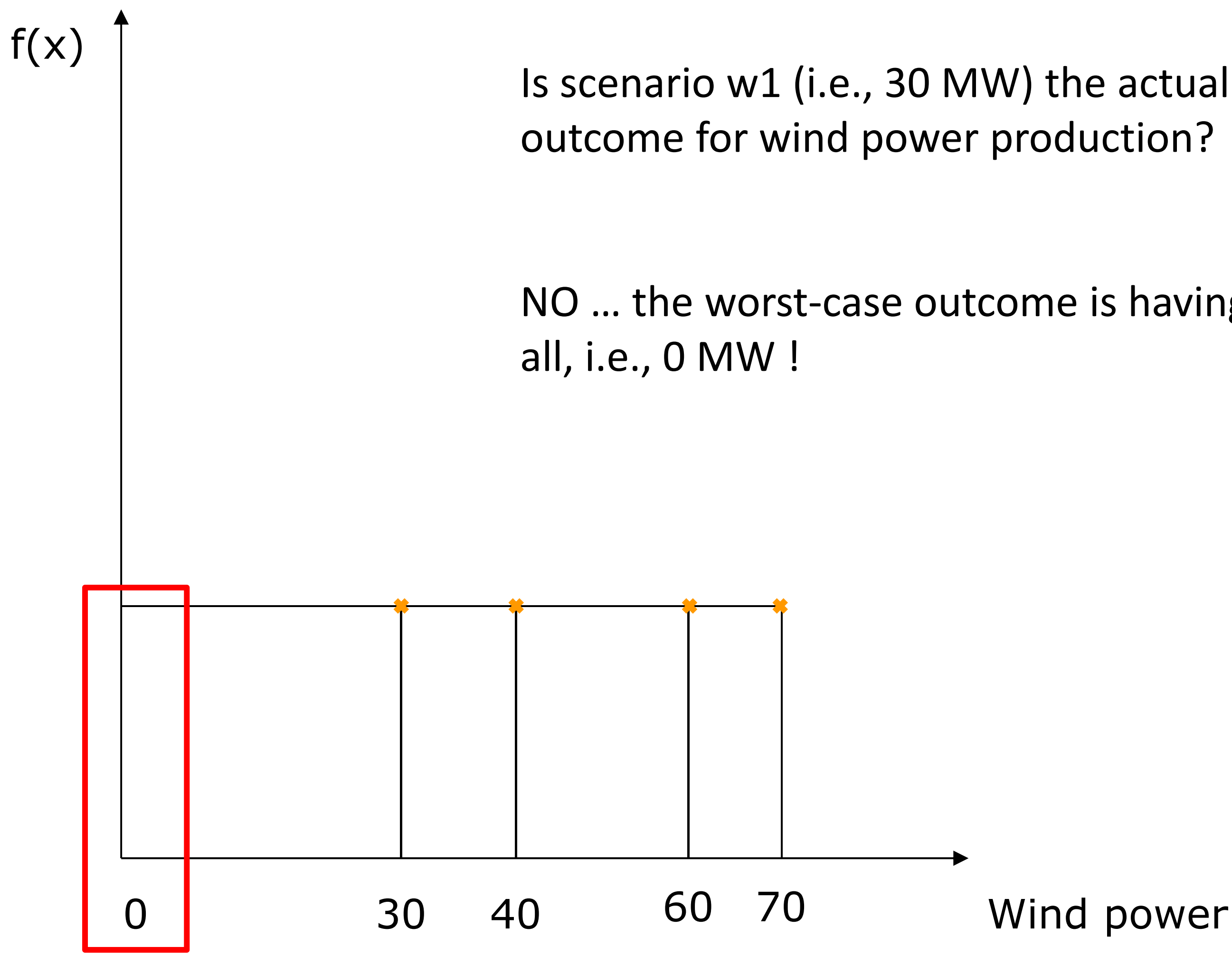


Uncertainty characterization



Is scenario w1 (i.e., 30 MW) the actual worst-case outcome for wind power production?

Uncertainty characterization



Is scenario w1 (i.e., 30 MW) the actual worst-case outcome for wind power production?

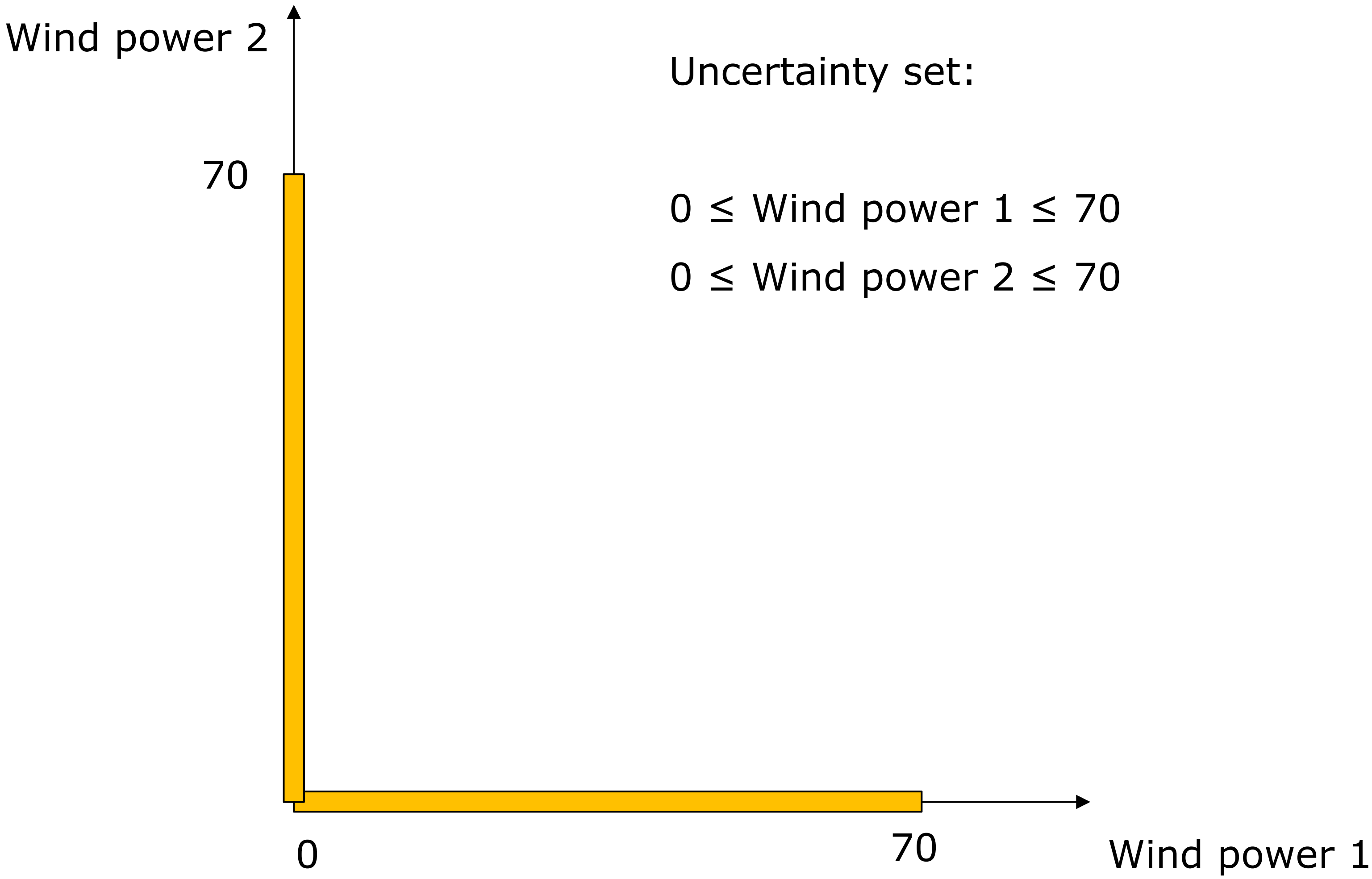
NO … the worst-case outcome is having no wind at all, i.e., 0 MW !

Uncertainty characterization

Uncertainty set:

0 ≤ Wind power ≤ 70

Uncertainty characterization

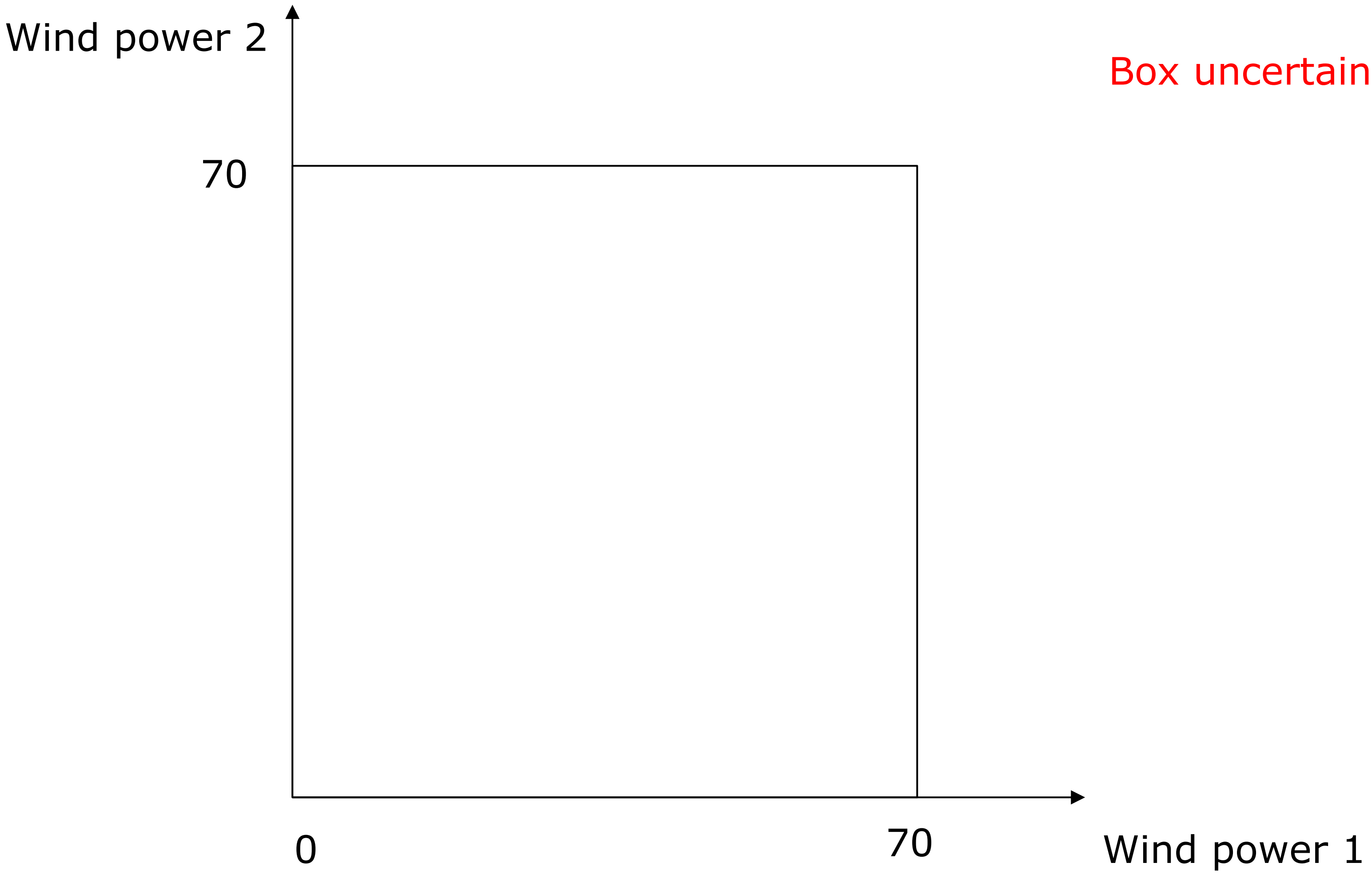


Uncertainty set:

0 ≤ Wind power 1 ≤ 70

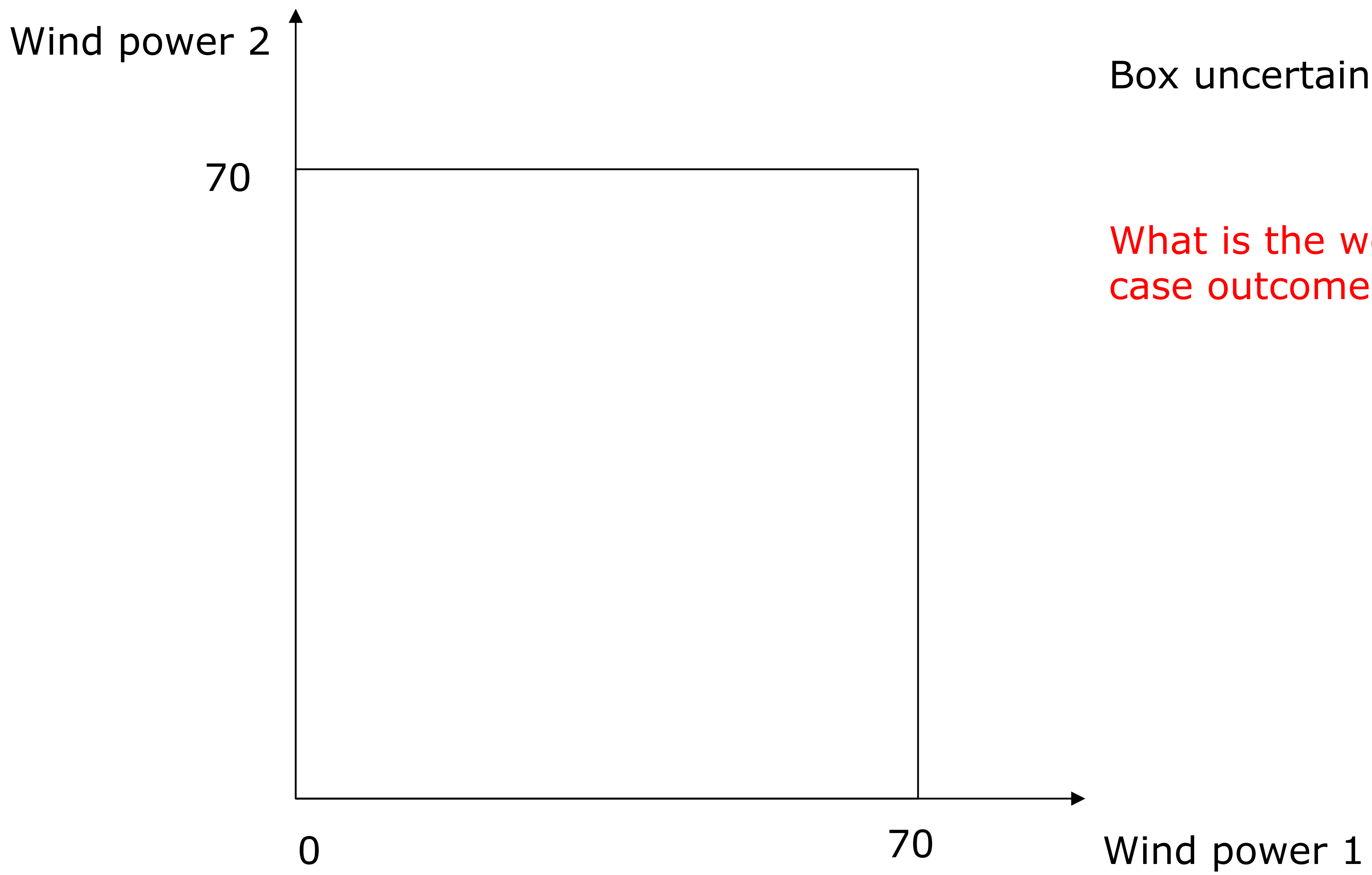
0 ≤ Wind power 2 ≤ 70

Uncertainty characterization



Box uncertainty set

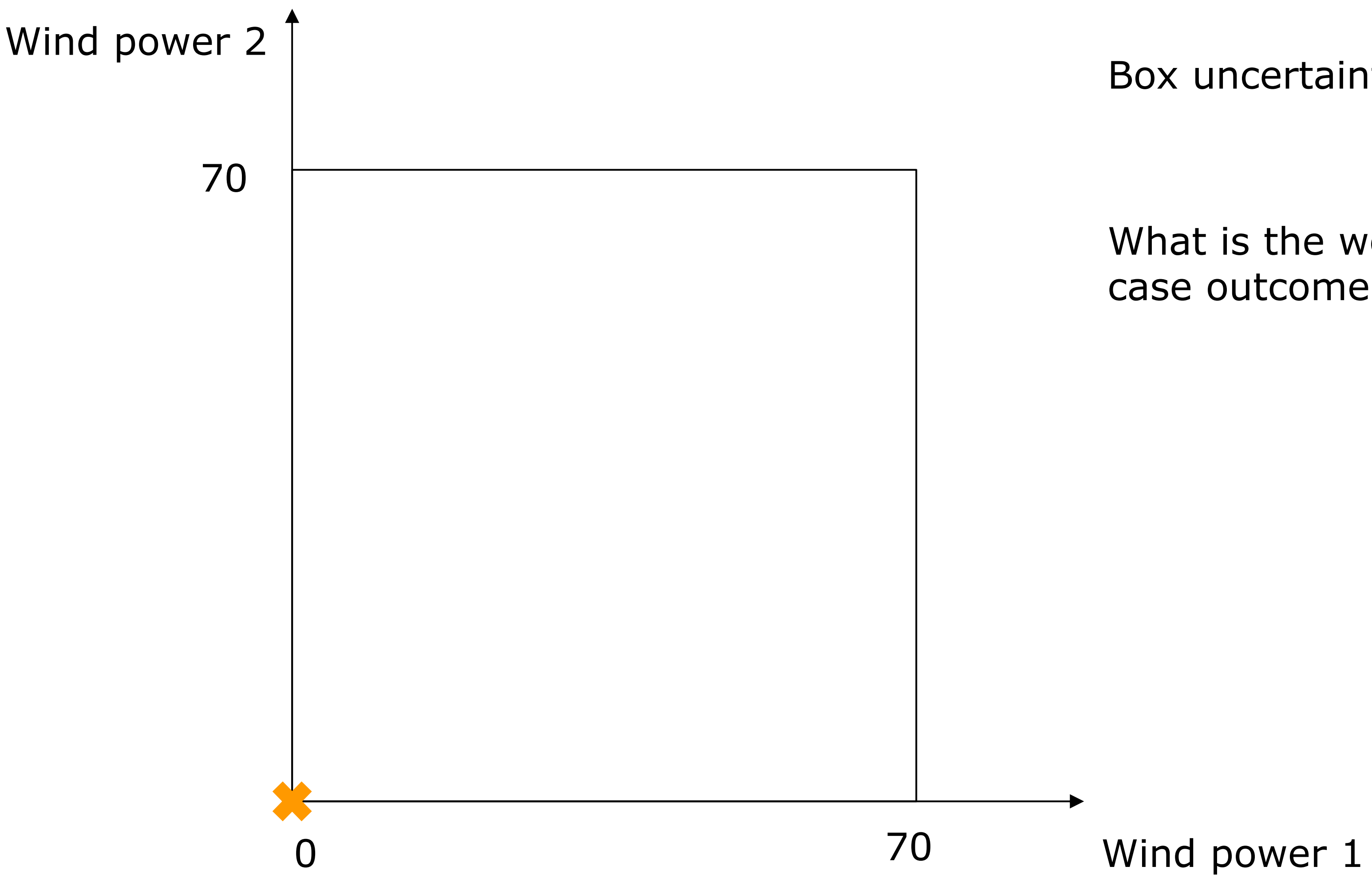
Uncertainty characterization



Box uncertainty set

What is the worst-case outcome?

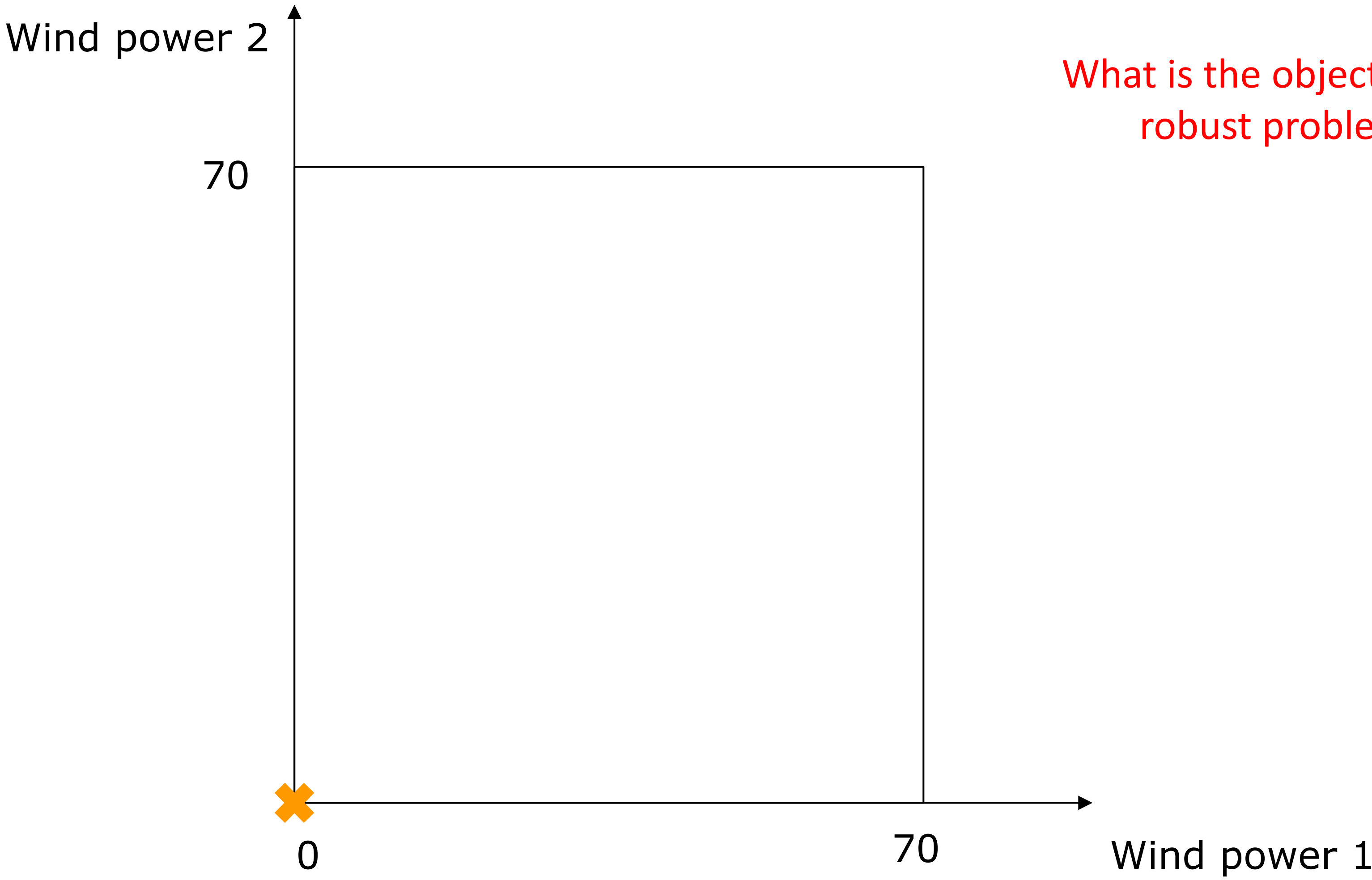
Uncertainty characterization



Box uncertainty set

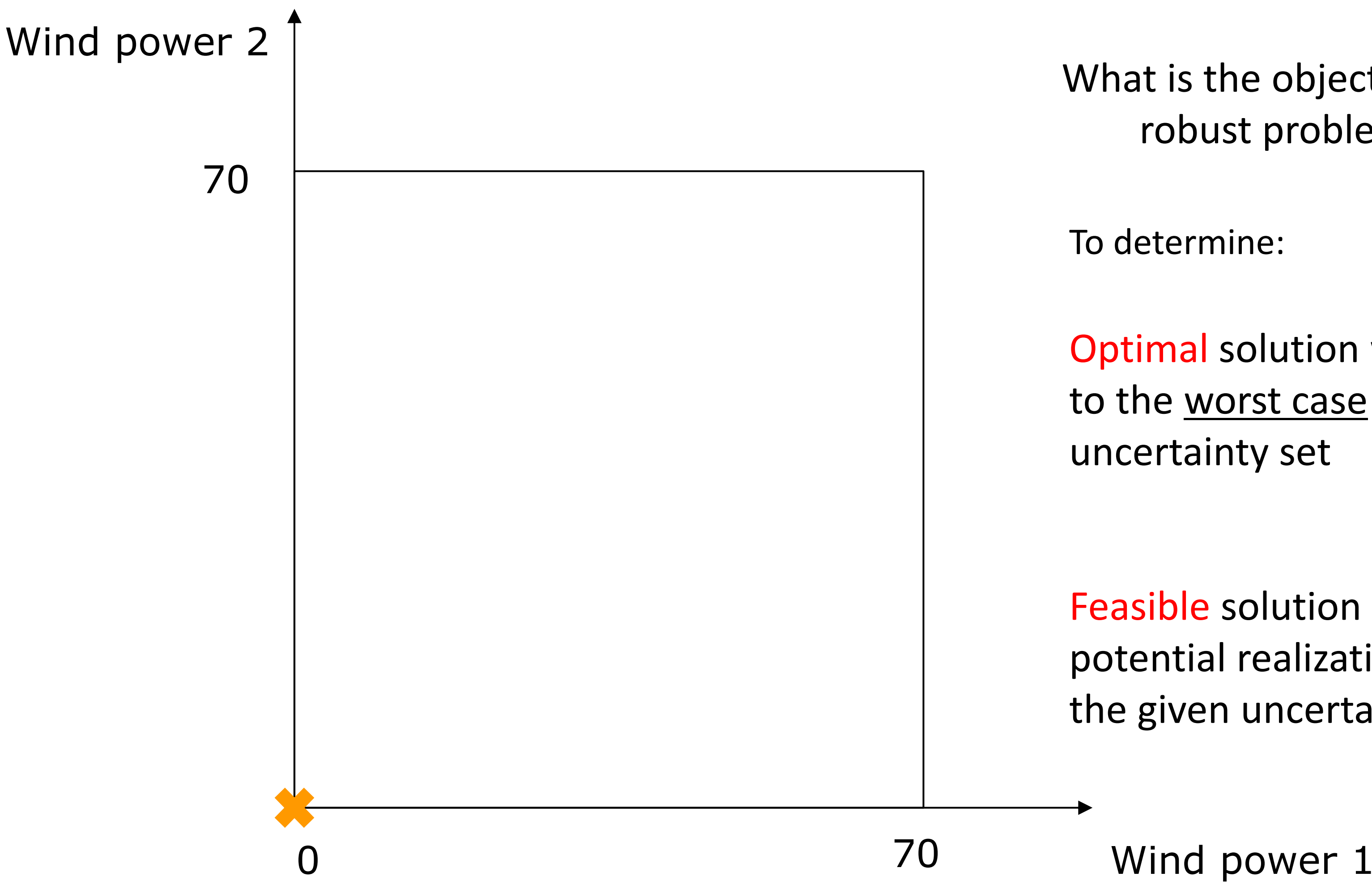
What is the worst-case outcome?

Uncertainty characterization



What is the objective of a robust problem?

Uncertainty characterization



What is the objective of a robust problem?

To determine:

Optimal solution with respect to the worst case of the given uncertainty set

Feasible solution for any other potential realization within the given uncertainty set

Jalal Kazempour

Uncertainty characterization

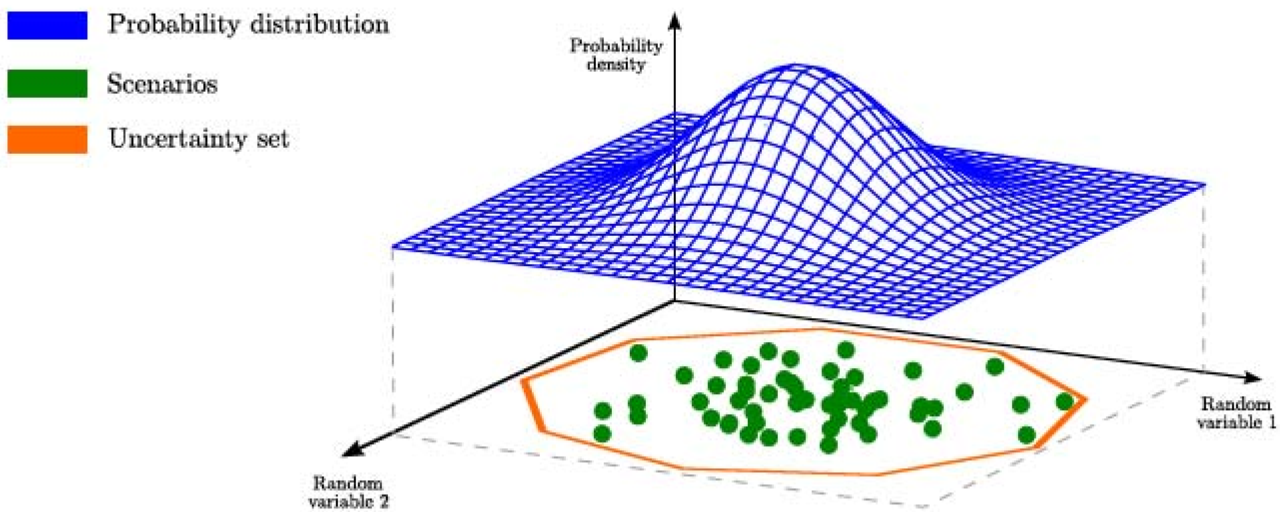


Figure from the Ph.D. thesis of Stefanos Delikaraoglou (DTU)

Uncertainty set in practice is usually more complicated than a simple box uncertainty!

Adaptive robust market clearing: optimization

Adaptive robust market clearing: optimization

Not a “scenario-based” approach!

Adaptive robust market clearing: optimization

Discussion: How can we formulate a robust optimization problem without having scenarios?

Minimize (total operational cost[1] of the system in DA) +

(total operational cost of the system in RT under the worst case)

subject to:

- Production limits (in DA and RT)
- Transmission network limits (in DA and RT)
- Load shedding limits (in RT)
- Nodal power balances (in DA and RT)

[1] We assume loads are inelastic to price, otherwise the social welfare should be maximized.

Adaptive robust market clearing: optimization

$$
\begin{aligned}
\min_{\substack{p^{\text{G2,RT}},\\ p^{\text{shed}},\\ p^{\text{spill}}}} \quad & [20p^{\text{G2,RT}} + 80p^{\text{shed}}] \\
\text{s. t.} \quad & 0 \leq (p^{\text{G2,DA}} + p^{\text{G2,RT}}) \leq 30 \\
& -30 \leq p^{\text{G2,RT}} \leq 30 \\
& 0 \leq p^{\text{spill}} \leq W \\
& 0 \leq p^{\text{shed}} \leq 120 \\
& p^{\text{G2,RT}} + (W - p^{\text{WP,DA}} - p^{\text{spill}}) + p^{\text{shed}} = 0
\end{aligned}
$$

Real-time problem

Adaptive robust market clearing: optimization

$$
\begin{aligned}
\min_{\substack{p^{\mathrm{G2,RT}},\\ p^{\mathrm{shed}},\\ p^{\mathrm{spill}}}} \quad & [20p^{\mathrm{G2,RT}} + 80p^{\mathrm{shed}}] \\
\text{s.t.} \quad & 0 \leq (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}) \leq 30 \\
& -30 \leq p^{\mathrm{G2,RT}} \leq 30 \\
& 0 \leq p^{\mathrm{spill}} \leq \boxed{W} \\
& 0 \leq p^{\mathrm{shed}} \leq 120 \\
& p^{\mathrm{G2,RT}} + (\boxed{W} - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}) + p^{\mathrm{shed}} = 0
\end{aligned}
$$

Real-time problem

Adaptive robust market clearing: optimization

$$
\begin{aligned}
\min_{\substack{p^{\mathrm{G2,RT}},\\ p^{\mathrm{shed}},\\ p^{\mathrm{spill}}}} \quad & [20p^{\mathrm{G2,RT}} + 80p^{\mathrm{shed}}] \\
\text{s. t.} \quad & 0 \leq (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}) \leq 30 \\
& -30 \leq p^{\mathrm{G2,RT}} \leq 30 \\
& 0 \leq p^{\mathrm{spill}} \leq \boxed{W} \\
& 0 \leq p^{\mathrm{shed}} \leq 120 \\
& p^{\mathrm{G2,RT}} + (\boxed{W} - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}) + p^{\mathrm{shed}} = 0
\end{aligned}
$$

How can we find the worst-case wind realization?

Adaptive robust market clearing: optimization

$$\max_{W} \quad \min_{\substack{p^{\mathrm{G2,RT}},\\ p^{\mathrm{shed}},\\ p^{\mathrm{spill}}}} \quad [20p^{\mathrm{G2,RT}} + 80p^{\mathrm{shed}}]$$

$$\text{s.t.} \quad 0 \leq (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}) \leq 30$$

$$-30 \leq p^{\mathrm{G2,RT}} \leq 30$$

$$0 \leq p^{\mathrm{spill}} \leq \boxed{W}$$

$$0 \leq p^{\mathrm{shed}} \leq 120$$

$$p^{\mathrm{G2,RT}} + (\boxed{W} - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}) + p^{\mathrm{shed}} = 0$$

$$\text{s.t.} \quad 0 \leq W \leq 70$$

How can we find the worst-case wind realization?

Adaptive robust market clearing: optimization

$$\max_{\boxed{W}} \quad \min_{\substack{p^{\text{G2,RT}},\\ p^{\text{shed}},\\ p^{\text{spill}}}} \quad [20p^{\text{G2,RT}} + 80p^{\text{shed}}]$$

$$\text{s.t.} \quad 0 \leq (p^{\text{G2,DA}} + p^{\text{G2,RT}}) \leq 30$$

$$-30 \leq p^{\text{G2,RT}} \leq 30$$

$$0 \leq p^{\text{spill}} \leq \boxed{W}$$

$$0 \leq p^{\text{shed}} \leq 120$$

$$p^{\text{G2,RT}} + (\boxed{W} - p^{\text{WP,DA}} - p^{\text{spill}}) + p^{\text{shed}} = 0$$

$$\text{s.t.} \quad 0 \leq \boxed{W} \leq 70$$

W is a <u>variable</u> for the outer maximization problem, but a <u>parameter</u> for the inner minimization problem!

Adaptive robust market clearing: optimization

$$
\begin{aligned}
\max_{\boxed{W}} \quad & \min_{\substack{p^{\mathrm{G2,RT}},\\ p^{\mathrm{shed}},\\ p^{\mathrm{spill}}}} \quad [20p^{\mathrm{G2,RT}} + 80p^{\mathrm{shed}}] \\
& \text{s.t.} \quad 0 \leq (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}) \leq 30 \\
& \qquad\quad -30 \leq p^{\mathrm{G2,RT}} \leq 30 \\
& \qquad\quad 0 \leq p^{\mathrm{spill}} \leq \boxed{W} \\
& \qquad\quad 0 \leq p^{\mathrm{shed}} \leq 120 \\
& \qquad\quad p^{\mathrm{G2,RT}} + (\boxed{W} - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}) + p^{\mathrm{shed}} = 0 \\
\text{s.t.} \quad & 0 \leq \boxed{W} \leq 70
\end{aligned}
$$

- Does this structure seem familiar?
- What is the objective function of the outer maximization problem?

Adaptive robust market clearing: optimization

$$\min_{p^{\mathrm{G1,DA}},\, p^{\mathrm{G2,DA}},\, p^{\mathrm{WP,DA}}} \quad [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] +$$

$$\text{s.t.} \quad 0 \le p^{\mathrm{G1,DA}} \le 100$$
$$0 \le p^{\mathrm{G2,DA}} \le 30$$
$$0 \le p^{\mathrm{WP,DA}} \le 70$$
$$p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120$$

$$\max_{W}$$

$$\text{s.t.} \quad 0 \le W \le 70$$

$$\min_{p^{\mathrm{G2,RT}},\, p^{\mathrm{shed}},\, p^{\mathrm{spill}}} \quad [20p^{\mathrm{G2,RT}} + 80p^{\mathrm{shed}}]$$

$$\text{s.t.} \quad 0 \le (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}) \le 30$$
$$-30 \le p^{\mathrm{G2,RT}} \le 30$$
$$0 \le p^{\mathrm{spill}} \le W$$
$$0 \le p^{\mathrm{shed}} \le 120$$
$$p^{\mathrm{G2,RT}} + (W - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}) + p^{\mathrm{shed}} = 0$$

Day-ahead problem

Nature reveals the worst outcome

Real-time problem

Adaptive robust market clearing: optimization

$$\min_{\substack{p^{\mathrm{G1,DA}},\\ p^{\mathrm{G2,DA}},\\ p^{\mathrm{WP,DA}}}} \quad [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \quad \max_{W} \quad \min_{\substack{p^{\mathrm{G2,RT}},\\ p^{\mathrm{shed}},\\ p^{\mathrm{spill}}}} \quad [20p^{\mathrm{G2,RT}} + 80p^{\mathrm{shed}}]$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \le (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}) \le 30 \\
& -30 \le p^{\mathrm{G2,RT}} \le 30 \\
& 0 \le p^{\mathrm{spill}} \le W \\
& 0 \le p^{\mathrm{shed}} \le 120 \\
& p^{\mathrm{G2,RT}} + (W - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}) + p^{\mathrm{shed}} = 0
\end{aligned}$$

$$\text{s.t.} \quad 0 \le W \le 70$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \le p^{\mathrm{G1,DA}} \le 100 \\
& 0 \le p^{\mathrm{G2,DA}} \le 30 \\
& 0 \le p^{\mathrm{WP,DA}} \le 70 \\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120
\end{aligned}$$

This **min-max-min** problem seems quite complex to solve!

Adaptive robust market clearing: optimization

$$\min_{\substack{p^{\mathrm{G1,DA}},\\ p^{\mathrm{G2,DA}},\\ p^{\mathrm{WP,DA}}}} \quad [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \max_{W} \quad \min_{\substack{p^{\mathrm{G2,RT}},\\ p^{\mathrm{shed}},\\ p^{\mathrm{spill}}}} \quad [20p^{\mathrm{G2,RT}} + 80p^{\mathrm{shed}}]$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \le (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}) \le 30 \\
& -30 \le p^{\mathrm{G2,RT}} \le 30 \\
& 0 \le p^{\mathrm{spill}} \le W \\
& 0 \le p^{\mathrm{shed}} \le 120 \\
& p^{\mathrm{G2,RT}} + (W - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}) + p^{\mathrm{shed}} = 0
\end{aligned}$$

$$\text{s.t.} \quad 0 \le W \le 70$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \le p^{\mathrm{G1,DA}} \le 100 \\
& 0 \le p^{\mathrm{G2,DA}} \le 30 \\
& 0 \le p^{\mathrm{WP,DA}} \le 70 \\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120
\end{aligned}$$

This min-max-min problem seems quite complex to solve!

Any idea how to solve it?

Duality …

The inner minimization problem:

$$
\begin{aligned}
\min_{p^{\mathrm{G2,RT}}, p^{\mathrm{shed}}, p^{\mathrm{spill}}} \quad & [20p^{\mathrm{G2,RT}} + 80p^{\mathrm{shed}}] \\
\text{s.t.} \quad & 0 \leq (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}) \leq 30 : \mu_1, \mu_2 \\
& -30 \leq p^{\mathrm{G2,RT}} \leq 30 : \mu_3, \mu_4 \\
& 0 \leq p^{\mathrm{spill}} \leq W : \mu_5, \mu_6 \\
& 0 \leq p^{\mathrm{shed}} \leq 120 : \mu_7, \mu_8 \\
& p^{\mathrm{G2,RT}} + (W - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}) + p^{\mathrm{shed}} = 0 : \lambda
\end{aligned}
$$

Duality …

The inner minimization problem:

$$
\begin{aligned}
\min_{p^{\mathrm{G2,RT}},p^{\mathrm{shed}},p^{\mathrm{spill}}} \quad & [20p^{\mathrm{G2,RT}} + 80p^{\mathrm{shed}}] \\
\mathrm{s.t.} \quad & 0 \leq (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}) \leq 30 : \mu_1, \mu_2 \\
& -30 \leq p^{\mathrm{G2,RT}} \leq 30 : \mu_3, \mu_4 \\
& 0 \leq p^{\mathrm{spill}} \leq W : \mu_5, \mu_6 \\
& 0 \leq p^{\mathrm{shed}} \leq 120 : \mu_7, \mu_8 \\
& p^{\mathrm{G2,RT}} + (W - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}) + p^{\mathrm{shed}} = 0 : \lambda
\end{aligned}
$$

Recall that W and DA dispatch are treated as parameters within the inner minimization problem!

Duality …

Dual of the inner minimization problem:

$$
\begin{aligned}
\max_{\mu_{1,\dots,8},\lambda} \quad & [-p^{\mathrm{G2,DA}}\mu_1 + (-30 + p^{\mathrm{G2,DA}})\mu_2 - 30\,\mu_3 - 30\,\mu_4 - W\,\mu_6 - 120\,\mu_8\,] - \lambda(W - p^{\mathrm{WP,DA}}\,)] \\
\text{s.t.} \quad & \mu_1 - \mu_2 + \mu_3 - \mu_4 + \lambda = 20 \\
& \mu_5 - \mu_6 - \lambda = 0 \\
& \mu_7 - \mu_8 + \lambda = 80 \\
& \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7, \mu_8 \geq 0 \\
& \lambda \quad free
\end{aligned}
$$

Duality …

Dual of the inner minimization problem:

$$\begin{aligned}
\max_{\mu_{1,\ldots,8},\lambda} \quad & [-p^{\text{G2,DA}}\mu_1 + (-30 + p^{\text{G2,DA}})\mu_2 - 30\,\mu_3 - 30\,\mu_4 - W\,\mu_6 - 120\,\mu_8\,] - \lambda(W - p^{\text{WP,DA}})] \\
\text{s.t.} \quad & \mu_1 - \mu_2 + \mu_3 - \mu_4 + \lambda = 20 \\
& \mu_5 - \mu_6 - \lambda = 0 \\
& \mu_7 - \mu_8 + \lambda = 80 \\
& \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7, \mu_8 \geq 0 \\
& \lambda \quad free
\end{aligned}$$

Let's now get back to the original min-max-min problem!

Solution technique

Primal inner problem

$$\min_{p^{G1,DA},\, p^{G2,DA},\, p^{WP,DA}} \quad [10p^{G1,DA} + 20p^{G2,DA}] + \quad \max_{W} \quad \min_{p^{G2,RT},\, p^{shed},\, p^{spill}} \quad [20p^{G2,RT} + 80p^{shed}]$$

$$\text{s.t.} \quad 0 \le (p^{G2,DA} + p^{G2,RT}) \le 30$$
$$-30 \le p^{G2,RT} \le 30$$
$$0 \le p^{spill} \le W$$
$$0 \le p^{shed} \le 120$$
$$p^{G2,RT} + (W - p^{WP,DA} - p^{spill}) + p^{shed} = 0$$

$$\text{s.t.} \quad 0 \le W \le 70$$

$$\text{s.t.} \quad 0 \le p^{G1,DA} \le 100$$
$$0 \le p^{G2,DA} \le 30$$
$$0 \le p^{WP,DA} \le 70$$
$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120$$

Solution technique

$$\min_{p^{\mathrm{G1,DA}},\, p^{\mathrm{G2,DA}},\, p^{\mathrm{WP,DA}}} \quad [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] +$$

$$\max_{W}$$

Primal inner problem

$$\begin{aligned}
\min_{p^{\mathrm{G2,RT}},\, p^{\mathrm{shed}},\, p^{\mathrm{spill}}} \quad & [20p^{\mathrm{G2,RT}} + 80p^{\mathrm{shed}}] \\
\text{s.t.} \quad & 0 \leq (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}) \leq 30 \\
& -30 \leq p^{\mathrm{G2,RT}} \leq 30 \\
& 0 \leq p^{\mathrm{spill}} \leq W \\
& 0 \leq p^{\mathrm{shed}} \leq 120 \\
& p^{\mathrm{G2,RT}} + (W - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}) + p^{\mathrm{shed}} = 0
\end{aligned}$$

$$\text{s.t.} \quad 0 \leq W \leq 70$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \leq p^{\mathrm{G1,DA}} \leq 100 \\
& 0 \leq p^{\mathrm{G2,DA}} \leq 30 \\
& 0 \leq p^{\mathrm{WP,DA}} \leq 70 \\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120
\end{aligned}$$

Equivalent problems!

Dual of the inner problem

$$\begin{aligned}
\max_{\mu_{1,\ldots,8},\lambda} \quad & [-p^{\mathrm{G2,DA}}\mu_1 + (-30 + p^{\mathrm{G2,DA}})\mu_2 - 30\,\mu_3 - 30\,\mu_4 - W\,\mu_6 - 120\,\mu_8\,] - \lambda(W - p^{\mathrm{WP,DA}}\,)] \\
\text{s.t.} \quad & \mu_1 - \mu_2 + \mu_3 - \mu_4 + \lambda = 20 \\
& \mu_5 - \mu_6 - \lambda = 0 \\
& \mu_7 - \mu_8 + \lambda = 80 \\
& \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7, \mu_8 \geq 0 \\
& \lambda \quad free
\end{aligned}$$

Solution technique

Discussion:
If we replace the primal inner minimization problem with its dual maximization counterpart, can we merge it with the intermediate-level maximization problem?

$$\begin{aligned} \text{s.t.} \quad & 0 \leq p^{\text{G1,DA}} \leq 100 \\ & 0 \leq p^{\text{G2,DA}} \leq 30 \\ & 0 \leq p^{\text{WP,DA}} \leq 70 \\ & p^{\text{G1,DA}} + p^{\text{G2,DA}} + p^{\text{WP,DA}} = 120 \end{aligned}$$

$$\max_{W}$$

$$\text{s.t.} \quad 0 \leq W \leq 70$$

Primal inner problem

$$\begin{aligned} \min_{p^{\text{G2,RT}},\, p^{\text{shed}},\, p^{\text{spill}}} \quad & [20p^{\text{G2,RT}} + 80p^{\text{shed}}] \\ \text{s.t.} \quad & 0 \leq (p^{\text{G2,DA}} + p^{\text{G2,RT}}) \leq 30 \\ & -30 \leq p^{\text{G2,RT}} \leq 30 \\ & 0 \leq p^{\text{spill}} \leq W \\ & 0 \leq p^{\text{shed}} \leq 120 \\ & p^{\text{G2,RT}} + (W - p^{\text{WP,DA}} - p^{\text{spill}}) + p^{\text{shed}} = 0 \end{aligned}$$

Dual of the inner problem

$$\begin{aligned} \max_{\mu_{1,\ldots,8},\lambda} \quad & [-p^{\text{G2,DA}}\mu_1 + (-30 + p^{\text{G2,DA}})\mu_2 - 30\,\mu_3 - 30\,\mu_4 - W\mu_6 - 120\,\mu_8] - \lambda(W - p^{\text{WP,DA}})] \\ \text{s.t.} \quad & \mu_1 - \mu_2 + \mu_3 - \mu_4 + \lambda = 20 \\ & \mu_5 - \mu_6 - \lambda = 0 \\ & \mu_7 - \mu_8 + \lambda = 80 \\ & \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7, \mu_8 \geq 0 \\ & \lambda \quad \mathit{free} \end{aligned}$$

Solution technique

Discussion:
If we replace the primal inner minimization problem with its dual maximization counterpart, can we merge it with the intermediate-level maximization problem? YES!

$$\begin{aligned} \text{s.t.} \quad & 0 \leq p^{\mathrm{G1,DA}} \leq 100 \\ & 0 \leq p^{\mathrm{G2,DA}} \leq 30 \\ & 0 \leq p^{\mathrm{WP,DA}} \leq 70 \\ & p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120 \end{aligned}$$

$$\max_{W} \qquad \text{s.t.} \quad 0 \leq W \leq 70$$

Primal inner problem

$$\begin{aligned} \min_{p^{\mathrm{G2,RT}},\, p^{\mathrm{shed}},\, p^{\mathrm{spill}}} \quad & [20 p^{\mathrm{G2,RT}} + 80 p^{\mathrm{shed}}] \\ \text{s.t.} \quad & 0 \leq (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}) \leq 30 \\ & -30 \leq p^{\mathrm{G2,RT}} \leq 30 \\ & 0 \leq p^{\mathrm{spill}} \leq W \\ & 0 \leq p^{\mathrm{shed}} \leq 120 \\ & p^{\mathrm{G2,RT}} + (W - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}) + p^{\mathrm{shed}} = 0 \end{aligned}$$

Dual of the inner problem

$$\begin{aligned} \max_{\mu_{1,\dots,8},\lambda} \quad & [-p^{\mathrm{G2,DA}} \mu_1 + (-30 + p^{\mathrm{G2,DA}}) \mu_2 - 30 \mu_3 - 30 \mu_4 - W \mu_6 - 120 \mu_8] - \lambda (W - p^{\mathrm{WP,DA}})] \\ \text{s.t.} \quad & \mu_1 - \mu_2 + \mu_3 - \mu_4 + \lambda = 20 \\ & \mu_5 - \mu_6 - \lambda = 0 \\ & \mu_7 - \mu_8 + \lambda = 80 \\ & \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7, \mu_8 \geq 0 \\ & \lambda \quad free \end{aligned}$$

Solution technique

$$\min_{p^{G1,DA},\, p^{G2,DA},\, p^{WP,DA}} \quad [10p^{G1,DA} + 20p^{G2,DA}] +$$

$$\max_{W}$$

$$\text{s.t.} \quad 0 \leq W \leq 70$$

$$\min_{p^{G2,RT},\, p^{shed},\, p^{spill}} \quad [20p^{G2,RT} + 80p^{shed}]$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \leq (p^{G2,DA} + p^{G2,RT}) \leq 30 \\
& -30 \leq p^{G2,RT} \leq 30 \\
& 0 \leq p^{spill} \leq W \\
& 0 \leq p^{shed} \leq 120 \\
& p^{G2,RT} + (W - p^{WP,DA} - p^{spill}) + p^{shed} = 0
\end{aligned}$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \leq p^{G1,DA} \leq 100 \\
& 0 \leq p^{G2,DA} \leq 30 \\
& 0 \leq p^{WP,DA} \leq 70 \\
& p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120
\end{aligned}$$

Dual of the inner problem

$$\max_{\mu_{1,\dots,8},\lambda} \quad [-p^{G2,DA}\mu_1 + (-30 + p^{G2,DA})\mu_2 - 30\,\mu_3 - 30\,\mu_4 - W\,\mu_6 - 120\,\mu_8\,] - \lambda(W - p^{WP,DA}\,)]$$

$$\begin{aligned}
\text{s.t.} \quad & \mu_1 - \mu_2 + \mu_3 - \mu_4 + \lambda = 20 \\
& \mu_5 - \mu_6 - \lambda = 0 \\
& \mu_7 - \mu_8 + \lambda = 80 \\
& \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7, \mu_8 \geq 0 \\
& \lambda \quad free
\end{aligned}$$

To be merged!

Solution technique

Merged maximization problem (Note that W is a variable)

$$\min_{\substack{p^{\mathrm{G1,DA}},\\ p^{\mathrm{G2,DA}},\\ p^{\mathrm{WP,DA}}}} \quad [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] +$$

$$\begin{aligned} \max_{\mu_{1,\dots,8},\lambda,W} \quad & [-p^{\mathrm{G2,DA}}\mu_1 + (-30 + p^{\mathrm{G2,DA}})\mu_2 - 30\,\mu_3 - 30\,\mu_4 - W\mu_6 - 120\,\mu_8\,] - \lambda(W - p^{\mathrm{WP,DA}})] \\ \text{s.t.} \quad & \mu_1 - \mu_2 + \mu_3 - \mu_4 + \lambda = 20 \\ & \mu_5 - \mu_6 - \lambda = 0 \\ & \mu_7 - \mu_8 + \lambda = 80 \\ & \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7, \mu_8 \geq 0 \\ & \lambda \quad free \\ & 0 \leq W \leq 70 \end{aligned}$$

$$\begin{aligned} \text{s.t.} \quad & 0 \leq p^{\mathrm{G1,DA}} \leq 100 \\ & 0 \leq p^{\mathrm{G2,DA}} \leq 30 \\ & 0 \leq p^{\mathrm{WP,DA}} \leq 70 \\ & p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120 \end{aligned}$$

Solution technique

Merged maximization problem (Note that W is a variable)

$$\min_{\substack{p^{\mathrm{G1,DA}},\\ p^{\mathrm{G2,DA}},\\ p^{\mathrm{WP,DA}}}} \quad [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \boxed{\begin{aligned} \max_{\mu_{1,\dots,8},\lambda,W} \quad & [-p^{\mathrm{G2,DA}}\mu_1 + (-30 + p^{\mathrm{G2,DA}})\mu_2 - 30\,\mu_3 - 30\,\mu_4 - W\,\mu_6 - 120\,\mu_8\,] - \lambda(W - p^{\mathrm{WP,DA}})] \\ \text{s.t.} \quad & \mu_1 - \mu_2 + \mu_3 - \mu_4 + \lambda = 20 \\ & \mu_5 - \mu_6 - \lambda = 0 \\ & \mu_7 - \mu_8 + \lambda = 80 \\ & \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7, \mu_8 \geq 0 \\ & \lambda \quad free \\ & 0 \leq W \leq 70 \end{aligned}}$$

$$\begin{aligned} \text{s.t.} \quad & 0 \leq p^{\mathrm{G1,DA}} \leq 100 \\ & 0 \leq p^{\mathrm{G2,DA}} \leq 30 \\ & 0 \leq p^{\mathrm{WP,DA}} \leq 70 \\ & p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120 \end{aligned}$$

We now have a min-max problem!

Solution technique

Merged maximization problem (Note that W is a variable)

$$\min_{p^{\mathrm{G1,DA}},\, p^{\mathrm{G2,DA}},\, p^{\mathrm{WP,DA}}} \quad [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] +$$

$$\begin{aligned}
\max_{\mu_{1,\dots,8},\lambda,W} \quad & [-p^{\mathrm{G2,DA}}\mu_1 + (-30 + p^{\mathrm{G2,DA}})\mu_2 - 30\,\mu_3 - 30\,\mu_4 - W\,\mu_6 - 120\,\mu_8\,] - \lambda(W - p^{\mathrm{WP,DA}}\,)] \\
\text{s.t.} \quad & \mu_1 - \mu_2 + \mu_3 - \mu_4 + \lambda = 20 \\
& \mu_5 - \mu_6 - \lambda = 0 \\
& \mu_7 - \mu_8 + \lambda = 80 \\
& \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7, \mu_8 \geq 0 \\
& \lambda \quad free \\
& 0 \leq W \leq 70
\end{aligned}$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \leq p^{\mathrm{G1,DA}} \leq 100 \\
& 0 \leq p^{\mathrm{G2,DA}} \leq 30 \\
& 0 \leq p^{\mathrm{WP,DA}} \leq 70 \\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120
\end{aligned}$$

We now have a min-max problem!

Is there any issue in this resulting maximization problem?

Solution technique

$$\min_{\substack{p^{\mathrm{G1,DA}},\\ p^{\mathrm{G2,DA}},\\ p^{\mathrm{WP,DA}}}} \quad [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \max_{\mu_{1,\dots,8},\lambda,W} [-p^{\mathrm{G2,DA}}\mu_1 + (-30 + p^{\mathrm{G2,DA}})\mu_2 - 30\,\mu_3 - 30\,\mu_4 - \boxed{W\,\mu_6} - 120\,\mu_8\,] - \boxed{\lambda(W} - p^{\mathrm{WP,DA}}\,)]$$

$$\begin{aligned}
\text{s.t.}\quad & \mu_1 - \mu_2 + \mu_3 - \mu_4 + \lambda = 20\\
& \mu_5 - \mu_6 - \lambda = 0\\
& \mu_7 - \mu_8 + \lambda = 80\\
& \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7, \mu_8 \geq 0\\
& \lambda \quad free\\
& 0 \leq W \leq 70
\end{aligned}$$

$$\begin{aligned}
\text{s.t.}\quad & 0 \leq p^{\mathrm{G1,DA}} \leq 100\\
& 0 \leq p^{\mathrm{G2,DA}} \leq 30\\
& 0 \leq p^{\mathrm{WP,DA}} \leq 70\\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120
\end{aligned}$$

Bilinear terms --> NLP

Solution technique

$$\min_{p^{\mathrm{G1,DA}},\, p^{\mathrm{G2,DA}},\, p^{\mathrm{WP,DA}}} \quad [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \max_{\mu_{1,\dots,8},\lambda,W} [-p^{\mathrm{G2,DA}}\mu_1 + (-30 + p^{\mathrm{G2,DA}})\mu_2 - 30\,\mu_3 - 30\,\mu_4 - \boxed{W\,\mu_6} - 120\,\mu_8\,] - \boxed{\lambda(W} - p^{\mathrm{WP,DA}}\,)]$$

$$\begin{aligned}
\text{s.t.} \quad & \mu_1 - \mu_2 + \mu_3 - \mu_4 + \lambda = 20 \\
& \mu_5 - \mu_6 - \lambda = 0 \\
& \mu_7 - \mu_8 + \lambda = 80 \\
& \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7, \mu_8 \geq 0 \\
& \lambda \quad free \\
& 0 \leq W \leq 70
\end{aligned}$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \leq p^{\mathrm{G1,DA}} \leq 100 \\
& 0 \leq p^{\mathrm{G2,DA}} \leq 30 \\
& 0 \leq p^{\mathrm{WP,DA}} \leq 70 \\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120
\end{aligned}$$

If we fix one of the variables in the bilinear terms, the resulting optimization problem will become linear!

Solution technique

$$\min_{p^{G1,DA},\, p^{G2,DA},\, p^{WP,DA}} [10p^{G1,DA} + 20p^{G2,DA}] + \max_{\mu_{1,\dots,8},\lambda,W} [-p^{G2,DA}\mu_1 + (-30 + p^{G2,DA})\mu_2 - 30\,\mu_3 - 30\,\mu_4 - \boxed{W\,\mu_6} - 120\,\mu_8\,] - \boxed{\lambda(W} - p^{WP,DA}\,)]$$

$$\begin{aligned}
\text{s.t.}\quad & \mu_1 - \mu_2 + \mu_3 - \mu_4 + \lambda = 20\\
& \mu_5 - \mu_6 - \lambda = 0\\
& \mu_7 - \mu_8 + \lambda = 80\\
& \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7, \mu_8 \geq 0\\
& \lambda \quad free\\
& 0 \leq W \leq 70
\end{aligned}$$

$$\begin{aligned}
\text{s.t.}\quad & 0 \leq p^{G1,DA} \leq 100\\
& 0 \leq p^{G2,DA} \leq 30\\
& 0 \leq p^{WP,DA} \leq 70\\
& p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120
\end{aligned}$$

If we fix one of the variables in the bilinear terms, the resulting optimization problem will become linear!

The optimal solution will be a **vertex** of the feasible region

Solution technique

$$\min_{p^{G1,DA},\, p^{G2,DA},\, p^{WP,DA}} \quad [10p^{G1,DA} + 20p^{G2,DA}] + \max_{\mu_{1,\dots,8},\lambda,W} [-p^{G2,DA}\mu_1 + (-30 + p^{G2,DA})\mu_2 - 30\,\mu_3 - 30\,\mu_4 - \boxed{W\,\mu_6} - 120\,\mu_8\,] - \boxed{\lambda(W} - p^{WP,DA}\,)]$$

$$\begin{aligned}
\text{s.t.}\quad & \mu_1 - \mu_2 + \mu_3 - \mu_4 + \lambda = 20 \\
& \mu_5 - \mu_6 - \lambda = 0 \\
& \mu_7 - \mu_8 + \lambda = 80 \\
& \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7, \mu_8 \geq 0 \\
& \lambda \quad free \\
& 0 \leq W \leq 70
\end{aligned}$$

$$\begin{aligned}
\text{s.t.}\quad & 0 \leq p^{G1,DA} \leq 100 \\
& 0 \leq p^{G2,DA} \leq 30 \\
& 0 \leq p^{WP,DA} \leq 70 \\
& p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120
\end{aligned}$$

If we fix one of the variables in the bilinear terms, the resulting optimization problem will become linear!

The optimal solution will be a **vertex** of the feasible region

Hint: Dual variables and W have disjoint feasible regions that are independent of the first-stage decisions!

Any other solution technique?

Let's get back to the original min-max-min problem!

Adaptive robust market clearing: optimization

$$\min_{\substack{p^{\mathrm{G1,DA}},\\ p^{\mathrm{G2,DA}},\\ p^{\mathrm{WP,DA}}}} \quad [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \quad \max_{W} \quad \min_{\substack{p^{\mathrm{G2,RT}},\\ p^{\mathrm{shed}},\\ p^{\mathrm{spill}}}} \quad [20p^{\mathrm{G2,RT}} + 80p^{\mathrm{shed}}]$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \leq (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}) \leq 30\\
& -30 \leq p^{\mathrm{G2,RT}} \leq 30\\
& 0 \leq p^{\mathrm{spill}} \leq W\\
& 0 \leq p^{\mathrm{shed}} \leq 120\\
& p^{\mathrm{G2,RT}} + (W - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}) + p^{\mathrm{shed}} = 0
\end{aligned}$$

$$\text{s.t.} \quad 0 \leq W \leq 70$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \leq p^{\mathrm{G1,DA}} \leq 100\\
& 0 \leq p^{\mathrm{G2,DA}} \leq 30\\
& 0 \leq p^{\mathrm{WP,DA}} \leq 70\\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120
\end{aligned}$$

Adaptive robust market clearing: optimization

$$\min_{\substack{p^{\mathrm{G1,DA}},\\ p^{\mathrm{G2,DA}},\\ p^{\mathrm{WP,DA}}}} \quad [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \quad \max_{W} \quad \min_{\substack{p^{\mathrm{G2,RT}},\\ p^{\mathrm{shed}},\\ p^{\mathrm{spill}}}} \quad [20p^{\mathrm{G2,RT}} + 80p^{\mathrm{shed}}]$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \leq (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}) \leq 30 \\
& -30 \leq p^{\mathrm{G2,RT}} \leq 30 \\
& 0 \leq p^{\mathrm{spill}} \leq W \\
& 0 \leq p^{\mathrm{shed}} \leq 120 \\
& p^{\mathrm{G2,RT}} + (W - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}) + p^{\mathrm{shed}} = 0
\end{aligned}$$

$$\text{s.t.} \quad 0 \leq W \leq 70$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \leq p^{\mathrm{G1,DA}} \leq 100 \\
& 0 \leq p^{\mathrm{G2,DA}} \leq 30 \\
& 0 \leq p^{\mathrm{WP,DA}} \leq 70 \\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120
\end{aligned}$$

We only need to explore these two vertices!

Adaptive robust market clearing: optimization

$$\min_{\substack{p^{\mathrm{G1,DA}},\\ p^{\mathrm{G2,DA}},\\ p^{\mathrm{WP,DA}}}} \quad [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \max_{W} \quad \min_{\substack{p^{\mathrm{G2,RT}},\\ p^{\mathrm{shed}},\\ p^{\mathrm{spill}}}} \quad [20p^{\mathrm{G2,RT}} + 80p^{\mathrm{shed}}]$$

$$\begin{aligned} \text{s.t.} \quad & 0 \leq (p^{\mathrm{G2,DA}} + p^{\mathrm{G2,RT}}) \leq 30 \\ & -30 \leq p^{\mathrm{G2,RT}} \leq 30 \\ & 0 \leq p^{\mathrm{spill}} \leq W \\ & 0 \leq p^{\mathrm{shed}} \leq 120 \\ & p^{\mathrm{G2,RT}} + (W - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}) + p^{\mathrm{shed}} = 0 \end{aligned}$$

$$\text{s.t.} \quad 0 \leq W \leq 70$$

$$\begin{aligned} \text{s.t.} \quad & 0 \leq p^{\mathrm{G1,DA}} \leq 100 \\ & 0 \leq p^{\mathrm{G2,DA}} \leq 30 \\ & 0 \leq p^{\mathrm{WP,DA}} \leq 70 \\ & p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120 \end{aligned}$$

$$\begin{aligned} \min_{p^{\mathrm{G2,RT}},p^{\mathrm{shed}},p^{\mathrm{spill}},\beta} \quad & \beta \\ \text{s.t.} \quad & 0 \leq (p^{\mathrm{G2,DA}} + p_{v_1}^{\mathrm{G2,RT}}) \leq 30 \\ & -30 \leq p_{v_1}^{\mathrm{G2,RT}} \leq 30 \\ & 0 \leq p_{v_1}^{\mathrm{spill}} \leq W_{v_1} \\ & 0 \leq p_{v_1}^{\mathrm{shed}} \leq 120 \\ & p_{v_1}^{\mathrm{G2,RT}} + (W_{v_1} - p^{\mathrm{WP,DA}} - p_{v_1}^{\mathrm{spill}}) + p_{v_1}^{\mathrm{shed}} = 0 \\ & 0 \leq (p^{\mathrm{G2,DA}} + p_{v_2}^{\mathrm{G2,RT}}) \leq 30 \\ & -30 \leq p_{v_2}^{\mathrm{G2,RT}} \leq 30 \\ & 0 \leq p_{v_2}^{\mathrm{spill}} \leq W_{v_2} \\ & 0 \leq p_{v_2}^{\mathrm{shed}} \leq 120 \\ & p_{v_2}^{\mathrm{G2,RT}} + (W_{v_2} - p^{\mathrm{WP,DA}} - p_{v_2}^{\mathrm{spill}}) + p_{v_2}^{\mathrm{shed}} = 0 \\ & [20p_{v_1}^{\mathrm{G2,RT}} + 80p_{v_1}^{\mathrm{shed}}] \leq \beta \\ & [20p_{v_2}^{\mathrm{G2,RT}} + 80p_{v_2}^{\mathrm{shed}}] \leq \beta \end{aligned}$$

Adaptive robust market clearing: optimization

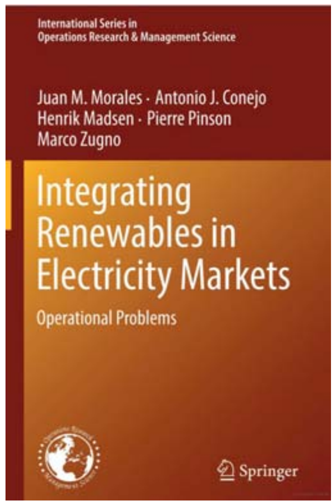
$$\min_{p^{G1,DA},\, p^{G2,DA},\, p^{WP,DA}} \quad [10p^{G1,DA} + 20p^{G2,DA}] + \quad \max_{W} \quad \min_{p^{G2,RT},\, p^{shed},\, p^{spill}} \quad [20p^{G2,RT} + 80p^{shed}]$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \leq (p^{G2,DA} + p^{G2,RT}) \leq 30 \\
& -30 \leq p^{G2,RT} \leq 30 \\
& 0 \leq p^{spill} \leq W \\
& 0 \leq p^{shed} \leq 120 \\
& p^{G2,RT} + (W - p^{WP,DA} - p^{spill}) + p^{shed} = 0
\end{aligned}$$

$$\text{s.t.} \quad 0 \leq W \leq 70$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \leq p^{G1,DA} \leq 100 \\
& 0 \leq p^{G2,DA} \leq 30 \\
& 0 \leq p^{WP,DA} \leq 70 \\
& p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120
\end{aligned}$$

More at pages 95-98 of



Adaptive robust market clearing: optimization

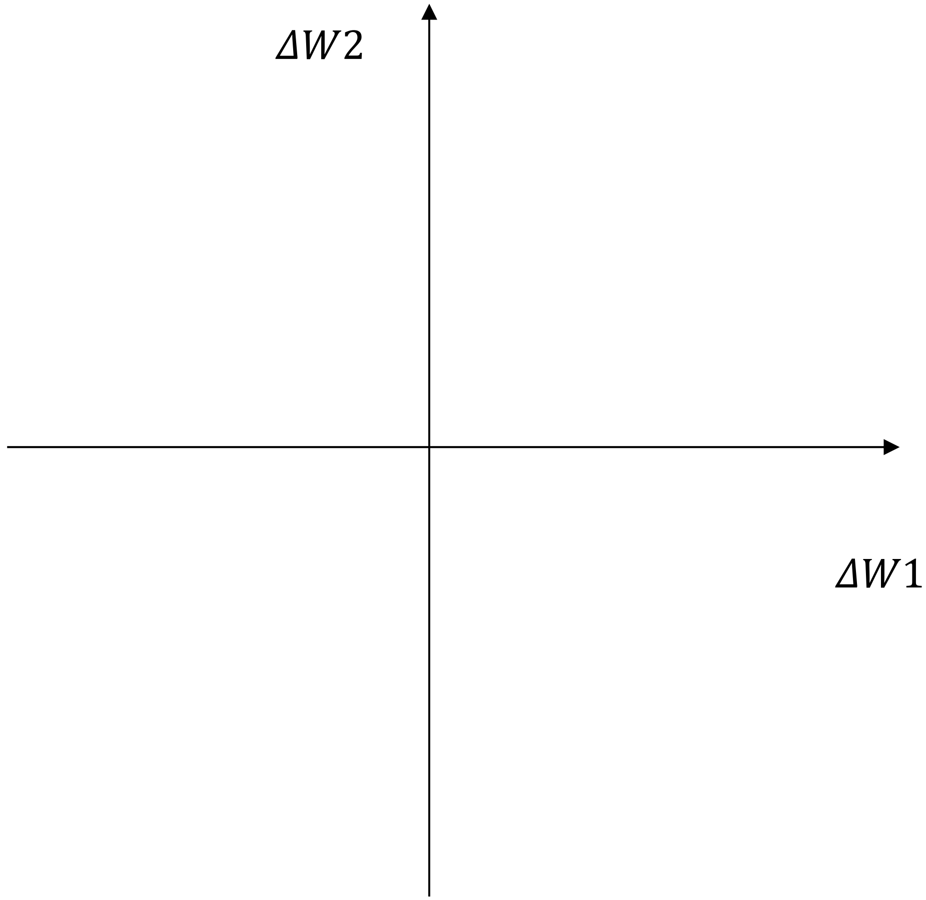
	DA	RT (scenario 1) Wind power realization: 30 MW	RT (scenario 2) Wind power realization: 60 MW	RT (scenario 3) Wind power realization: 70 MW	RT (scenario 4) Wind power realization: 40 MW
G1 [MW]	100	Unchanged	Unchanged	Unchanged	Unchanged
G2 [MW]	0	Unchanged	Unchanged	Unchanged	Unchanged
WP [MW]	20	Unchanged (10 spilled)	Unchanged (40 spilled)	Unchanged (50 spilled)	Unchanged (20 spilled)
Load [MW]	120	Not curtailed	Not curtailed	Not curtailed	Not curtailed
Corresponding operational cost [$]	1000	0	0	0	0

- Total <u>expected</u> operational cost of the system = 1000 – 0.25*[0+0+0+0] = $1000.
- If either w1, w2, w3 or w4 realizes, the total operational cost of the system will be $1000.

Uncertainty sets

Wind power = $\widehat{W} + \Delta W$

Uncertainty sets



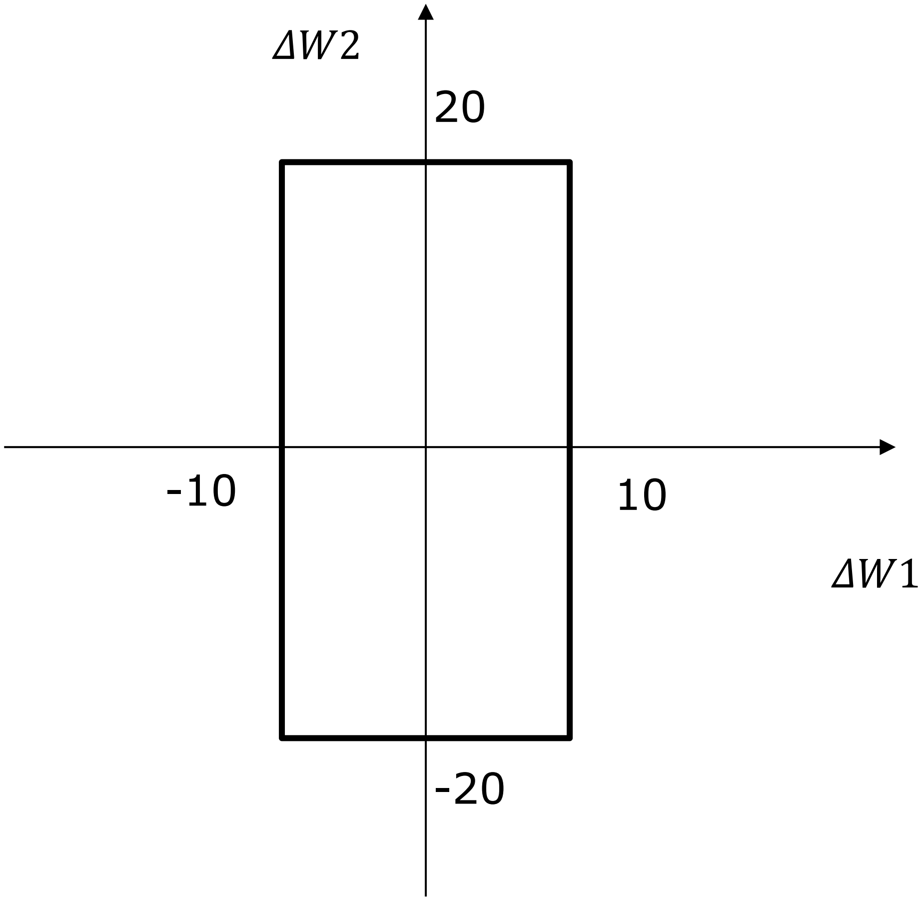
Wind power = $\widehat{W} + \Delta W$

Box uncertainty set with 2 wind farms

$|\Delta W1| \leq 10$

$|\Delta W2| \leq 20$

Uncertainty sets



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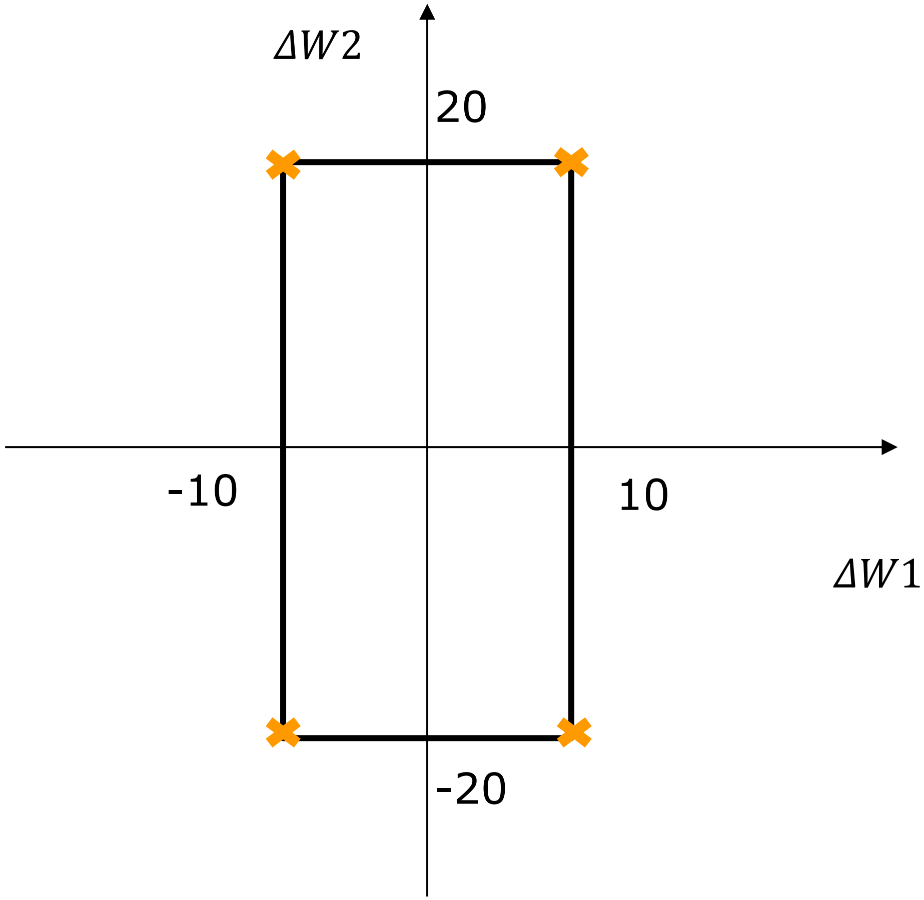
Uncertainty sets



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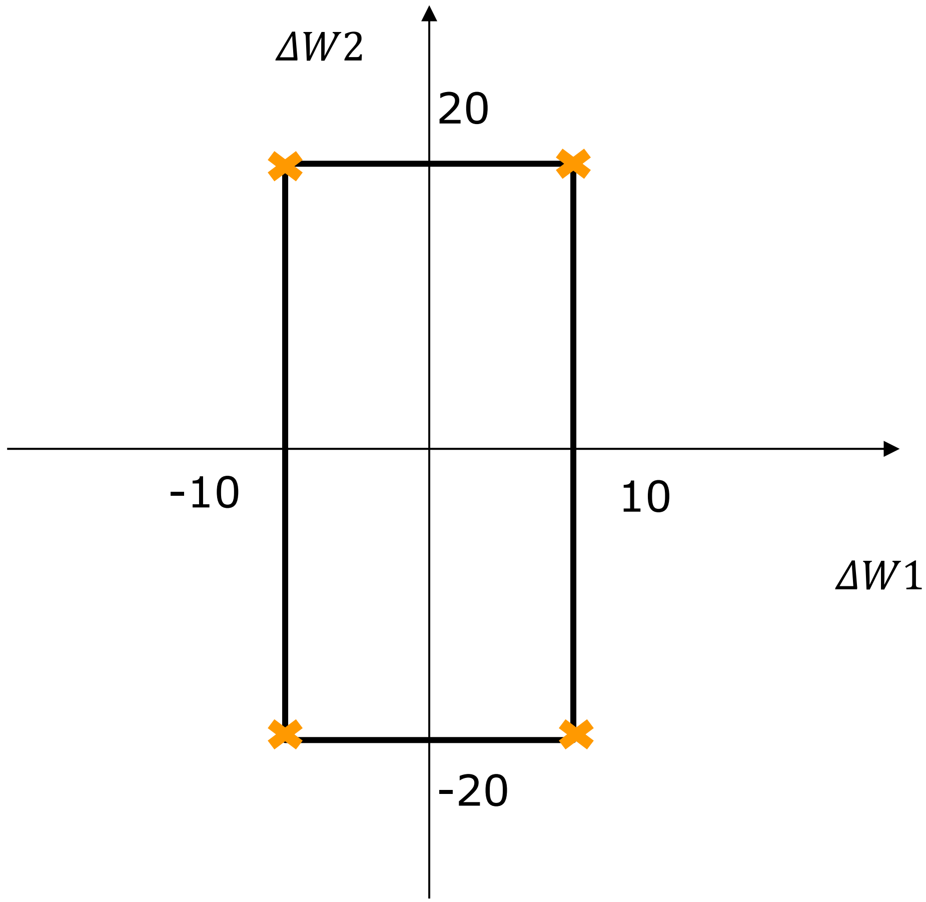


Wind power = $\widehat{W} + \Delta W$

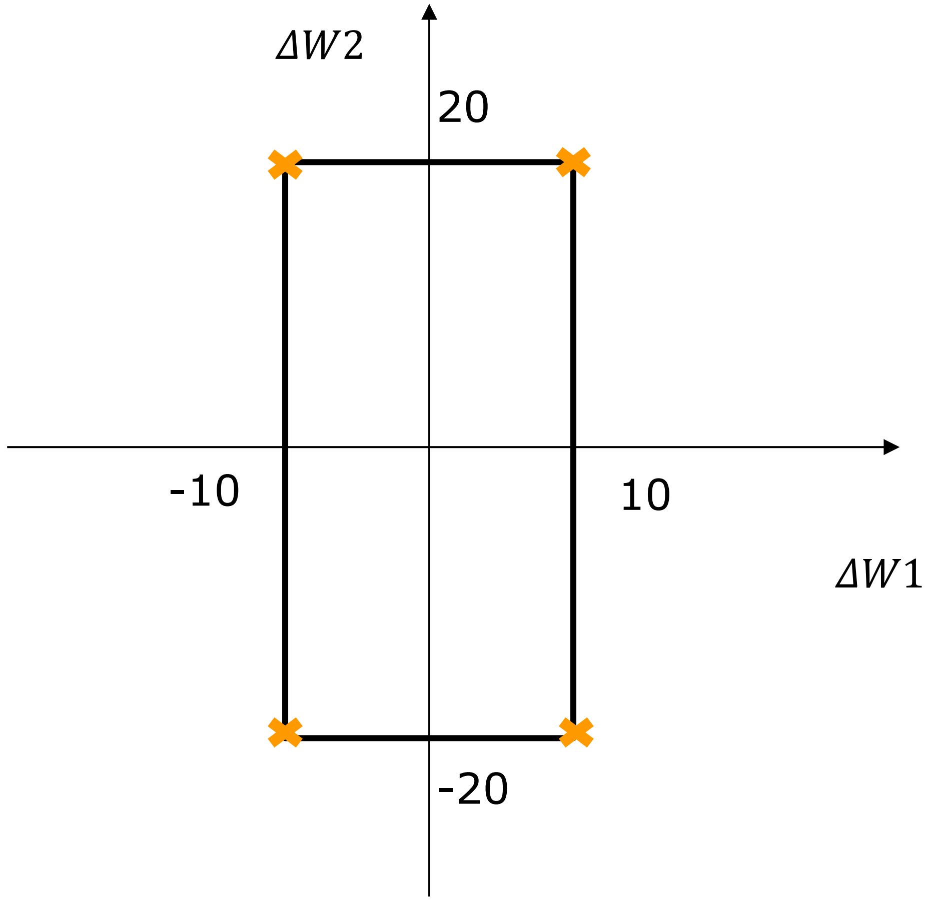
Box uncertainty set with 2 wind farms

$|\Delta W1| \leq 10$

$|\Delta W2| \leq 20$

Number of vertices as a function of the number of wind farms (N)?

Uncertainty sets



Wind power = $\widehat{W} + \Delta W$

Box uncertainty set with 2 wind farms

$|\Delta W1| \leq 10$

$|\Delta W2| \leq 20$

Number of vertices as a function of the number of wind farms (N)?

2^N

Uncertainty sets

Wind power = $\widehat{W} + \Delta W$

Budget type uncertainty set with 2 wind farms

$$|\Delta W1| \leq 10$$

$$|\Delta W2| \leq 20$$

$$\frac{|\Delta W1|}{10} + \frac{|\Delta W2|}{20} \leq 1.4$$

Budget constraint reflecting the correlation of two farms

Uncertainty sets

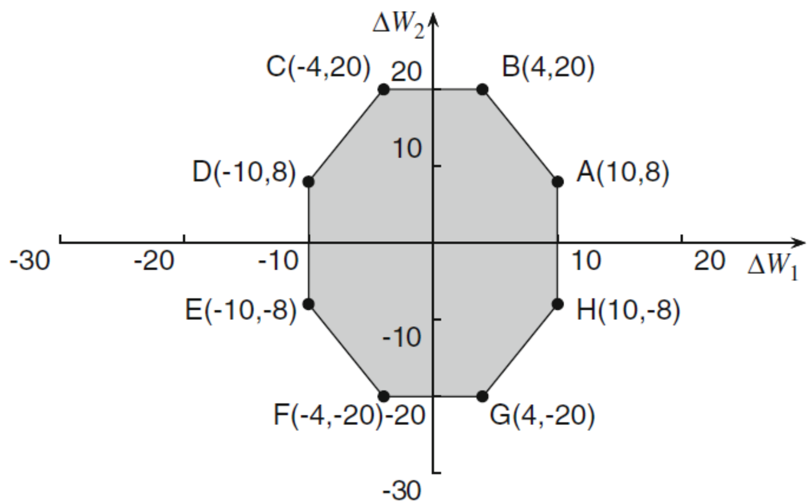
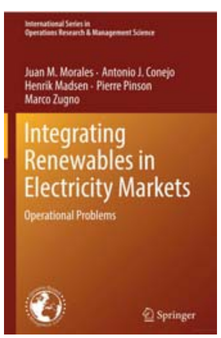


Figure from



Wind power = $\widehat{W} + \Delta W$

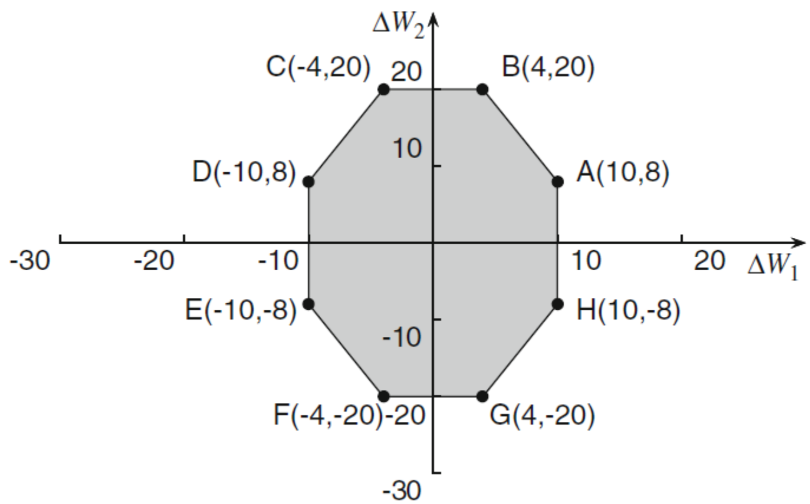
Budget type uncertainty set with 2 wind farms

$$|\Delta W1| \leq 10$$

$$|\Delta W2| \leq 20$$

$$\frac{|\Delta W1|}{10} + \frac{|\Delta W2|}{20} \leq 1.4$$

Uncertainty sets



Wind power = $\widehat{W} + \Delta W$

Budget type uncertainty set with 2 wind farms

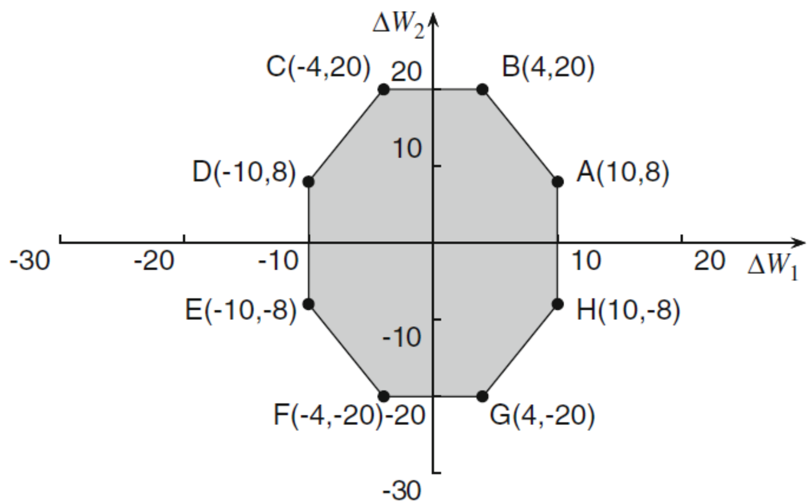
$$|\Delta W1| \leq 10$$

$$|\Delta W2| \leq 20$$

$$\frac{|\Delta W1|}{10} + \frac{|\Delta W2|}{20} \leq 1.4$$

Number of vertices as a function of the number of wind farms (N)?

Uncertainty sets



Wind power = $\widehat{W} + \Delta W$

Budget type uncertainty set with 2 wind farms

$$|\Delta W1| \leq 10$$

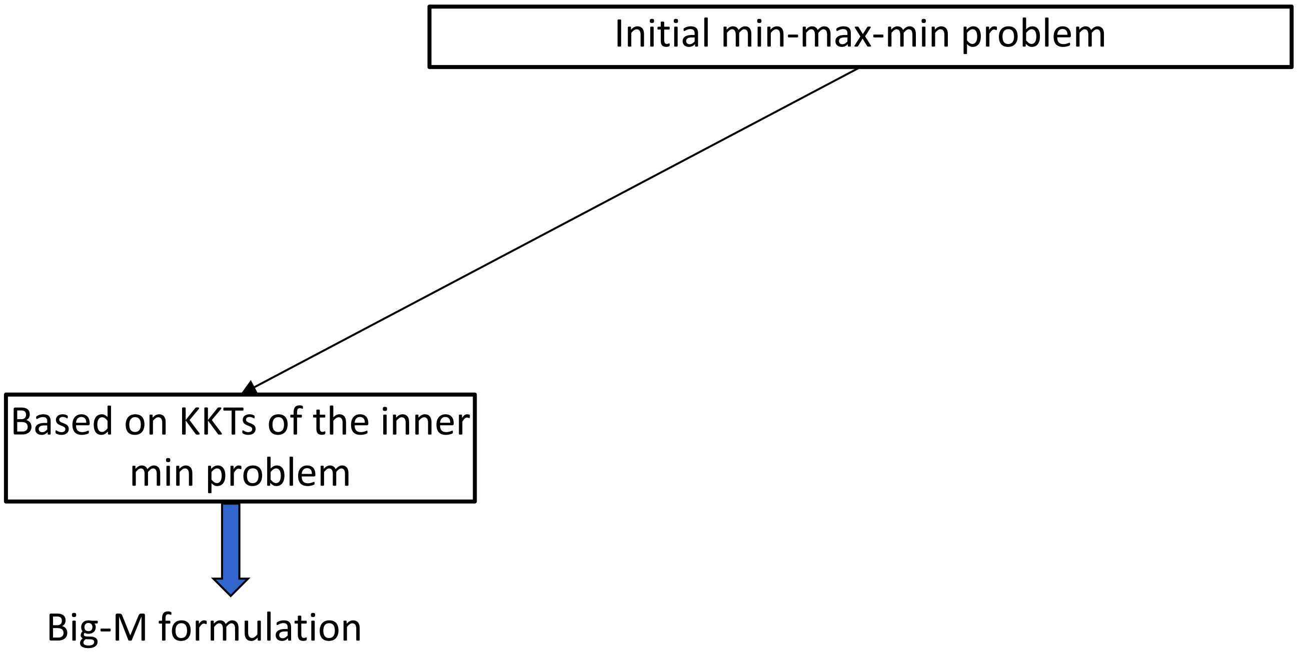
$$|\Delta W2| \leq 20$$

$$\frac{|\Delta W1|}{10} + \frac{|\Delta W2|}{20} \leq 1.4$$

Number of vertices as a function of the number of wind farms (N)?

$2^N N$

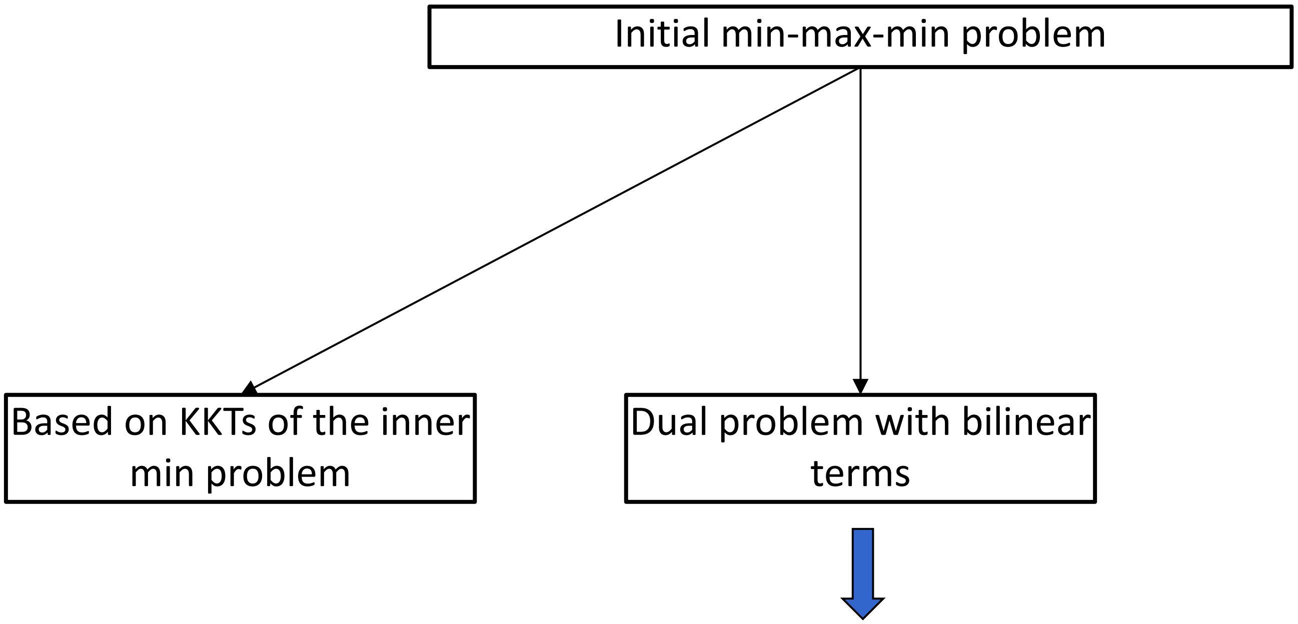
Three different solution techniques



Relevant reference:

Ruiz, C., & Conejo, A.J.,(2015). Robust transmission expansion planning. *European Journal of Operational Research*, 242, 390-401.

Three different solution techniques



Relevant references:

- Bertsimas, D., Litvinov, E., Sun, X., Zhao, J., & Zheng, T. (2012). Adaptive robust optimization for the security constrained unit commitment problem. *IEEE Transactions on Power Systems*, 28(1), 52-63.
- Jian, R., Wang, J., & Guan, Y. (2012). Robust unit commitment with wind power and pumped storage hydro. *IEEE Transactions on Power Systems*, 27(2), 800-810.
- Zugno, M., & Conejo, A.J.,(2015). A robust optimization approach to energy and reserve dispatch in electricity markets. *European Journal of Operational Research,* 247, 659-671.

Three different solution techniques

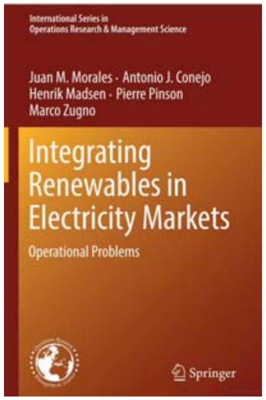
Initial min-max-min problem

Based on KKTs of the inner min problem

Dual problem with bilinear terms

Enumeration of vertices

LP



Recap

In robust optimization, under which outcome of uncertainty do we get the optimal solution?

Recap

In robust optimization, under which outcome of uncertainty do we get the optimal solution?

- Worst case

Recap

In robust optimization, under which outcome of uncertainty do we get the optimal solution?

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What about the other outcomes of uncertainty? Do we know something about these cases?

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In robust optimization, under which outcome of uncertainty do we get the optimal solution?

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What about the other outcomes of uncertainty? Do we know something about these cases?

- Feasibility

How can we reduce conservativeness of the solution?

Recap

In robust optimization, under which outcome of uncertainty do we get the optimal solution?

- Worst case

What about the other outcomes of uncertainty? Do we know something about these cases?

- Feasibility

How can we reduce conservativeness of the solution?

- Well structured uncertainty sets

Chance-constrained programming

DTU Electrical Engineering, Technical University of Denmark

Jalal Kazempour

Chance-constrained programming

The optimization problem is formulated in such a way that the probability of meeting a certain constraint is above a certain level ε.

The feasibility region is extended with respect to robust optimization by allowing a violation of constraints in ε times of the cases.

Chance-constrained programming

The optimization problem is formulated in such a way that the probability of meeting a certain constraint is above a certain level ε.

The feasibility region is extended with respect to robust optimization by allowing a violation of constraints in ε times of the cases.

$$\mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30) \geq 1 - \epsilon$$

Chance-constrained programming

The optimization problem is formulated in such a way that the probability of meeting a certain constraint is above a certain level ε.

The feasibility region is extended with respect to robust optimization by allowing a violation of constraints in ε times of the cases.

$$\mathbb{P}(\underbrace{p^{\mathrm{G2,RT}}(\Delta W)} \leq 30) \geq 1 - \epsilon$$

Risk parameter (given) whose value is lying between zero and one

Probability distribution of ΔW (known)

Recourse action (variable) depends on uncertainty, e.g., via a linear function!

Chance-constrained programming

The optimization problem is formulated in such a way that the probability of meeting a certain constraint is above a certain level ε.

The feasibility region is extended with respect to robust optimization by allowing a violation of constraints in ε times of the cases.

$$\mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30) \geq 1 - \epsilon$$

Probability distribution of ΔW **(known)**

This assumption (perfect knowledge about the probability distribution of the uncertainty) will be relaxed later!

Chance-constrained programming

The optimization problem is formulated in such a way that the probability of meeting a certain constraint is above a certain level ε.

The feasibility region is extended with respect to robust optimization by allowing a violation of constraints in ε times of the cases.

$$\mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30) \geq 1 - \epsilon$$

If ε = 0, then it is like solving the problem with a robust constraint!

Chance-constrained programming

$$
\begin{aligned}
\min_{p^{\mathrm{G1,DA}},p^{\mathrm{G2,DA}},p^{\mathrm{WP,DA}},p^{\mathrm{G2,RT}},p^{\mathrm{shed}},p^{\mathrm{spill}}} \quad & [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \mathbb{E}^{\mathbb{P}}(20p^{\mathrm{G2,RT}}(\Delta W) + 80p^{\mathrm{shed}}(\Delta W)) \\
\text{s. t.} \quad & 0 \le p^{\mathrm{G1,DA}} \le 100 \\
& 0 \le p^{\mathrm{G2,DA}} \le 30 \\
& 0 \le p^{\mathrm{WP,DA}} \le 70 \\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120 \\
& \mathbb{P}(-p^{\mathrm{G2,DA}} \le p^{\mathrm{G2,RT}}(\Delta W)) \ge 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \le 30 - p^{\mathrm{G2,DA}}) \ge 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \le 30) \ge 1-\epsilon \\
& \mathbb{P}(-30 \le p^{\mathrm{G2,RT}}(\Delta W)) \ge 1-\epsilon \\
& \mathbb{P}(0 \le p^{\mathrm{spill}}(\Delta W)) \ge 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{spill}}(\Delta W) \le 30) \ge 1-\epsilon \\
& \mathbb{P}(0 \le p^{\mathrm{shed}}(\Delta W)) \ge 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{shed}}(\Delta W) \le 120) \ge 1-\epsilon \\
& p^{\mathrm{G2,RT}}(\Delta W) + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}(\Delta W)) + p^{\mathrm{shed}}(\Delta W) = 0
\end{aligned}
$$

Chance-constrained programming

$$
\begin{aligned}
\min_{\boxed{p^{\mathrm{G1,DA}},p^{\mathrm{G2,DA}},p^{\mathrm{WP,DA}}},p^{\mathrm{G2,RT}},p^{\mathrm{shed}},p^{\mathrm{spill}}} \quad & [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \mathbb{E}^{\mathbb{P}}(20p^{\mathrm{G2,RT}}(\Delta W) + 80p^{\mathrm{shed}}(\Delta W)) \\
\text{s. t.} \quad & 0 \le p^{\mathrm{G1,DA}} \le 100 \\
& 0 \le p^{\mathrm{G2,DA}} \le 30 \\
& 0 \le p^{\mathrm{WP,DA}} \le 70 \\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120 \\
& \mathbb{P}(-p^{\mathrm{G2,DA}} \le p^{\mathrm{G2,RT}}(\Delta W)) \ge 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \le 30 - p^{\mathrm{G2,DA}}) \ge 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \le 30) \ge 1-\epsilon \\
& \mathbb{P}(-30 \le p^{\mathrm{G2,RT}}(\Delta W)) \ge 1-\epsilon \\
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& \mathbb{P}(p^{\mathrm{spill}}(\Delta W) \le 30) \ge 1-\epsilon \\
& \mathbb{P}(0 \le p^{\mathrm{shed}}(\Delta W)) \ge 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{shed}}(\Delta W) \le 120) \ge 1-\epsilon \\
& p^{\mathrm{G2,RT}}(\Delta W) + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}(\Delta W)) + p^{\mathrm{shed}}(\Delta W) = 0
\end{aligned}
$$

DA variables: schedules of G1, G2 and WP

Chance-constrained programming

$$\min_{p^{\mathrm{G1,DA}},p^{\mathrm{G2,DA}},p^{\mathrm{WP,DA}},\boxed{p^{\mathrm{G2,RT}},p^{\mathrm{shed}},p^{\mathrm{spill}}}} [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \mathbb{E}^{\mathbb{P}}(20p^{\mathrm{G2,RT}}(\Delta W) + 80p^{\mathrm{shed}}(\Delta W))$$

$$\begin{aligned}
\text{s. t.}\quad & 0 \leq p^{\mathrm{G1,DA}} \leq 100\\
& 0 \leq p^{\mathrm{G2,DA}} \leq 30\\
& 0 \leq p^{\mathrm{WP,DA}} \leq 70\\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120\\
& \mathbb{P}(-p^{\mathrm{G2,DA}} \leq p^{\mathrm{G2,RT}}(\Delta W)) \geq 1-\epsilon\\
& \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30 - p^{\mathrm{G2,DA}}) \geq 1-\epsilon\\
& \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30) \geq 1-\epsilon\\
& \mathbb{P}(-30 \leq p^{\mathrm{G2,RT}}(\Delta W)) \geq 1-\epsilon\\
& \mathbb{P}(0 \leq p^{\mathrm{spill}}(\Delta W)) \geq 1-\epsilon\\
& \mathbb{P}(p^{\mathrm{spill}}(\Delta W) \leq 30) \geq 1-\epsilon\\
& \mathbb{P}(0 \leq p^{\mathrm{shed}}(\Delta W)) \geq 1-\epsilon\\
& \mathbb{P}(p^{\mathrm{shed}}(\Delta W) \leq 120) \geq 1-\epsilon\\
& p^{\mathrm{G2,RT}}(\Delta W) + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}(\Delta W)) + p^{\mathrm{shed}}(\Delta W) = 0
\end{aligned}$$

RT variables: dependent on uncertainty, i.e., a function of ΔW

Chance-constrained programming

$$
\begin{aligned}
\min_{p^{\mathrm{G1,DA}},p^{\mathrm{G2,DA}},p^{\mathrm{WP,DA}},p^{\mathrm{G2,RT}},p^{\mathrm{shed}},p^{\mathrm{spill}}} \quad & \boxed{[10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}]} + \mathbb{E}^{\mathbb{P}}(20p^{\mathrm{G2,RT}}(\Delta W) + 80p^{\mathrm{shed}}(\Delta W)) \\
\text{s.t.} \quad & 0 \leq p^{\mathrm{G1,DA}} \leq 100 \\
& 0 \leq p^{\mathrm{G2,DA}} \leq 30 \\
& 0 \leq p^{\mathrm{WP,DA}} \leq 70 \\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120 \\
& \mathbb{P}(-p^{\mathrm{G2,DA}} \leq p^{\mathrm{G2,RT}}(\Delta W)) \geq 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30 - p^{\mathrm{G2,DA}}) \geq 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30) \geq 1-\epsilon \\
& \mathbb{P}(-30 \leq p^{\mathrm{G2,RT}}(\Delta W)) \geq 1-\epsilon \\
& \mathbb{P}(0 \leq p^{\mathrm{spill}}(\Delta W)) \geq 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{spill}}(\Delta W) \leq 30) \geq 1-\epsilon \\
& \mathbb{P}(0 \leq p^{\mathrm{shed}}(\Delta W)) \geq 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{shed}}(\Delta W) \leq 120) \geq 1-\epsilon \\
& p^{\mathrm{G2,RT}}(\Delta W) + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}(\Delta W)) + p^{\mathrm{shed}}(\Delta W) = 0
\end{aligned}
$$

Total operational cost of the system in DA (deterministic)

Chance-constrained programming

$$
\begin{aligned}
\min_{p^{\text{G1,DA}},p^{\text{G2,DA}},p^{\text{WP,DA}},p^{\text{G2,RT}},p^{\text{shed}},p^{\text{spill}}} \quad & [10p^{\text{G1,DA}} + 20p^{\text{G2,DA}}] + \boxed{\mathbb{E}^{\mathbb{P}}(20p^{\text{G2,RT}}(\Delta W) + 80p^{\text{shed}}(\Delta W))} \\
\text{s. t.} \quad & 0 \leq p^{\text{G1,DA}} \leq 100 \\
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& 0 \leq p^{\text{WP,DA}} \leq 70 \\
& p^{\text{G1,DA}} + p^{\text{G2,DA}} + p^{\text{WP,DA}} = 120 \\
& \mathbb{P}(-p^{\text{G2,DA}} \leq p^{\text{G2,RT}}(\Delta W)) \geq 1-\epsilon \\
& \mathbb{P}(p^{\text{G2,RT}}(\Delta W) \leq 30 - p^{\text{G2,DA}}) \geq 1-\epsilon \\
& \mathbb{P}(p^{\text{G2,RT}}(\Delta W) \leq 30) \geq 1-\epsilon \\
& \mathbb{P}(-30 \leq p^{\text{G2,RT}}(\Delta W)) \geq 1-\epsilon \\
& \mathbb{P}(0 \leq p^{\text{spill}}(\Delta W)) \geq 1-\epsilon \\
& \mathbb{P}(p^{\text{spill}}(\Delta W) \leq 30) \geq 1-\epsilon \\
& \mathbb{P}(0 \leq p^{\text{shed}}(\Delta W)) \geq 1-\epsilon \\
& \mathbb{P}(p^{\text{shed}}(\Delta W) \leq 120) \geq 1-\epsilon \\
& p^{\text{G2,RT}}(\Delta W) + (30 - p^{\text{WP,DA}} - p^{\text{spill}}(\Delta W)) + p^{\text{shed}}(\Delta W) = 0
\end{aligned}
$$

Total expected operational cost of the system in RT

Chance-constrained programming

$$
\begin{aligned}
\min_{p^{\text{G1,DA}},p^{\text{G2,DA}},p^{\text{WP,DA}},p^{\text{G2,RT}},p^{\text{shed}},p^{\text{spill}}} \quad & [10p^{\text{G1,DA}} + 20p^{\text{G2,DA}}] + \boxed{\mathbb{E}^{\mathbb{P}}(20p^{\text{G2,RT}}(\Delta W) + 80p^{\text{shed}}(\Delta W))} \\
\text{s. t.} \quad & 0 \leq p^{\text{G1,DA}} \leq 100 \\
& 0 \leq p^{\text{G2,DA}} \leq 30 \\
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& p^{\text{G1,DA}} + p^{\text{G2,DA}} + p^{\text{WP,DA}} = 120 \\
& \mathbb{P}(-p^{\text{G2,DA}} \leq p^{\text{G2,RT}}(\Delta W)) \geq 1-\epsilon \\
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& \mathbb{P}(p^{\text{G2,RT}}(\Delta W) \leq 30) \geq 1-\epsilon \\
& \mathbb{P}(-30 \leq p^{\text{G2,RT}}(\Delta W)) \geq 1-\epsilon \\
& \mathbb{P}(0 \leq p^{\text{spill}}(\Delta W)) \geq 1-\epsilon \\
& \mathbb{P}(p^{\text{spill}}(\Delta W) \leq 30) \geq 1-\epsilon \\
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& \mathbb{P}(p^{\text{shed}}(\Delta W) \leq 120) \geq 1-\epsilon \\
& p^{\text{G2,RT}}(\Delta W) + (30 - p^{\text{WP,DA}} - p^{\text{spill}}(\Delta W)) + p^{\text{shed}}(\Delta W) = 0
\end{aligned}
$$

Total expected operational cost of the system in RT

Challenge: How can we get rid of the "expectation" operator?

- Remember we don't have scenarios!

Chance-constrained programming

$$
\begin{aligned}
\min_{p^{\mathrm{G1,DA}}, p^{\mathrm{G2,DA}}, p^{\mathrm{WP,DA}}, p^{\mathrm{G2,RT}}, p^{\mathrm{shed}}, p^{\mathrm{spill}}} \quad & [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \mathbb{E}^{\mathbb{P}}(20p^{\mathrm{G2,RT}}(\Delta W) + 80p^{\mathrm{shed}}(\Delta W)) \\
\text{s.t.} \quad & 0 \leq p^{\mathrm{G1,DA}} \leq 100 \\
& 0 \leq p^{\mathrm{G2,DA}} \leq 30 \\
& 0 \leq p^{\mathrm{WP,DA}} \leq 70 \\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120 \\
& \mathbb{P}(-p^{\mathrm{G2,DA}} \leq p^{\mathrm{G2,RT}}(\Delta W)) \geq 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30 - p^{\mathrm{G2,DA}}) \geq 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30) \geq 1-\epsilon \\
& \mathbb{P}(-30 \leq p^{\mathrm{G2,RT}}(\Delta W)) \geq 1-\epsilon \\
& \mathbb{P}(0 \leq p^{\mathrm{spill}}(\Delta W)) \geq 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{spill}}(\Delta W) \leq 30) \geq 1-\epsilon \\
& \mathbb{P}(0 \leq p^{\mathrm{shed}}(\Delta W)) \geq 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{shed}}(\Delta W) \leq 120) \geq 1-\epsilon \\
& p^{\mathrm{G2,RT}}(\Delta W) + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}(\Delta W)) + p^{\mathrm{shed}}(\Delta W) = 0
\end{aligned}
$$

DA constraints

RT constraints

Chance-constrained programming

$$\min_{p^{\mathrm{G1,DA}}, p^{\mathrm{G2,DA}}, p^{\mathrm{WP,DA}}, p^{\mathrm{G2,RT}}, p^{\mathrm{shed}}, p^{\mathrm{spill}}} \quad [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \mathbb{E}^{\mathbb{P}}(20p^{\mathrm{G2,RT}}(\Delta W) + 80p^{\mathrm{shed}}(\Delta W))$$

s. t.

DA constraints

$$\begin{aligned}
&0 \leq p^{\mathrm{G1,DA}} \leq 100\\
&0 \leq p^{\mathrm{G2,DA}} \leq 30\\
&0 \leq p^{\mathrm{WP,DA}} \leq 70\\
&p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120
\end{aligned}$$

RT constraints

$$\begin{aligned}
&\mathbb{P}(-p^{\mathrm{G2,DA}} \leq p^{\mathrm{G2,RT}}(\Delta W)) \geq 1-\epsilon\\
&\mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30 - p^{\mathrm{G2,DA}}) \geq 1-\epsilon\\
&\mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30) \geq 1-\epsilon\\
&\mathbb{P}(-30 \leq p^{\mathrm{G2,RT}}(\Delta W)) \geq 1-\epsilon\\
&\mathbb{P}(0 \leq p^{\mathrm{spill}}(\Delta W)) \geq 1-\epsilon\\
&\mathbb{P}(p^{\mathrm{spill}}(\Delta W) \leq 30) \geq 1-\epsilon\\
&\mathbb{P}(0 \leq p^{\mathrm{shed}}(\Delta W)) \geq 1-\epsilon\\
&\mathbb{P}(p^{\mathrm{shed}}(\Delta W) \leq 120) \geq 1-\epsilon\\
&p^{\mathrm{G2,RT}}(\Delta W) + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}(\Delta W)) + p^{\mathrm{shed}}(\Delta W) = 0
\end{aligned}$$

Note: Here, we have a set of <u>individual</u> chance constraints (but not a single <u>joint</u> chance constraints).

Chance-constrained programming

$$
\begin{aligned}
\min_{p^{\mathrm{G1,DA}},p^{\mathrm{G2,DA}},p^{\mathrm{WP,DA}},p^{\mathrm{G2,RT}},p^{\mathrm{shed}},p^{\mathrm{spill}}} \quad & [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \mathbb{E}^{\mathbb{P}}(20p^{\mathrm{G2,RT}}(\Delta W) + 80p^{\mathrm{shed}}(\Delta W)) \\
\text{s.t.} \quad & 0 \le p^{\mathrm{G1,DA}} \le 100 \\
& 0 \le p^{\mathrm{G2,DA}} \le 30 \\
& 0 \le p^{\mathrm{WP,DA}} \le 70 \\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120 \\
& \mathbb{P}(-p^{\mathrm{G2,DA}} \le p^{\mathrm{G2,RT}}(\Delta W)) \ge 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \le 30 - p^{\mathrm{G2,DA}}) \ge 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \le 30) \ge 1-\epsilon \\
& \mathbb{P}(-30 \le p^{\mathrm{G2,RT}}(\Delta W)) \ge 1-\epsilon \\
& \mathbb{P}(0 \le p^{\mathrm{spill}}(\Delta W)) \ge 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{spill}}(\Delta W) \le 30) \ge 1-\epsilon \\
& \mathbb{P}(0 \le p^{\mathrm{shed}}(\Delta W)) \ge 1-\epsilon \\
& \mathbb{P}(p^{\mathrm{shed}}(\Delta W) \le 120) \ge 1-\epsilon \\
& p^{\mathrm{G2,RT}}(\Delta W) + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}(\Delta W)) + p^{\mathrm{shed}}(\Delta W) = 0
\end{aligned}
$$

DA constraints

RT constraints

Challenge: How can we get rid of chance constraints?

Chance-constrained programming

1) Analytical reformulation approach

2) Sampling-based approach

Chance-constrained programming

1) Analytical reformulation approach

In specific cases under which the underlying probability distribution has **certain properties**, see [1] and [2], there exist analytical (exact) reformulations for the objective function (expected term) and for chance constraints!

[1] A. Nemirovski and A. Shapiro, "Convex approximations of chance constrained programs," *SIAM Journal on Optimization*, vol. 17, no. 4, pp. 969–996, 2007.

[2] L. Roald, "Optimization methods to manage uncertainty and risk in power systems operation," *PhD thesis, ETH Zurich*, 2016 (available online).

2) Sampling-based approach

Chance-constrained programming

1) Analytical reformulation approach

In specific cases under which the underlying probability distribution has **certain properties**, see [1] and [2], there exist analytical (exact) reformulations for the objective function (expected term) and for chance constraints!

- Example: Gaussian distribution
- This analytical reformulation results in a second-order code program (SOCP). Under some assumptions, it reduces to a linear program.

[1] A. Nemirovski and A. Shapiro, "Convex approximations of chance constrained programs," *SIAM Journal on Optimization*, vol. 17, no. 4, pp. 969–996, 2007.

[2] L. Roald, "Optimization methods to manage uncertainty and risk in power systems operation," *PhD thesis, ETH Zurich*, 2016 (available online).

2) Sampling-based approach

Chance-constrained programming

1) **Analytical reformulation approach**

2) **Sampling-based approach**

- **<u>No</u>** assumption on the type and properties of the underlying distribution
- Objective function (expected term): Sample average approximation (similar to the scenario-based stochastic programming)
- Chance constraints: Approximation via samples

Sampling approach for chance constraints

Approximate the original chance constraint

$$\mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30) \geq 1 - \epsilon$$

Sampling approach for chance constraints

Approximate the original chance constraint

$$\mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30) \geq 1 - \epsilon$$

by N sampled constraints based on the following formula [1]

$$N \geq \left\lceil \frac{1}{\epsilon} \frac{e}{e-1} \left(n_x - 1 + ln\left(\frac{1}{\beta}\right)\right)\right\rceil$$

where

- e is the Euler number
- n_x is the number of decision variables
- β is the level of confidence ensuring the solution to be an ε-level solution
- ⌈ ⌉ picks the smallest integer higher than the argument

[1] K. Margellos, P. Goulart, and J. Lygeros, “On the road between robust optimization and the scenario approach for chance constrained optimization problems,” *IEEE Transactions on Automatic Control*, vol. 59, no. 8, pp. 2258–2263, 2014.

Sampling approach for chance constraints

Approximate the original chance constraint

$$\mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30) \geq 1 - \epsilon$$

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For a standard test case of the IEEE 24-node RTS system with 6 wind farms, β=0.0001, ε=0.05 and 24 time periods:

N = 86,956 samples

Sampling approach for chance constraints

Approximate the original chance constraint

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For a standard test case of the IEEE 24-node RTS system with 6 wind farms, β=0.0001, ε=0.05 and 24 time periods:

N = 86,956 samples →

We are guaranteed to satisfy constraints for predefined ε in **out-of-sample** analysis!

Sampling approach for chance constraints

Approximate the original chance constraint

$$\mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30) \geq 1 - \epsilon$$

by N sampled constraints based on the following formula [1]

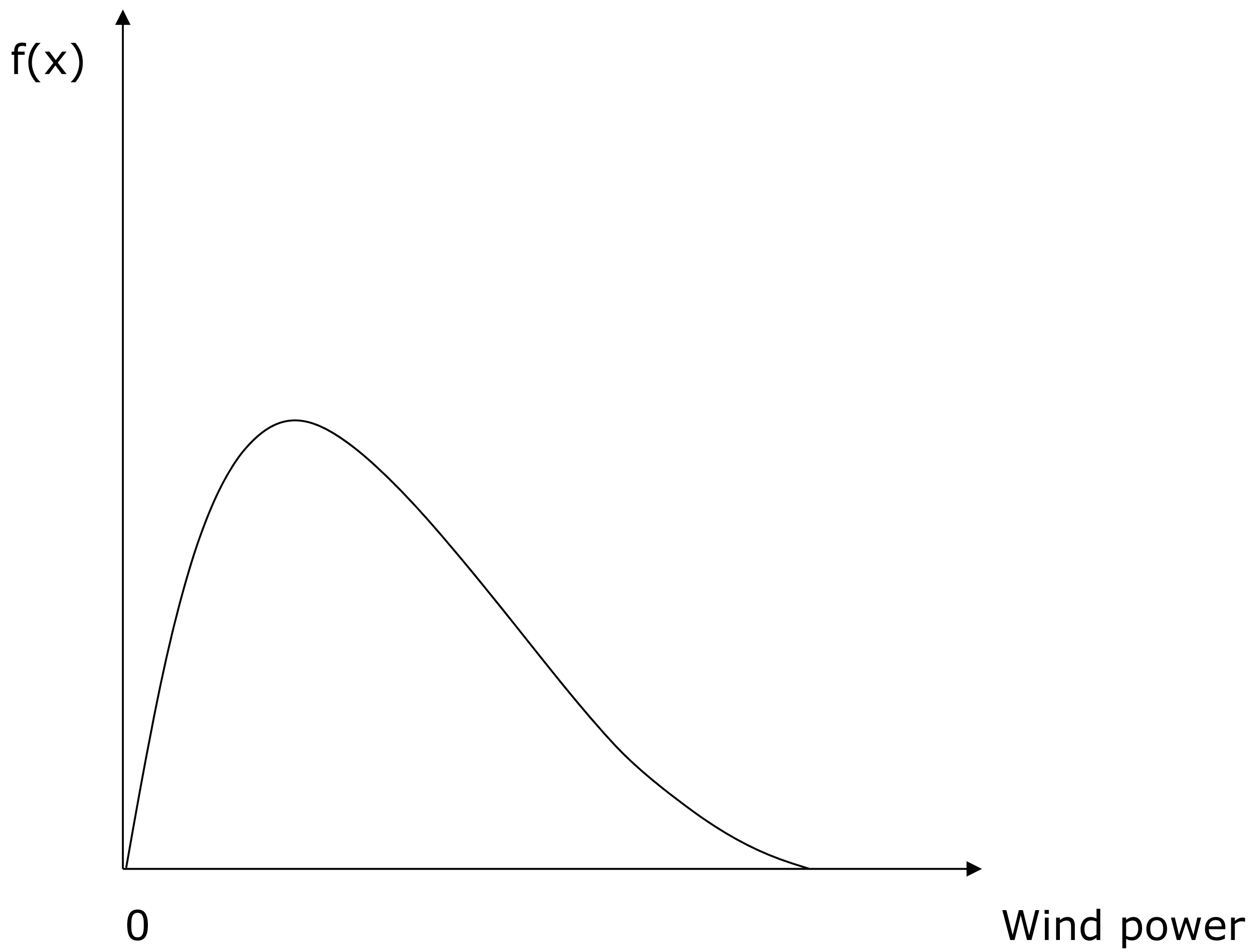
$$N \geq \left\lceil \frac{1}{\epsilon} \frac{e}{e-1} \left(n_x - 1 + ln\left(\frac{1}{\beta}\right)\right)\right\rceil$$

For a standard test case of the IEEE 24-node RTS system with 6 wind farms, β=0.0001, ε=0.05 and 24 time periods:

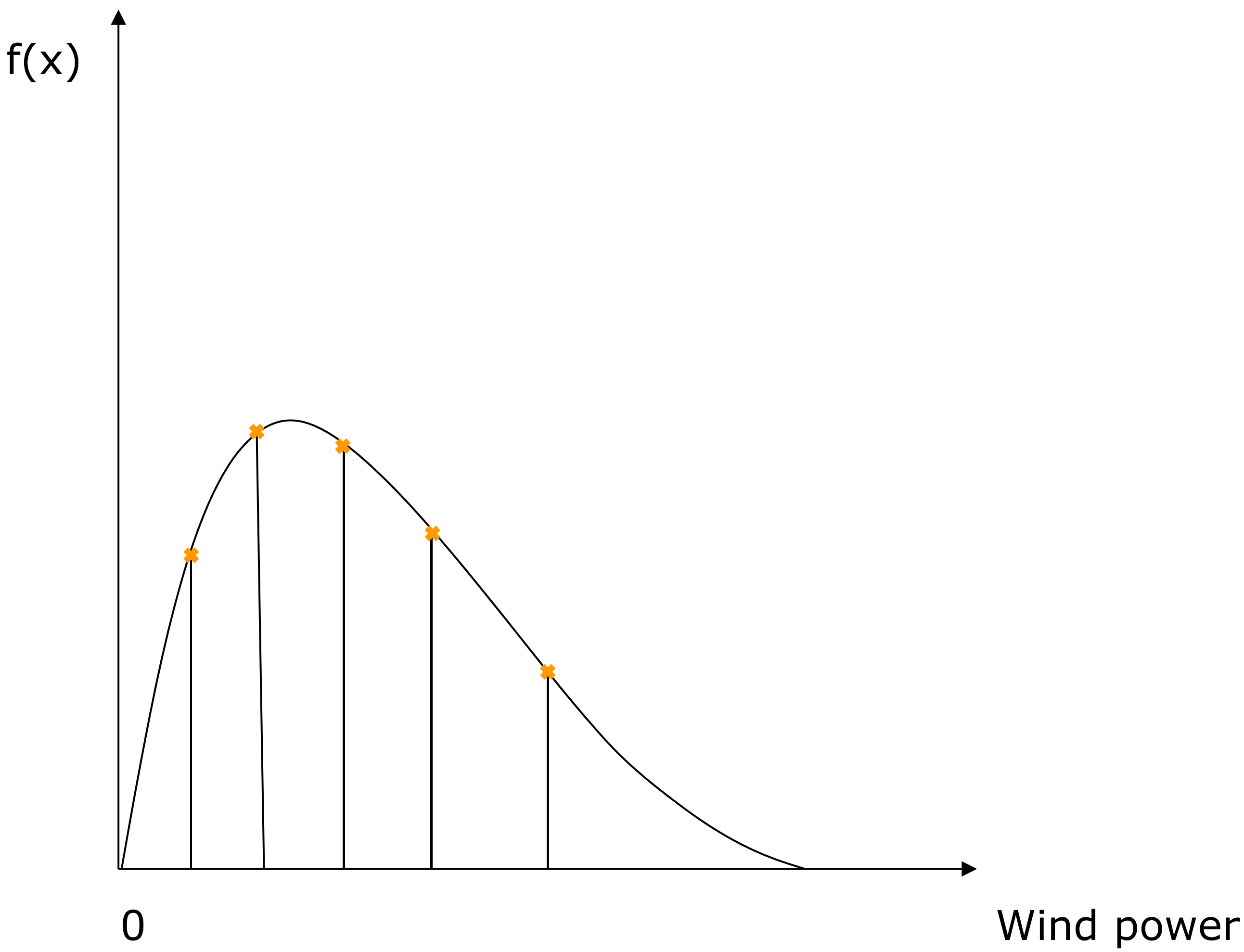
N = 86,956 samples → Computationally hard to solve …

future lectures: decomposition techniques!

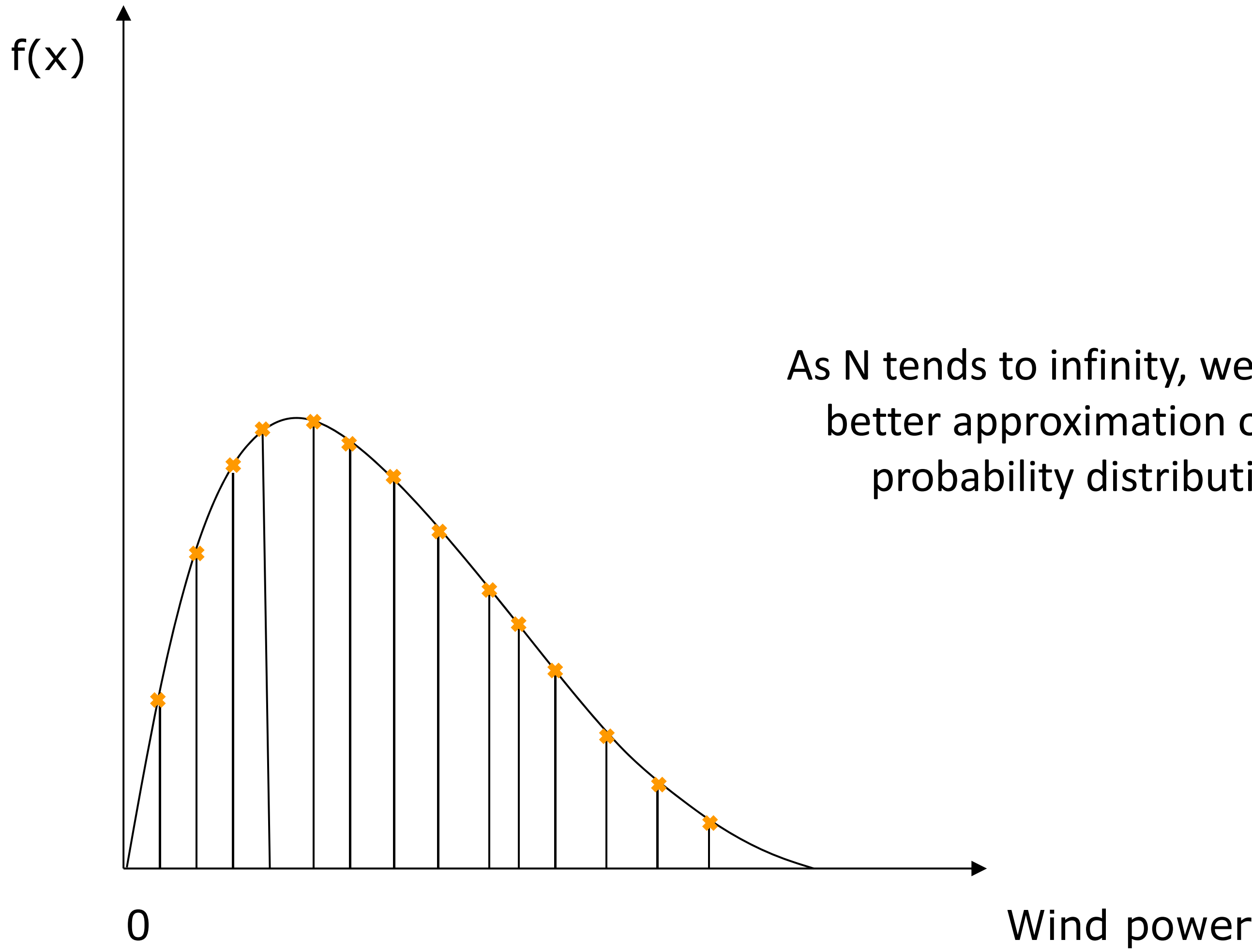
Sampling approach for chance constraints



Sampling approach for chance constraints



Sampling approach for chance constraints



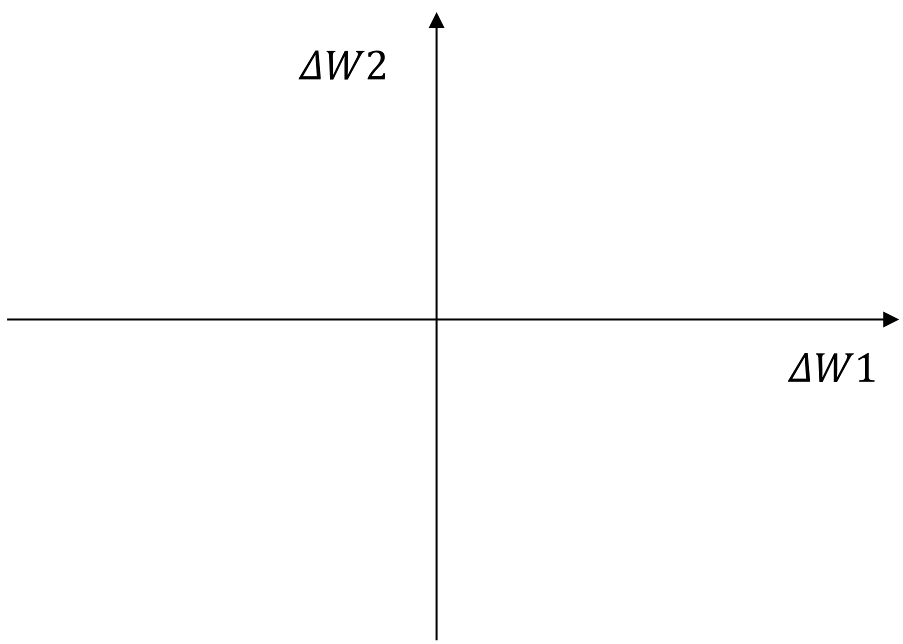
As N tends to infinity, we have a better approximation of the probability distribution

Robust approach for chance constraints

Use the same formula to calculate the number of samples N

$$N \geq \left\lceil \frac{1}{\epsilon} \frac{e}{e-1} \left(n_x - 1 + ln\left(\frac{1}{\beta}\right)\right)\right\rceil$$

Use the samples



Robust approach for chance constraints

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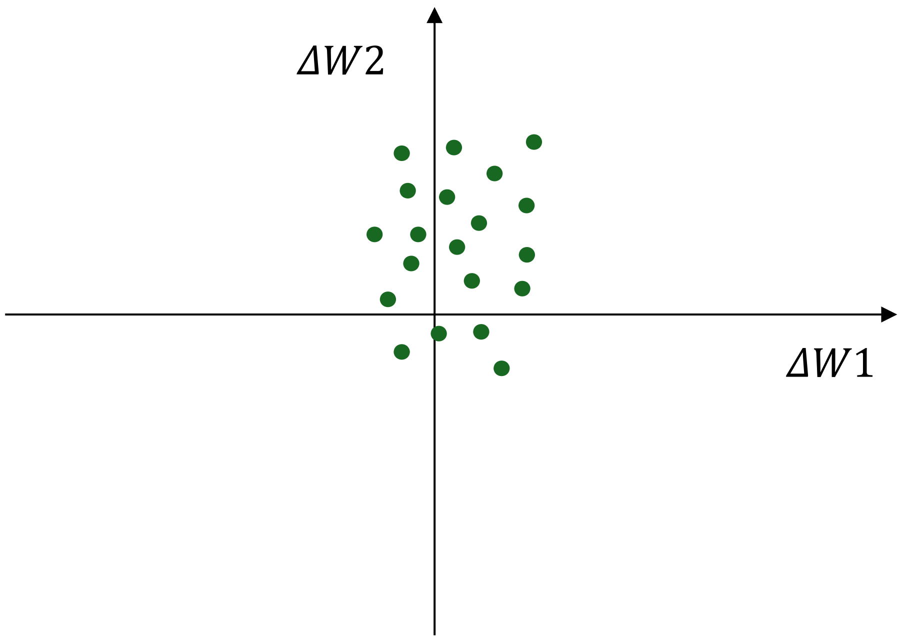


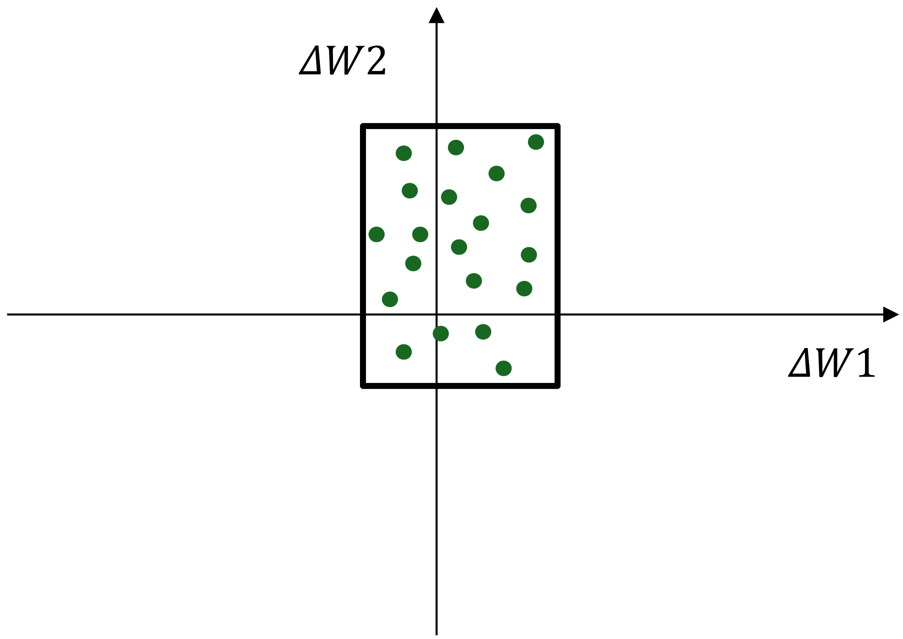
Jalal Kazempour

Robust approach for chance constraints

Use the same formula to calculate the number of samples N

$$N \geq \left\lceil \frac{1}{\epsilon} \frac{e}{e-1} \left(n_x - 1 + ln\left(\frac{1}{\beta}\right) \right) \right\rceil$$

Use the samples and fit a box (or an ellipsoid) uncertainty set around the samples



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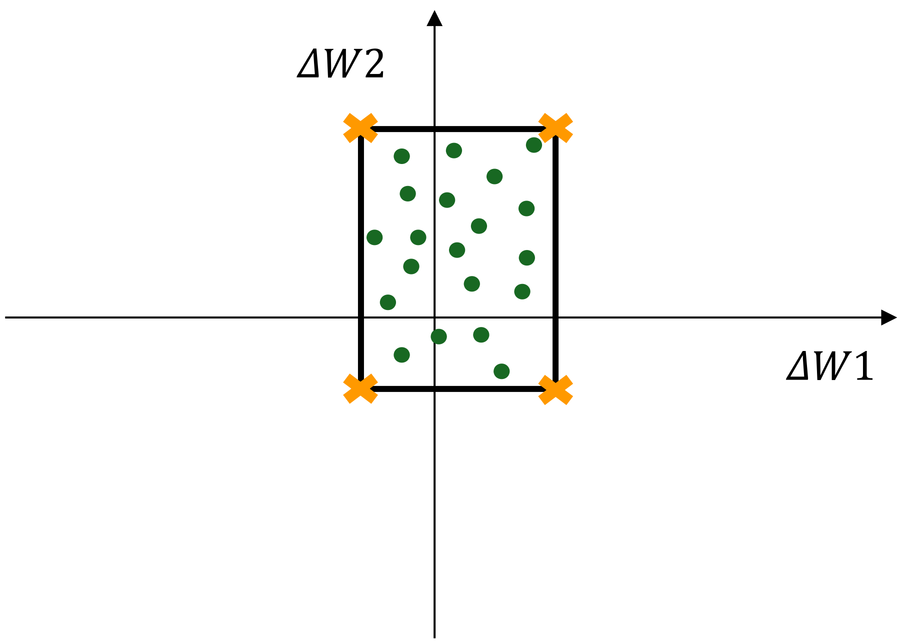

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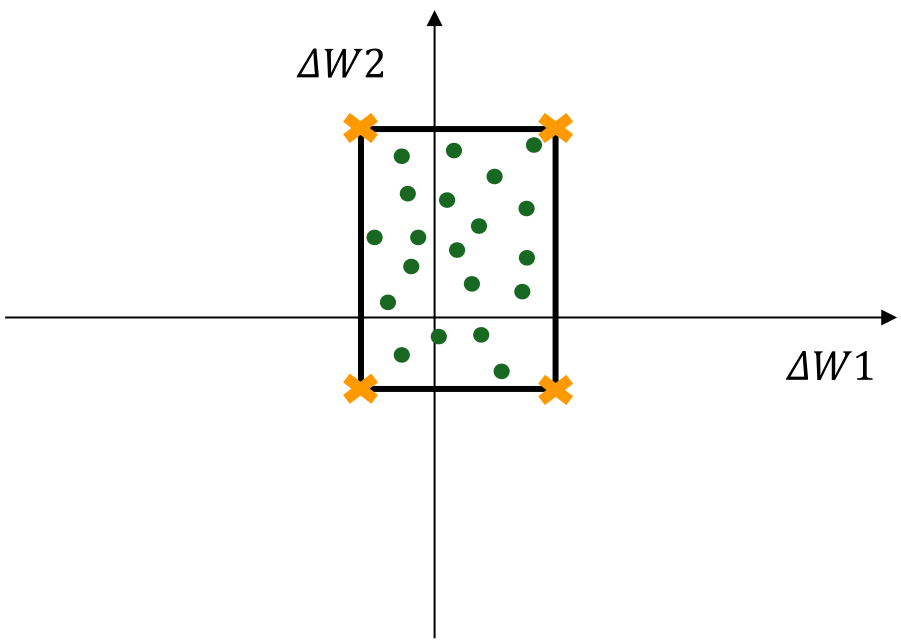

Usually yields conservative solutions as robust optimization

Additional references on chance-constrained programming

- D. Bienstock, M. Chertkov, and S. Harnett, "Chance-Constrained Optimal Power Flow: Risk-Aware Network Control under Uncertainty," *SIAM Review*, vol. 56, no. 3, pp. 461–495, 2014.

- Ben-Tal, L. El Ghaoui, and A. Nemirovski, Robust Optimization, Princeton University Press, 2009.

- Presentations by Prof. Shabbir Ahmed:

https://www2.isye.gatech.edu/people/faculty/Shabbir_Ahmed/talks.html

Recap

For a given uncertainty set, how can we reduce the conservativeness of robust optimization?

Recap

For a given uncertainty set, how can we reduce the conservativeness of robust optimization?

- By using chance constraints

Recap

For a given uncertainty set, how can we reduce the conservativeness of robust optimization?

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- Usually yes!

How to deal with uncertainty without making an assumption of probability distribution and when N is relatively small?

- Distributionally robust optimization

Distributionally robust optimization (DRO)

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Robust optimization	Distributionally robust optimization
Worst-case realization in a given uncertainty set	Worst-case distribution in a given ambiguity set
Worst-case cost	Worst-case expected cost

Distributionally robust optimization (DRO)

$$
\begin{aligned}
\min_{p^{\text{G1,DA}},p^{\text{G2,DA}},p^{\text{WP,DA}},p^{\text{G2,RT}},p^{\text{shed}},p^{\text{spill}}} \quad & [10p^{\text{G1,DA}} + 20p^{\text{G2,DA}}] + \max_{\mathbb{P}\in\mathcal{P}} \mathbb{E}^{\mathbb{P}}(20p^{\text{G2,RT}}(\Delta W) + 80p^{\text{shed}}(\Delta W)) \\
\text{s.t.} \quad & 0 \leq p^{\text{G1,DA}} \leq 100 \\
& 0 \leq p^{\text{G2,DA}} \leq 30 \\
& 0 \leq p^{\text{WP,DA}} \leq 70 \\
& p^{\text{G1,DA}} + p^{\text{G2,DA}} + p^{\text{WP,DA}} = 120 \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(-p^{\text{G2,DA}} \leq p^{\text{G2,RT}}(\Delta W)) \geq 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\text{G2,RT}}(\Delta W) \leq 30 - p^{\text{G2,DA}}) \geq 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\text{G2,RT}}(\Delta W) \leq 30) \geq 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(-30 \leq p^{\text{G2,RT}}(\Delta W)) \geq 1-\epsilon \\
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& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\text{shed}}(\Delta W) \leq 120) \geq 1-\epsilon \\
& p^{\text{G2,RT}}(\Delta W) + (30 - p^{\text{WP,DA}} - p^{\text{spill}}(\Delta W)) + p^{\text{shed}}(\Delta W) = 0
\end{aligned}
$$

Distributionally robust optimization (DRO)

$$\min_{p^{\text{G1,DA}},p^{\text{G2,DA}},p^{\text{WP,DA}},p^{\text{G2,RT}},p^{\text{shed}},p^{\text{spill}}} \quad [10p^{\text{G1,DA}} + 20p^{\text{G2,DA}}] + \max_{\mathbb{P}\in\boxed{\mathcal{P}}} \mathbb{E}^{\mathbb{P}}(20p^{\text{G2,RT}}(\Delta W) + 80p^{\text{shed}}(\Delta W))$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \leq p^{\text{G1,DA}} \leq 100 \\
& 0 \leq p^{\text{G2,DA}} \leq 30 \\
& 0 \leq p^{\text{WP,DA}} \leq 70 \\
& p^{\text{G1,DA}} + p^{\text{G2,DA}} + p^{\text{WP,DA}} = 120 \\
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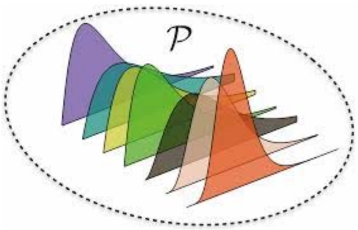
\end{aligned}$$

Ambiguity set: a <u>set</u> of potential probability distributions for modeling uncertainty



Source: MIT.edu/vanparys

Distributionally robust optimization (DRO)

$$
\begin{aligned}
\min_{p^{\mathrm{G1,DA}},p^{\mathrm{G2,DA}},p^{\mathrm{WP,DA}},p^{\mathrm{G2,RT}},p^{\mathrm{shed}},p^{\mathrm{spill}}} \quad & [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \max_{\mathbb{P}\in\boxed{\mathcal{P}}} \mathbb{E}^{\mathbb{P}}(20p^{\mathrm{G2,RT}}(\Delta W) + 80p^{\mathrm{shed}}(\Delta W)) \\
\text{s.t.} \quad & 0 \le p^{\mathrm{G1,DA}} \le 100 \\
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& 0 \le p^{\mathrm{WP,DA}} \le 70 \\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120 \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(-p^{\mathrm{G2,DA}} \le p^{\mathrm{G2,RT}}(\Delta W)) \ge 1-\epsilon \\
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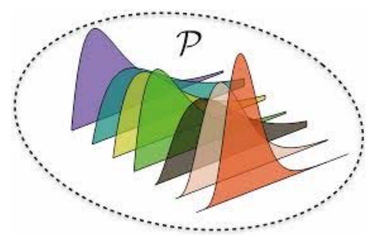

Source: MIT.edu/vanparys

Why do we consider a set of distributions instead of a single one to model an uncertainty?

Distributionally robust optimization (DRO)

$$
\begin{aligned}
\min_{p^{\text{G1,DA}},p^{\text{G2,DA}},p^{\text{WP,DA}},p^{\text{G2,RT}},p^{\text{shed}},p^{\text{spill}}} \quad & [10p^{\text{G1,DA}} + 20p^{\text{G2,DA}}] + \max_{\mathbb{P}\in\boxed{\mathcal{P}}} \mathbb{E}^{\mathbb{P}}(20p^{\text{G2,RT}}(\Delta W) + 80p^{\text{shed}}(\Delta W)) \\
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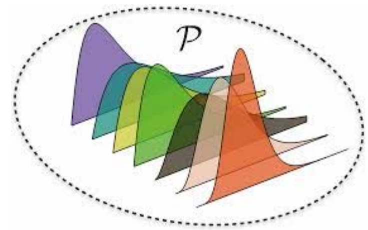
Ambiguity set: a set of potential probability distributions for modeling uncertainty



Source: MIT.edu/vanparys

Why do we consider a set of distributions instead of a single one to model an uncertainty?
Because we don't know the "true" distribution!

Distributionally robust optimization (DRO)

$$
\begin{aligned}
\min_{p^{\mathrm{G1,DA}},p^{\mathrm{G2,DA}},p^{\mathrm{WP,DA}},p^{\mathrm{G2,RT}},p^{\mathrm{shed}},p^{\mathrm{spill}}} \quad & [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \max_{\mathbb{P}\in\boxed{\mathcal{P}}} \mathbb{E}^{\mathbb{P}}(20p^{\mathrm{G2,RT}}(\Delta W) + 80p^{\mathrm{shed}}(\Delta W)) \\
\text{s. t.} \quad & 0 \leq p^{\mathrm{G1,DA}} \leq 100 \\
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\end{aligned}
$$

Ambiguity set: a set of potential probability distributions for modeling uncertainty

We will talk soon how to build an ambiguity set!

Distributionally robust optimization (DRO)

$$
\begin{aligned}
\min_{p^{\mathrm{G1,DA}}, p^{\mathrm{G2,DA}}, p^{\mathrm{WP,DA}}, p^{\mathrm{G2,RT}}, p^{\mathrm{shed}}, p^{\mathrm{spill}}} \quad & [10 p^{\mathrm{G1,DA}} + 20 p^{\mathrm{G2,DA}}] + \max_{\boxed{\mathbb{P}} \in \mathcal{P}} \mathbb{E}^{\mathbb{P}} (20 p^{\mathrm{G2,RT}}(\Delta W) + 80 p^{\mathrm{shed}}(\Delta W)) \\
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& \min_{\mathbb{P} \in \mathcal{P}} \mathbb{P}(-p^{\mathrm{G2,DA}} \leq p^{\mathrm{G2,RT}}(\Delta W)) \geq 1 - \epsilon \\
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& p^{\mathrm{G2,RT}}(\Delta W) + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}(\Delta W)) + p^{\mathrm{shed}}(\Delta W) = 0
\end{aligned}
$$

Worst probability distribution within the ambiguity set (to be determined endogenously within the problem)

Distributionally robust optimization (DRO)

$$
\begin{aligned}
\min_{p^{\mathrm{G1,DA}},p^{\mathrm{G2,DA}},p^{\mathrm{WP,DA}},p^{\mathrm{G2,RT}},p^{\mathrm{shed}},p^{\mathrm{spill}}} \quad & [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \boxed{\max_{\mathbb{P}\in\mathcal{P}} \mathbb{E}^{\mathbb{P}}(20p^{\mathrm{G2,RT}}(\Delta W) + 80p^{\mathrm{shed}}(\Delta W))} \\
\text{s.t.} \quad & 0 \le p^{\mathrm{G1,DA}} \le 100 \\
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& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\mathrm{shed}}(\Delta W) \le 120) \ge 1-\epsilon \\
& p^{\mathrm{G2,RT}}(\Delta W) + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}(\Delta W)) + p^{\mathrm{shed}}(\Delta W) = 0
\end{aligned}
$$

What is this?

Distributionally robust optimization (DRO)

$$
\begin{aligned}
\min_{p^{\text{G1,DA}},p^{\text{G2,DA}},p^{\text{WP,DA}},p^{\text{G2,RT}},p^{\text{shed}},p^{\text{spill}}} \quad & [10p^{\text{G1,DA}} + 20p^{\text{G2,DA}}] + \boxed{\max_{\mathbb{P}\in\mathcal{P}} \mathbb{E}^{\mathbb{P}}(20p^{\text{G2,RT}}(\Delta W) + 80p^{\text{shed}}(\Delta W))} \\
\text{s.t.} \quad & 0 \leq p^{\text{G1,DA}} \leq 100 \\
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& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(-p^{\text{G2,DA}} \leq p^{\text{G2,RT}}(\Delta W)) \geq 1-\epsilon \\
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& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\text{G2,RT}}(\Delta W) \leq 30) \geq 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(-30 \leq p^{\text{G2,RT}}(\Delta W)) \geq 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(0 \leq p^{\text{spill}}(\Delta W)) \geq 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\text{spill}}(\Delta W) \leq 30) \geq 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(0 \leq p^{\text{shed}}(\Delta W)) \geq 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\text{shed}}(\Delta W) \leq 120) \geq 1-\epsilon \\
& p^{\text{G2,RT}}(\Delta W) + (30 - p^{\text{WP,DA}} - p^{\text{spill}}(\Delta W)) + p^{\text{shed}}(\Delta W) = 0
\end{aligned}
$$

Worst-case “expected” operational cost of the system in the RT stage

Distributionally robust optimization (DRO)

$$
\begin{aligned}
\min_{p^{\mathrm{G1,DA}},p^{\mathrm{G2,DA}},p^{\mathrm{WP,DA}},p^{\mathrm{G2,RT}},p^{\mathrm{shed}},p^{\mathrm{spill}}} \quad & [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \boxed{\max_{\mathbb{P}\in\mathcal{P}} \mathbb{E}^{\mathbb{P}}(20p^{\mathrm{G2,RT}}(\Delta W) + 80p^{\mathrm{shed}}(\Delta W))} \\
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& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30 - p^{\mathrm{G2,DA}}) \geq 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30) \geq 1-\epsilon \\
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& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\mathrm{shed}}(\Delta W) \leq 120) \geq 1-\epsilon \\
& p^{\mathrm{G2,RT}}(\Delta W) + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}(\Delta W)) + p^{\mathrm{shed}}(\Delta W) = 0
\end{aligned}
$$

Worst-case "expected" operational cost of the system in the RT stage

So, we determine the **worst** distribution within the ambiguity set, and optimize the problem in **expectation** with respect to such a worst distribution!

Distributionally robust optimization (DRO)

$$
\begin{aligned}
\min_{p^{\mathrm{G1,DA}},p^{\mathrm{G2,DA}},p^{\mathrm{WP,DA}},p^{\mathrm{G2,RT}},p^{\mathrm{shed}},p^{\mathrm{spill}}} \quad & [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \max_{\mathbb{P}\in\mathcal{P}} \mathbb{E}^{\mathbb{P}}(20p^{\mathrm{G2,RT}}(\Delta W) + 80p^{\mathrm{shed}}(\Delta W)) \\
\text{s. t.} \quad & 0 \leq p^{\mathrm{G1,DA}} \leq 100 \\
& 0 \leq p^{\mathrm{G2,DA}} \leq 30 \\
& 0 \leq p^{\mathrm{WP,DA}} \leq 70 \\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120 \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(-p^{\mathrm{G2,DA}} \leq p^{\mathrm{G2,RT}}(\Delta W)) \geq 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \leq 30 - p^{\mathrm{G2,DA}}) \geq 1-\epsilon \\
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& p^{\mathrm{G2,RT}}(\Delta W) + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}(\Delta W)) + p^{\mathrm{shed}}(\Delta W) = 0
\end{aligned}
$$

Distributionally robust chance constraints (DRCCs)

Distributionally robust optimization (DRO)

$$\min_{p^{\mathrm{G1,DA}}, p^{\mathrm{G2,DA}}, p^{\mathrm{WP,DA}}, p^{\mathrm{G2,RT}}, p^{\mathrm{shed}}, p^{\mathrm{spill}}} \quad [10p^{\mathrm{G1,DA}} + 20p^{\mathrm{G2,DA}}] + \max_{\mathbb{P}\in\mathcal{P}} \mathbb{E}^{\mathbb{P}}(20p^{\mathrm{G2,RT}}(\Delta W) + 80p^{\mathrm{shed}}(\Delta W))$$

$$\begin{aligned}
\text{s.t.} \quad & 0 \le p^{\mathrm{G1,DA}} \le 100\\
& 0 \le p^{\mathrm{G2,DA}} \le 30\\
& 0 \le p^{\mathrm{WP,DA}} \le 70\\
& p^{\mathrm{G1,DA}} + p^{\mathrm{G2,DA}} + p^{\mathrm{WP,DA}} = 120\\
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& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\mathrm{G2,RT}}(\Delta W) \le 30 - p^{\mathrm{G2,DA}}) \ge 1-\epsilon\\
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& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(-30 \le p^{\mathrm{G2,RT}}(\Delta W)) \ge 1-\epsilon\\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(0 \le p^{\mathrm{spill}}(\Delta W)) \ge 1-\epsilon\\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\mathrm{spill}}(\Delta W) \le 30) \ge 1-\epsilon\\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(0 \le p^{\mathrm{shed}}(\Delta W)) \ge 1-\epsilon\\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\mathrm{shed}}(\Delta W) \le 120) \ge 1-\epsilon\\
& p^{\mathrm{G2,RT}}(\Delta W) + (30 - p^{\mathrm{WP,DA}} - p^{\mathrm{spill}}(\Delta W)) + p^{\mathrm{shed}}(\Delta W) = 0
\end{aligned}$$

Distributionally robust chance constraints (DRCCs)

Discussion: What does a DRCC enforce?

Distributionally robust optimization (DRO)

$$
\begin{aligned}
\min_{p^{\text{G1,DA}},p^{\text{G2,DA}},p^{\text{WP,DA}},p^{\text{G2,RT}},p^{\text{shed}},p^{\text{spill}}} \quad & [10p^{\text{G1,DA}} + 20p^{\text{G2,DA}}] + \max_{\mathbb{P}\in\mathcal{P}} \mathbb{E}^{\mathbb{P}}(20p^{\text{G2,RT}}(\Delta W) + 80p^{\text{shed}}(\Delta W)) \\
\text{s. t.} \quad & 0 \le p^{\text{G1,DA}} \le 100 \\
& 0 \le p^{\text{G2,DA}} \le 30 \\
& 0 \le p^{\text{WP,DA}} \le 70 \\
& p^{\text{G1,DA}} + p^{\text{G2,DA}} + p^{\text{WP,DA}} = 120 \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(-p^{\text{G2,DA}} \le p^{\text{G2,RT}}(\Delta W)) \ge 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\text{G2,RT}}(\Delta W) \le 30 - p^{\text{G2,DA}}) \ge 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\text{G2,RT}}(\Delta W) \le 30) \ge 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(-30 \le p^{\text{G2,RT}}(\Delta W)) \ge 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(0 \le p^{\text{spill}}(\Delta W)) \ge 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\text{spill}}(\Delta W) \le 30) \ge 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(0 \le p^{\text{shed}}(\Delta W)) \ge 1-\epsilon \\
& \min_{\mathbb{P}\in\mathcal{P}} \mathbb{P}(p^{\text{shed}}(\Delta W) \le 120) \ge 1-\epsilon \\
& p^{\text{G2,RT}}(\Delta W) + (30 - p^{\text{WP,DA}} - p^{\text{spill}}(\Delta W)) + p^{\text{shed}}(\Delta W) = 0
\end{aligned}
$$

Distributionally robust chance constraints (DRCCs)

<u>Note</u>: The worst distribution obtained for a DRCC might **<u>not</u>** be necessarily the same as that of other DRCCs or that of the objective function!

Discussion

- By **enlarging** the ambiguity set, does the solution of DRO converges towards the solution of a robust optimization?

- By **shrinking** the ambiguity set, does the solution of DRO converges towards the solution of a stochastic program?

Discussion

- By **enlarging** the ambiguity set, does the solution of DRO converges towards the solution of a robust optimization?

- By **shrinking** the ambiguity set, does the solution of DRO converges towards the solution of a stochastic program?

Yes and Yes! DRO somehow provided a generalized model, linking robust optimization and stochastic programming!

How to build an ambiguity set?

How to build an ambiguity set?

Two common approaches:

1) Metric-based ambiguity sets

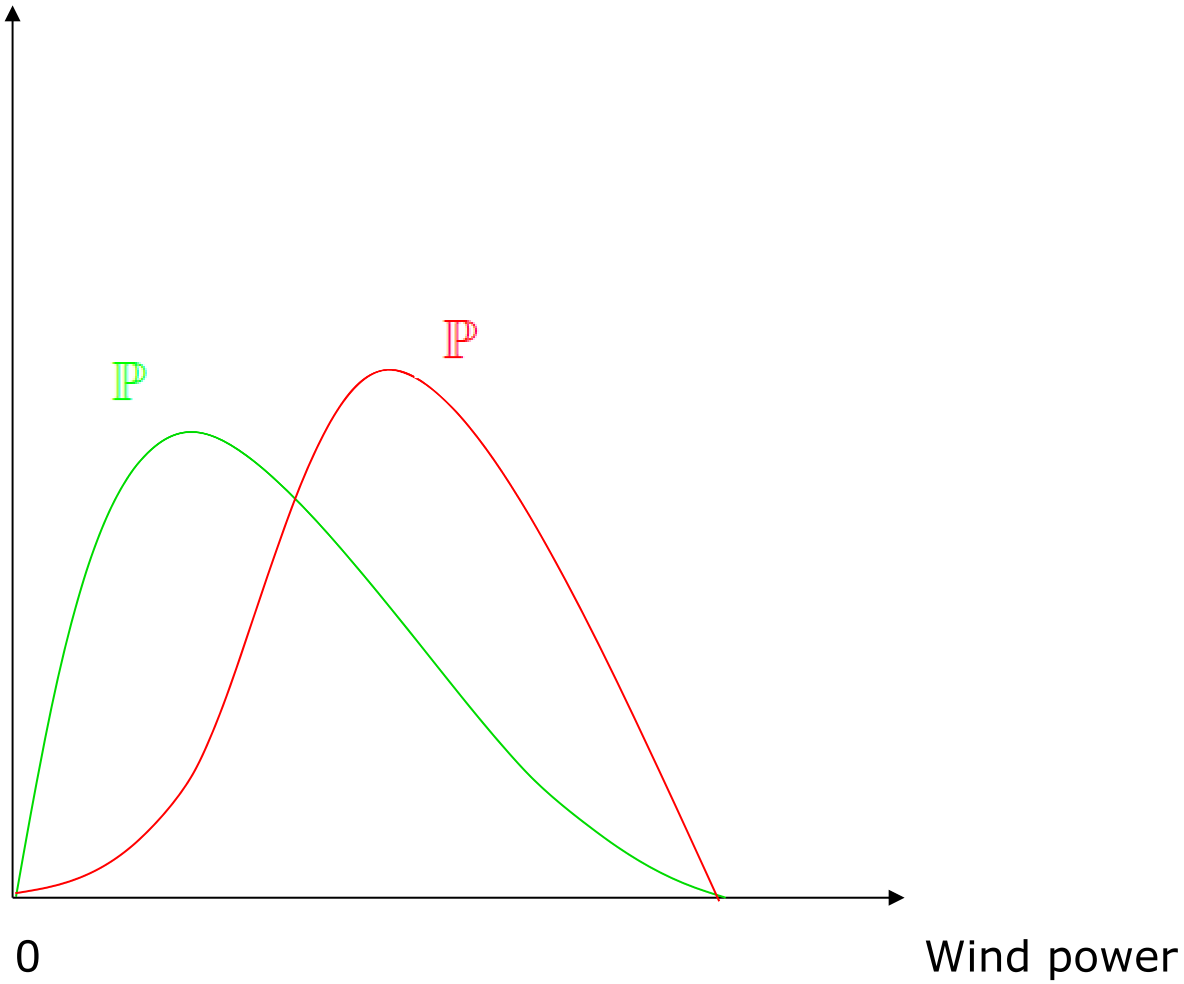
- Use historical data to build an empirical probability distribution to approximate the true probability distribution
- Use probability metrics, e.g., Wasserstein distance, to define ambiguity set

2) Moment-based ambiguity sets

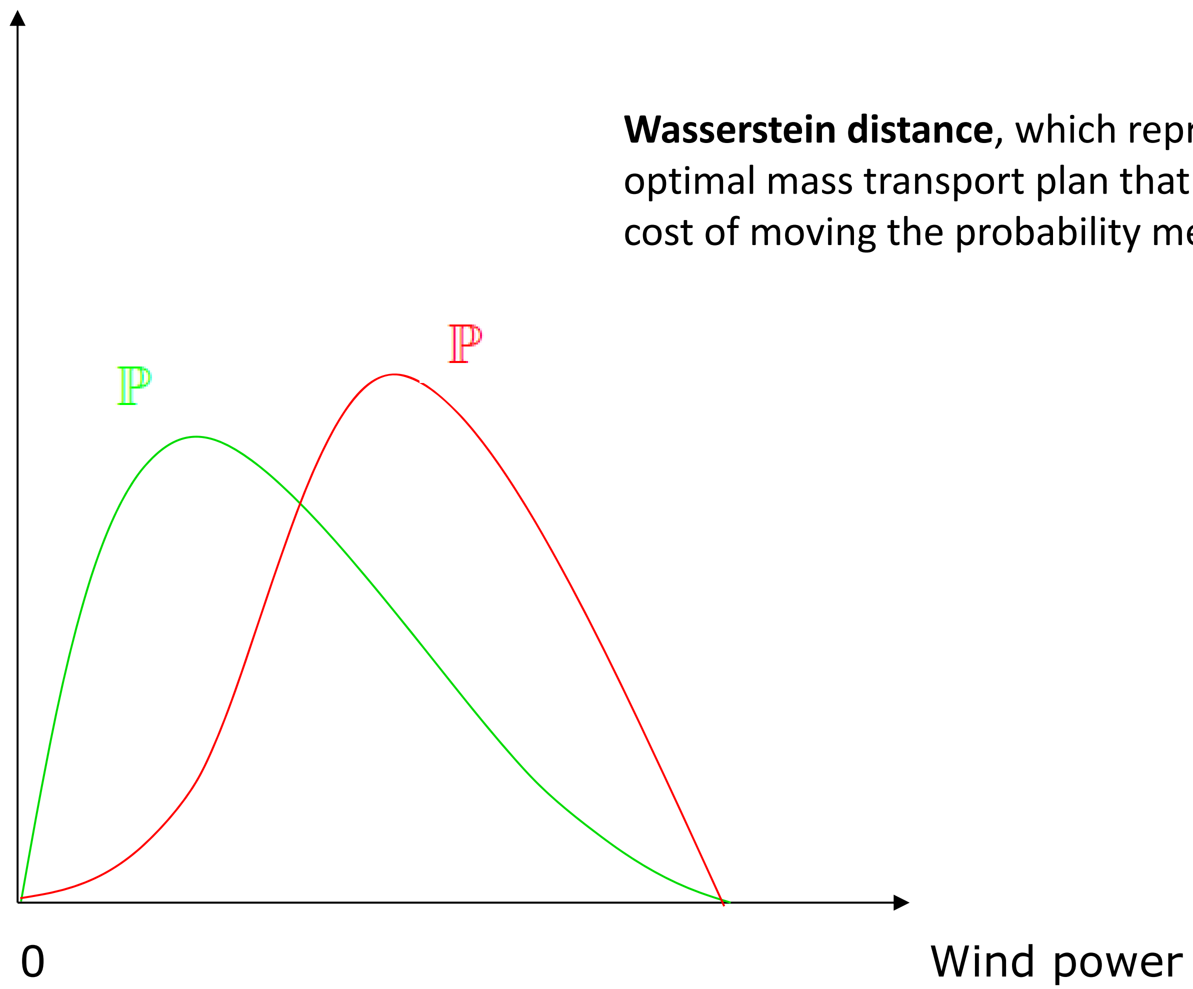
- Use historical data to infer moments, e.g., mean and covariance
- Build ambiguity set based on these moments

Metric-based ambiguity set

Metric-based ambiguity set



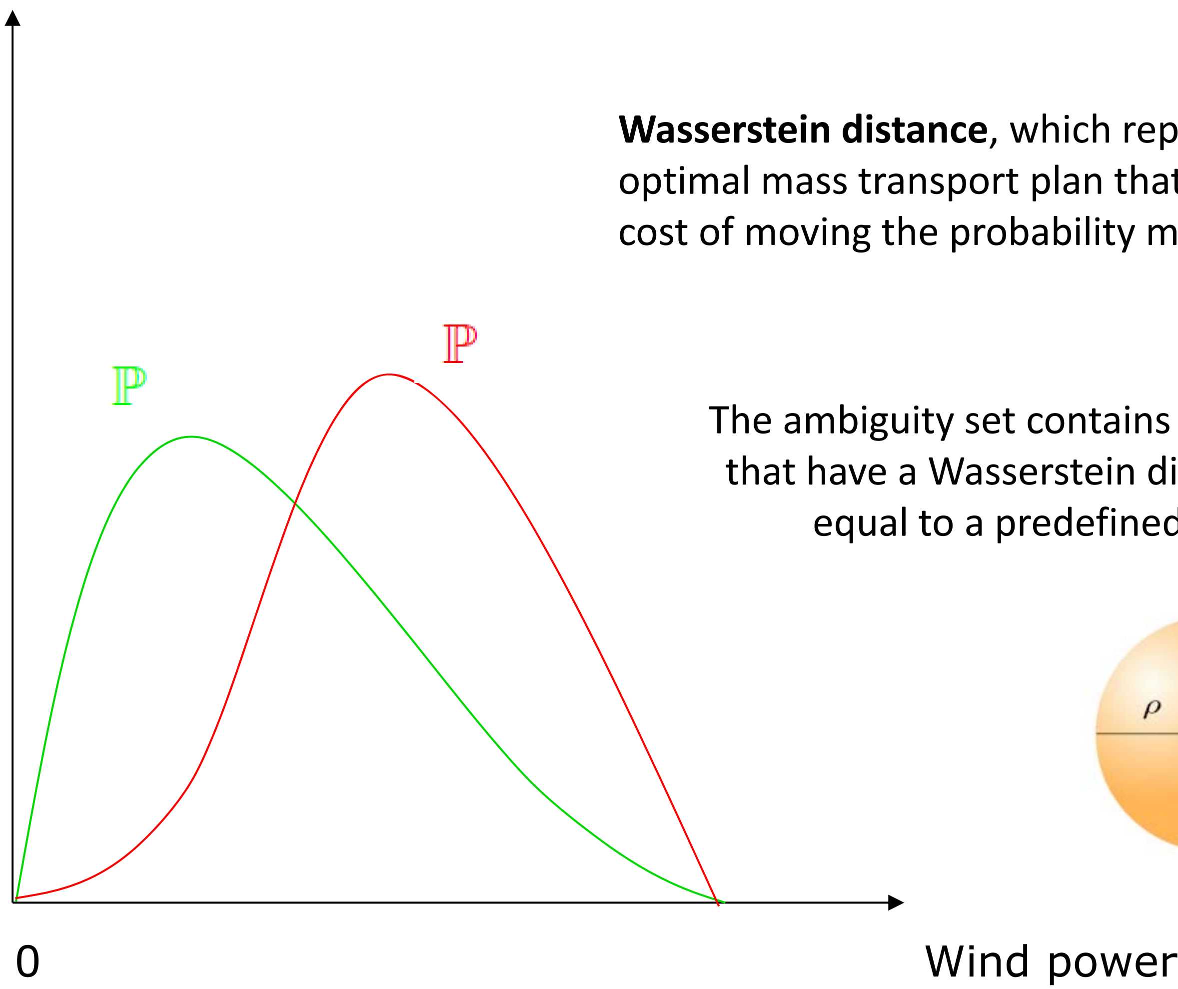
Metric-based ambiguity set



Wasserstein distance, which represents an optimal mass transport plan that minimizes the cost of moving the probability measure $\mathbb{P}$ to $\mathbb{P}$

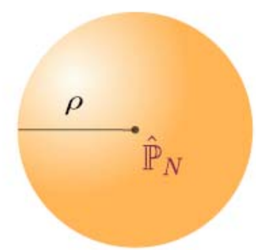
Video online for Wasserstein metric: https://youtu.be/ymWDGzpQdls

Metric-based ambiguity set

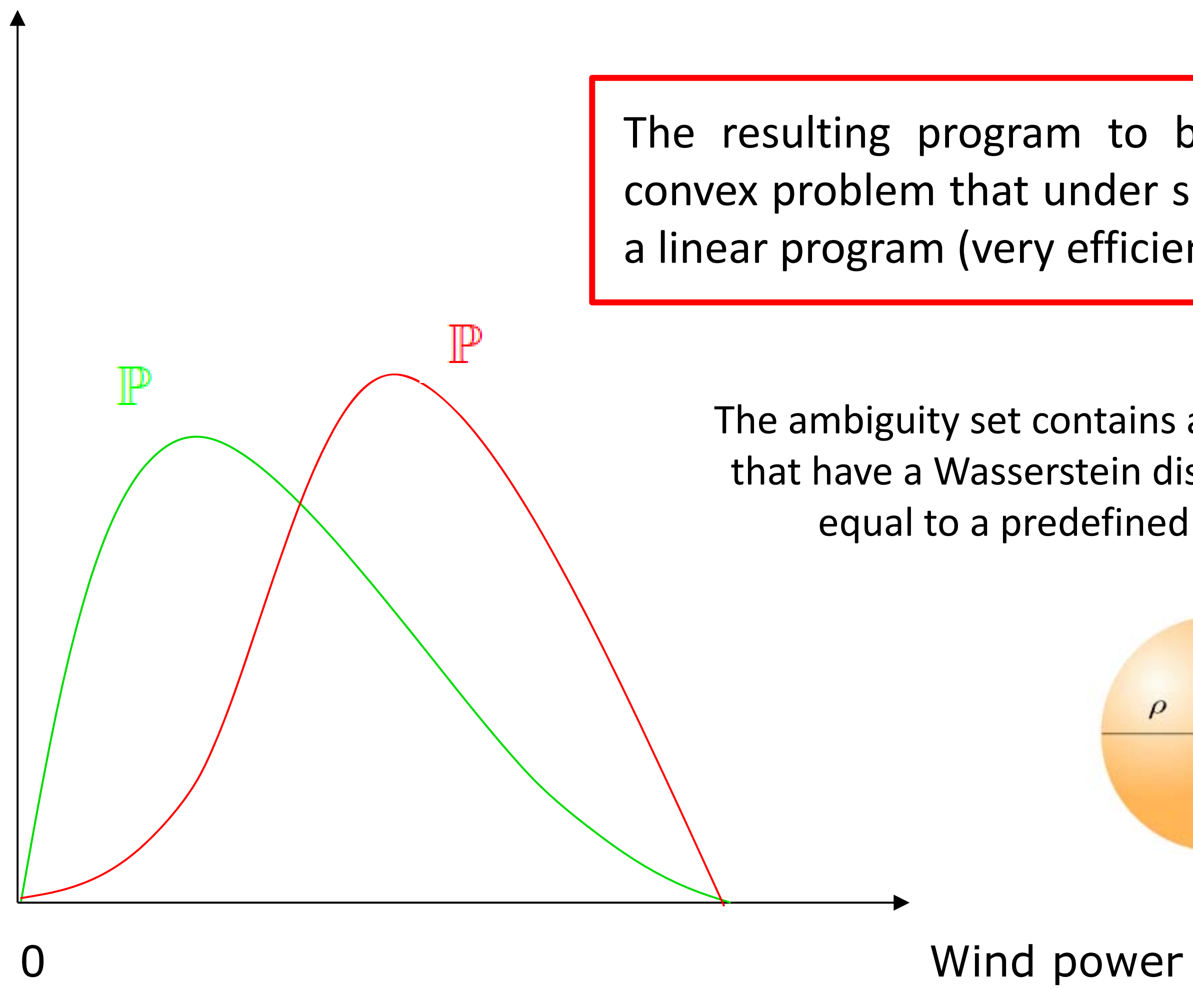


Wasserstein distance, which represents an optimal mass transport plan that minimizes the cost of moving the probability measure $\mathbb{P}$ to $\mathbb{P}$

The ambiguity set contains all distributions that have a Wasserstein distance at most equal to a predefined value (ρ)

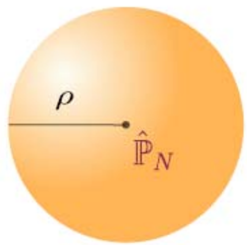


Metric-based ambiguity set



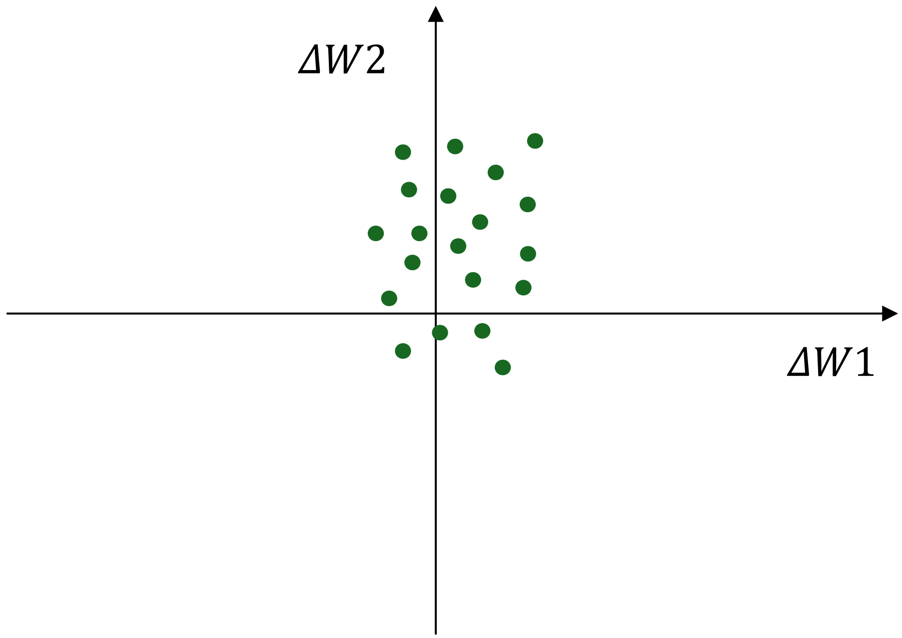
The resulting program to be solved is a convex problem that under specific cases is a linear program (very efficient to solve)

The ambiguity set contains all distributions that have a Wasserstein distance at most equal to a predefined value (ρ)



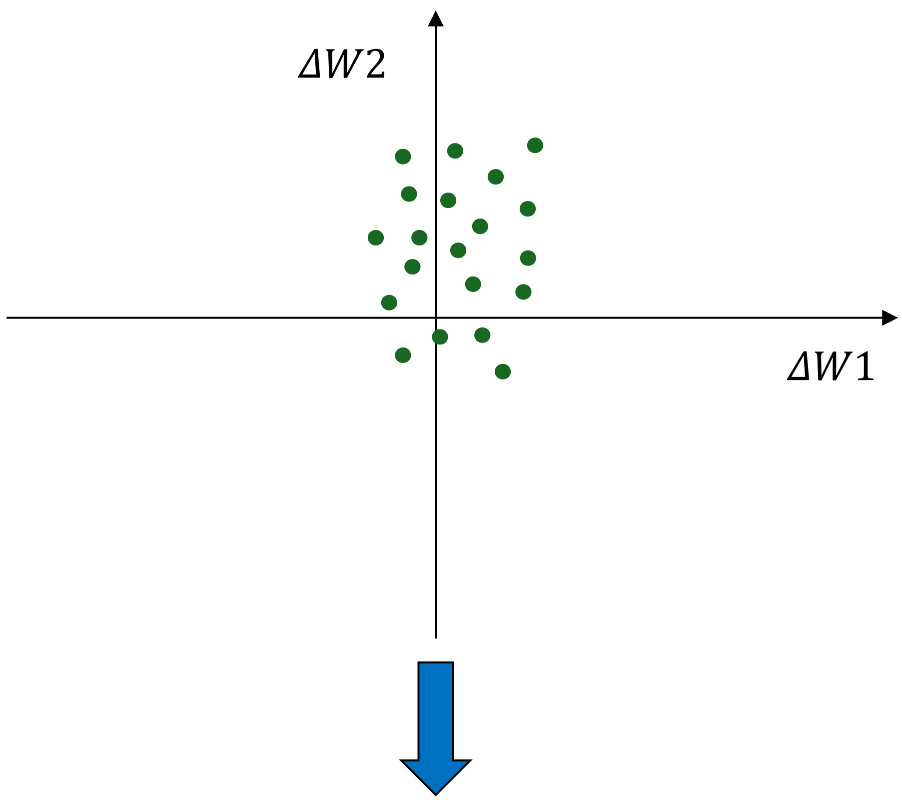
Moment-based ambiguity set

Moment-based ambiguity set



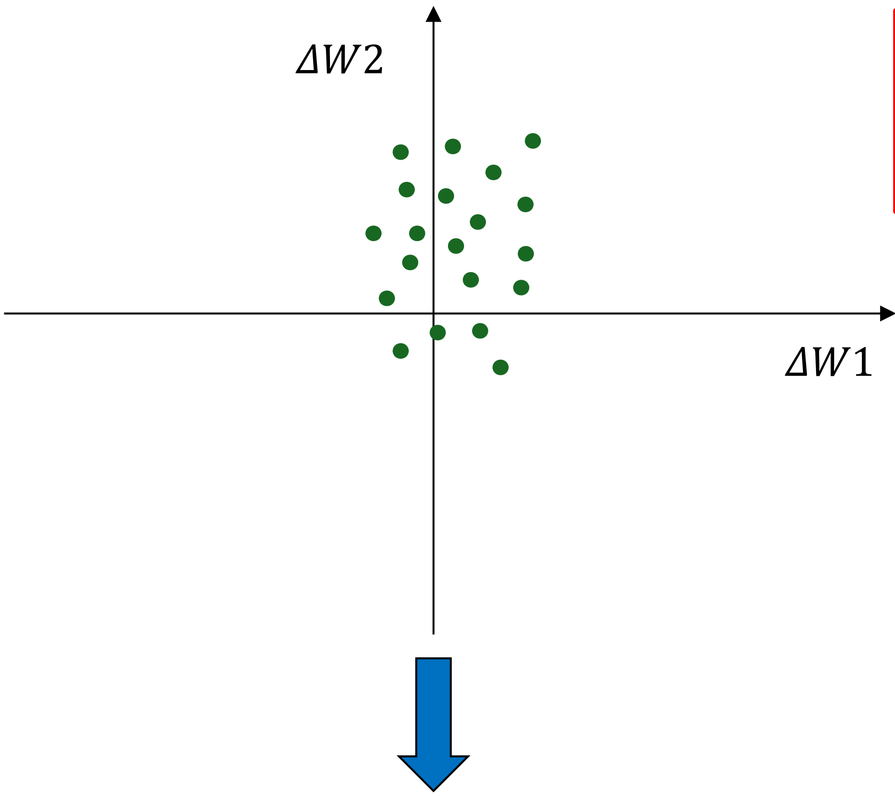
Jalal Kazempour

Moment-based ambiguity set



Calculate mean and covariance

Moment-based ambiguity set



Calculate mean and covariance

The resulting program to be solved is a second-order cone program (very efficient to solve)

The ambiguity set contains all distributions that have the specified mean and covariance!

References on distributionally robust optimization

- W. Xie and S. Ahmed, "Distributionally robust chance constrained optimal power flow with renewables: A conic reformulation," *IEEE Transactions on Power Systems*, vol. 33, no. 2, pp. 1860–1867, 2018.
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- Y. Zhang, S. Shen, and J. L. Mathieu, "Distributionally robust chance constrained optimal power flow with uncertain renewables and uncertain reserves provided by loads," *IEEE Transactions on Power Systems*, vol. 32, no. 2, pp. 1378–1388, 2017.
- Online video lectures by Prof. Daniel Kuhn:
  - https://youtu.be/b4lJENGAeEA
  - https://youtu.be/M2NcdKwn824

Thanks for your attention!

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